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Spatio-Temporal Modeling of Equity Markets Using Graph Neural Networks

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Abstract

This thesis investigates the efficacy of spatio-temporal Graph Neural Networks (GNNs) in capturing the complex, non-linear interdependencies of the Indian equity market for cross-sectional stock return ranking. Traditional forecasting models often treat assets as independent time series, thereby failing to account for the persistent liquidity dynamics and cross-stock spillover effects that are fundamental to short-term price formation. To address this limitation, I propose a quantamental graph framework that integrates market-cap-scaled volume shocks as node-level features alongside price-correlation and sectoral edges. The framework is optimized using a novel Hybrid Quantile-Ranking Loss, which simultaneously enforces cross-sectional ordinal ranking and calibrates downside risk bounds (Q_{10}, Q_{50}, Q_{90}). I conduct a rigorous comparative evaluation of three architectural paradigms: a temporal-only LSTM baseline, a parallel-fusion GAT-LSTM hybrid, and a stateful dynamic ROLAND architecture. Empirical results on a 103-stock universe from the NSE indicate that the GAT-Hybrid framework significantly outperforms the temporal baseline, achieving an annualized Sharpe ratio of 2.341 and identifying statistically significant alpha in the long leg. Conversely, although the dynamic ROLAND architecture demonstrated the highest validation-set predictive power, it exhibited substantial out-of-sample degradation, highlighting the tension between high-capacity sequential modeling and temporal distribution shifts in financial regimes. These findings suggest that incorporating microstructure-aware liquidity features and graph-based relational topology is essential for robust alpha generation, while also providing a foundation for future research into cross-market transfer learning and dynamic quantile calibration.

Chapter 1

Introduction

Forecasting in stock market is a highly complex, dynamic, and a non-linear system which is driven by a lot of interconnected factors, such as market volatility, non-linearity, and sensitivity to various factors, which makes accurate price forecasting one of the most challenging problems in financial analysis. Traditionally, quantitative researchers and investment professionals have relied on linear regression or statistical autoregressive time-series models (such as ARIMA or GARCH), or fundamental analysis to predict stock price movements. The traditional models fail to capture complex dependencies and temporal dynamics in financial data (Gandhmal & Kannan, 2019). Their decisions have mostly been based on historical data of specific stock segments or with linearity assumptions. Early machine learning models such as Support Vector Machines (SVM) and Artificial Neural Networks (ANN) improved on traditional methods by identifying historical patterns but struggled with the sequential nature of financial time series. Convolutional neural networks (CNNs) excel in capturing spatial patterns but are less effective with the temporal dependencies essential for stock forecasting as highlighted by (Senapati et al., 2018). Recently, we witnessed a paradigm shift towards "Quantamental" investing, which is a hybrid approach, leveraging machine learning to synthesize massive datasets of human-driven fundamental research and computer-driven quantitative signals.

While advanced deep learning models, particularly Long Short-Term Memory (LSTM) networks, have shown immense promise in capturing sequential, long-term temporal dependencies in historical stock prices, they possess a critical limitation: they largely treat individual stocks in isolation. However, the financial market

operates as a deeply interconnected network. The performance of a single stock is rarely independent; and continuously influenced by cross-market correlations, sector trends, and shared investor behaviour.

Graph Neural Networks (GNNs) have emerged as a powerful solution to bridge the gap between technical and fundamental analyses. By conceptualizing the stock market as an interconnected graph, where individual stocks act as "Nodes" and their complex interdependencies act as "Edges", GNNs can capture both the temporal dynamics of individual assets and the corresponding spatial (relational) interactions between them. However, market relationships are not static; the flow of capital and the correlation between stocks evolve continuously. Advanced dynamic frameworks, such as the ROLAND architecture, address this by viewing node embedding as hierarchical states that update recurrently over time, enabling the model to scale and adapt to dynamic graphs efficiently

The primary objective of this thesis is to forecast stock prices and rank stocks over a 7-day target horizon by evaluating and comparing the predictive capabilities of purely temporal models against spatial-temporal and dynamic graph architectures. Specifically, this research conducts a rigorous comparative analysis between a standalone LSTM model, static hybrid graph models (GAT integrated with LSTM), and a dynamic, evolving graph framework (ROLAND). Combining LSTM and GNN models into hybrid architectures has shown promise in analysing both temporal and relational aspects of stock data (Thakkar and Chaudhari 2021; Ran et al. 2024).

This thesis constructs a multi-dimensional dataset focusing on the stocks from liquid Futures & Options (F&O) security segment of the National Stock Exchange (NSE) of India. The research scope filters out recently listed equities to ensure data completeness and structural integrity across the entire historical timeframe of the analysis.

Chapter 2

Literature Review

During my research, I explored the evolution of financial forecasting that moved from traditional statistical and fundamental analysis to an integrated quantamental approach over the years. Former approaches relied on time-series econometrics and financial ratios and often treated the assets as independent entities. Recent research addresses this limitation by modelling the stock market as a complex network rather than isolated time series. The implementation of Graph Neural Networks (GNNs) enables simultaneous learning of both individual stock dynamics and cross-sectional dependencies. The model acts as a complex interconnected system with stocks acting as graph nodes, passing information to each other through connected edges that represent relationships between the stocks.

Quantamental Research and Multi-Factor Modelling

Quantamental Investing is a paradigm shift in portfolio management, which is defined as a combination of statistical technical indicators and quantitative analysis with human-derived fundamental research. This approach leverages AI and ML to fill the gap where humans fail to evaluate the thousands of combinations required for portfolio construction with a core objective to generate alpha (excess returns) by using data-driven insights in the investment process as presented by Tadoori and Yakaiah [5].

Various techniques involve multi-factor regression models that quantify linear relationships between a company's financial ratios and expected returns. Economic intuition suggests that the fundamental factors and financial ratios about profitability, revenues, assets and liabilities directly explain changes in stock prices. The

key signals include valuation factors (e.g. P/E, P/S, P/B), growth factors, and other financial factors (Total Asset Turnover, Asset Liability Ratios etc.) [6]. The quantamental strategies have advantages with increased discipline in investment decisions with the ability to analyze large groups of investments concurrently, and excellent risk control through hedging [5]. For e.g., a multi-factor regression strategy implemented on the Chinese-A-share market achieved an annualized return of 14% to 28% in backtests while maintaining a market-neutral profile [6]. However, the authors used quarterly rebalancing of the portfolio and good financial data in a quarter may not directly lead to the next good quarter.

GNN Implementations

When quantamental research focusses on a stock's internal health, GNN implementations address limitations of traditional models (e.g. ARIMA, GARCH, or standalone LSTMs) that treat stocks in isolation and ignore the relationship network between companies and industries. With GNN the market constituents are seen as a graph-structured system with stocks as nodes and their relationship (sectorial dependency, price-correlation, news, value-chain, investor holdings etc.) as edges.

There are four predominant models and studies in literature as discussed below:

Hybrid and Spatial-Temporal Architecture The research by Rahi et al. [7] emphasized that hybrid LSTM-GNN or GCN-GRU models capture both temporal patterns and spatial relationships. They introduced a non-IID Spatial-Temporal Graph Neural Network (NIST-GNN) model to capture the spatial relationships of the companies as well as patterns in sequences of the stock. In this paper, historical stock-prices and trading volume were collected over a period of 5 years across different sectors (including technology, healthcare, finance, and energy), Named Entity Recognition (NER) was applied to financial news articles to extract entity names like companies, sectors, financial terms to help with linkage of specific events to relevant stocks or industries. The temporal dynamics were captured using lagged variables for prior day/week, and moving averages were applied over different time windows to model trends and smooth short-term volatility. The study represented stocks as nodes, with stock-specific features such as price, volume, volatility as

node features. Further, edges were created based on inter-stock relationships such as historical price correlation and sectoral/industry membership, and the edge features quantified the strength of relationship to capture any spillover effects. It also included interaction features combining heterogenous signals (e.g. sentiment measures) to model cross-domain dependencies. All these factors are then combined together to reflect upon broader economic conditions and company fundamentals influencing stock performance. The model used a Semantic Company Relationship Graph (SCRG) based on financial news embeddings to achieve a Sharpe Ratio of 0.40, thus significantly outperforming benchmark models.

Similarly, the LSTM-GNN hybrid model for stock-price prediction using historical market data for 10 large-cap US companies across multiple sectors has demonstrated a 10.6% reduction in Mean Squared Error (MSE) compared to standalone LSTM by integrating relational data, thus addressing the effectiveness of integrating relational data through the GNN component. The study involved data preprocessing using min-max normalization to ensure numerical stability and prevent scale dominance. The sequences of normalized closing prices were structured as time-series batches and then processed using a stacked LSTM architecture to capture sequential dependencies for temporal modeling. A graph structure $G = (V, E)$ was used to model cross-stock interdependencies where stocks were represented as nodes (V), and relationships between stocks represented as edges (E). Linear relationships were quantified using Pearson correlation coefficient which was computed on daily returns and edges were retained when $i, j > 0.70$, where i, j represents different stocks. The study also disclosed the usage of Apriori algorithm to capture non-linear dependencies and identify co-movement patterns exceeding random expectation. The model was trained using an expanding window validation strategy, simulating real-world forecasting combined with early-stopping and grid-based hyperparameter tuning. The performance of the model was evaluated using MSE, and the results demonstrated that hybrid LSTM-GNN model outperformed standalone LSTM and other baseline models, thus highlighting the benefit of jointly modelling temporal dynamics and inter-stock relational structures [8].

Evolving and Dynamic Graph Modeling Since markets are dynamic, a significant development in GNN implementation is a shift from static to evolving

graph structures. The study "Stock market price forecasting using evolving graph neural network" by Korablyov et al. [9] addressed the limitation of traditional GNN applications in finance, namely the reliance on static relationship graphs. The financial markets are dynamic and stock relationships change constantly across different time scales. They proposed a framework that automatically updates graph structures for every trading day. They used a dendritic artificial immune network (DaiNet) to generate Evolving Learning Graphs (EGL). DaiNet used biologically inspired clustering principles to capture changing stock correlations for each trading day. They used a natural affinity threshold (NAT) to build edges between stocks, so connections falling below the threshold were pruned to reduce noise and retain stronger relationships. Furthermore, they used a two-stage graph attention layer to adaptively calculate the importance of neighbouring stocks and aggregate messages accordingly. They compared the evolving graph model against a static graph model and the evolving graph outperformed the static one by 3–5% and achieving approximately 62% forecasting accuracy. However, while this accuracy improvement is statistically significant, the study does not account for transaction costs or slippage, leaving it ambiguous whether the 62% accuracy translates into economically meaningful trading profitability. Furthermore, the computational overhead of recalculating distances and reconstructing the relationship graph every trading day poses scalability challenges, and the model's performance remains highly sensitive to the dynamic calculation of the NAT threshold.

The interactions between stocks change across time scales due to changes in price levels, investor interest, market policies etc. A separate framework called ROLAND also addresses this issue where node embeddings are viewed as hierarchical states and are updated on a rolling basis, allowing the model to scale to dynamic graphs with even millions of edges. The authors introduced a live-update evaluation setting that predicts future links on a rolling basis and utilizes incremental training to remain scalable for massive datasets. They incorporated meta-learning to ensure that the model adapts to evolving data patterns and achieved significant performance improvements over state-of-the-art baseline models.

To overcome the massive computational costs and memory bottlenecks associated with dynamic graph reconstruction, ROLAND utilizes an incremental training

strategy that retains only historical node states and the latest graph snapshot in memory, allowing it to easily scale to datasets with up to 56 million edges [10]. This approach if implemented in financial markets can uncover hidden insights and may outperform the traditional graph-based models.

Cross-Market and Cross-Chart Analysis The scope of graphical analysis has been expanded to include cross-border relationships and multi-indicator charts. MEIG (Macro-Event Driven Inter-Intra Graph) model explores interactions between developed and emerging economies, by incorporating macroeconomic indicators such as GDP, interest rates and unemployment rates taken as node features. A key innovation is the Cross-Graph Attention Layer (CGAT), which dynamically weighs the importance of local "intra-market" connections versus global "inter-market" networks during the prediction process [11]. This model relied heavily on intra- & inter-event sentiment analysis that required authors to scrape and process massive corpus of over 42.74 million tweets, i.e. huge data curation and natural language processing overhead. Further, the model struggled to forecast accurately when dealing with highly volatile macroeconomics variable across vastly different economies.

Another study "CIRGIN" (Chart Indicator Relationship Graph Neural Network) [12] implemented a framework that transformed technical indicators (SMA, EMA, Momentum) into visibility graphs, learning asymmetric and directional relationships between different indicator charts to improve efficiency. They tested the model in a live trading simulation that accounted for real-world trading frictions with 0.05% transaction costs and slippage. The major limitation being that CIRGIN only mapped relationship of indicators within a single stock but ignored how technical indicators of one stock might influence another stock (*cross-stock indicator influence*). Further, they highlighted the fact that increasing graph convolution depth beyond 3 leads to overfitting, thus model being very sensitive to hyperparameter tuning.

Multimodal Agentic Systems GNNs have also been deployed as a decentralized system in multimodal agentic architectures specifically designed for High-Frequency Trading (HFT). The trading challenge is decomposed into specialized

independent AI agents managing distinct capital pools and handling different data modalities. For example, News agent (LLM based) for sentiment analysis, a Spike Agent (GNN-based) for order book snapshots, and a Strategy Agent (RL-based) for technical vectors. This modularity ensured that system is resilient and allows GNN component to focus specifically on detection of institutional flow spikes during GAT networks. This architecture has demonstrated substantial economic viability; live integration testing with retail brokers yielded execution latencies under 200 milliseconds and annualized returns between 20% and 28%. But at the same time achieving this performance introduces immense data and infrastructure dependencies. The GNN component requires the continuous, real-time ingestion of granular high-frequency data (such as L2 order book depth), which is computationally intensive to process and often subject to strict broker API rate limits. Furthermore, while the modular design isolates individual agent failures, the overarching system's stability relies heavily on the assumption that a higher-order "meta-agent" can flawlessly calculate and execute dynamic capital rebalancing during sudden, highly volatile market regime shifts [13].

Trading Volume, Institutional Bulk Deals & Market Impact

Trading Volume as a Proxy for Information Flow Empirical studies have established a robust, positive correlation between trading volume and absolute value of price changes, one prominent research being "Mixture of Distributions Hypothesis" (MDH) which postulated that both price variance and trading volume are jointly driven by a latent directing process i.e. rate at which new information flows into the market [14]. Furthermore, Andersen [16] also studied that abnormal trading volume (e.g. execution of bulk deals) acts as an observable proxy for unobservable arrival of new information. These theoretical models partition volume into "noise" (or liquidity) trading and "informed" trading, and the systematic trading volume are driven almost entirely by informed investors acting on private signals or fundamental analysis. Thus, incorporating bulk deal volume into predictive models captures the "effective time" or actual informational evolution of the asset, rather than just its calendar time evolution.

Market Impact and Capitalization Scaling Lillo et al. [17] analyses direct market impact of bulk transactions and complex dynamic between "liquidity takers" (informed institutional investors placing market orders) and "liquidity providers" (market makers). The price-impact function shows that price shift caused by a transaction is dependent on both the volume of trade and specific stock's market capitalization. They found that higher-capitalization stocks tend to have smaller price responses for the same normalized transaction size. Their empirical econ-physics research demonstrated that to create a uniform, universal curve for price impact across different stocks, the transaction volume must be rescaled by the firm's market capitalization. They defined bulk-deal intensity as a dimensionless ratio of deal value to market capitalization, so predictive models can effectively normalize the "liquidity shock" across the entire cross-section of the market, ensuring structurally comparable signals for both small-cap and large-cap equities. Furthermore, the price-impact function was found to be strongly concave, meaning large volumes impact prices less in relative terms than smaller volumes, justifying logarithmic transformations of scaled volume.

Temporal Persistence and Decay of Order Flow Bouchaud et al. [15] addressed a separate concern where sudden massive bulk deals would trigger immediate price jumps and exorbitant trading costs. Institutional investors often divide their large block trades into smaller, sequential chunks to avoid detection. This strategic behavior creates powerful, long-range temporal correlations in the sign and volume of trades, where persistent directional volume signals underlying institutional accumulation or distribution. Simultaneously, liquidity providers continuously update their reference prices based on these volume shocks. The market's "bare propagator" (the function describing the impact of a trade over time) exhibits a decaying behavior as market makers attempt to slowly mean-revert the price against the trading shock to optimize their gains. This mean reversion can be mathematically modeled as an exponential decay toward a finite asymptotic value. Therefore, applying an exponential decay function to bulk deal signals in predictive modeling perfectly simulates this "price memory" and information assimilation phase. It allows models to perceive not just the isolated shock of a single bulk deal, but the persistent, residual footprint of institutional accumulation

over a rolling time window.

While trading volume and bulk deals offer valuable predictive signals, their interpretation in financial modeling is constrained by several market dynamics. The "costly short sales hypothesis" highlighted an asymmetry where volume is typically higher during price increases than price decreases; because shorting is more restricted and costly than going long, models that treat volume symmetrically may overestimate bullish momentum and underestimate bearish signals. Furthermore, predictive architectures must distinguish structurally informed trades from noise-driven liquidity demands, such as hedging or index rebalancing, since incorrectly treating all large volume spikes as informed bulk deals degrades forecasting accuracy. The predictable, exponential decay of market impact relied on a delicate equilibrium between liquidity takers and providers that often breaks down during market regime shifts or crises; when sudden cooperativity among market orders outweighs stabilizing forces, fixed historical decay parameters fail, necessitating highly adaptive or meta-learning layers to prevent severe drawdowns. Finally, the predictive viability of public bulk deal data is limited by reporting lags; because market prices adjust to new information extremely quickly, the initial price impact is often fully absorbed before the transaction becomes public knowledge, requiring ultra-low latency processing to be effectively utilized.

Information Coefficient as Quantitative Tool

The Information Coefficient (IC) is a quantitative metric which is used to assess the predictive power of investment signals/ factors or "alpha factor". It measures the correlation between predicted scores of a factor (e.g., value or momentum) and the actual returns of the assets. According to Zhang et al. [18], the IC indicates how effectively a model's rankings forecast future performance. It is equivalent to the OLS slope coefficient in a regression of normalized actual returns on normalized predicted returns, giving it a firm statistical foundation. It serves as a direct gauge of forecasting skill, independent of the magnitude of the predictions. For a signal to be tradeable, it simply needs to rank securities correctly by predicted performance without the need to predict exact price targets. IC values range from -1 to 1. The positive values show that predictions align with actual returns, while negative values suggest an inverse relationship. The values close to zero indicate

no predictive power.

The Pearson IC measures linear correlation between raw scores and returns, while the Spearman Rank IC operates on ranked values, making it more robust to outliers and better suited to stock-ranking applications. For a ranking-and-trading model, Spearman IC is a more relevant measure, as the signal only needs to correctly order securities without predicting precise return levels [21, 20].

Empirically, IC values are far smaller than intuition suggests. Values of 0.05–0.15 represent strong, actionable signals. IC links directly to portfolio construction and shows that Quintile Spread $\approx 2.80 \times$ IC, and that any signal with a negative IC will produce a negative quintile spread, setting a theoretical lower bound for ICs of all permissible models [21, 18]. IC is statistically noisy, particularly in small universes. Zhang et al. [18] show the estimated IC $\sim \mathcal{N}\left(0, \frac{1}{N-3}\right)$, meaning recovery bias is severe at low N. Realized ICs are also highly volatile over time, with the time-varying component driving $\approx 95\%$ of total IC variability, making snapshot IC values largely uninformative and necessitating rolling-window evaluation. Kirsten [20] further documents IC decay rates of 20–50% over six months, reinforcing the need to match signal horizon to holding period. The key limitations for a trading application includes instability over short samples, look-ahead and survivorship biases in backtests, and the implementation gap, IC does not account for transaction costs or liquidity, so a positive IC does not guarantee positive net returns [21, 22].

Zhang et al. [18] formalized the relationship between IC and portfolio returns using the Quintile Spread metric:

$$\text{Quintile Spread} = (E[x > 80\text{th percentile}] - E[x < 20\text{th percentile}]) \approx 2.80 \times \text{IC}$$

Under normality assumptions, it shows that even modest IC values translate into economically meaningful return spreads between top and bottom quintile portfolios, a finding that directly motivates the ranking-based portfolio construction approach adopted in this study.

Model Validation and Resolution of Data Leakage in Time Series Forecasting

The model validation in machine learning uses techniques suck as k-fold cross validation based on the assumption that the observations are independent and identically distributed (i.i.d). In financial or any other temporal datasets, the data points are intrinsically correlated across time. The standard randomized or sequential time split allows the model to inadvertently access future information during the training phase, thus causing look-ahead bias (i.e. data leakage). This structural flaw causes the model to memorize overlapping sequential patterns rather than generalizing thus resulting in inflated out-of-sample performance metrics and ultimately failing to translate into real-world efficacy. To preserve chronological integrity of data and mitigate direct look-ahead bias the standardized splits are replaced with walk-forward validation. The training set is then set to precede validation set, which in turn precedes the test set. By ensuring that the information flows exclusively in forward direction the model evaluates only unseen future data, thus accurately mirroring the constraints of live trading data. While the chronological splitting of data prevents look-ahead bias, there is still a more subtle look-ahead bias persists at the boundary between contiguous training, validation and test datasets. This boundary leakage is a mathematical consequence of predictive architecture's dimension, the backward looking feature window (look-back) and the forward looking target window (horizon). To prevent the model from evaluating historical features derived from the exact temporal-period as its training targets, the overlapping records must be "purged" from the boundary. Furthermore, as the markets exhibit slow-moving autocorrelation and momentum, an "embargo" period, a temporal dead-zone after the training phase allows the market memory to reset before out-of-sample validation begins [24].

Research Gap

While recent Graph Neural Network (GNN) architectures have demonstrated efficacy in capturing statistical co-movements across equities, current literature exhibits three critical limitations regarding target optimization, liquidity integration, and temporal-topological fusion.

First, the vast majority of deep learning forecasting models optimize for the

conditional mean via Mean Squared Error (MSE). This standard formulation is fundamentally misaligned with market-neutral portfolio construction, as it treats all absolute errors equally, failing to prioritize relative cross-sectional ranking. Furthermore, mean-centric models ignore the fat-tailed nature of financial returns, failing to provide downside risk bounds or quantify predictive uncertainty.

Second, prior GNN-based stock models typically incorporate trading volume strictly on the input side as contemporaneous node features. While volume-based inputs are valuable for pattern recognition, relying solely on input features fails to address the optimization side of the problem: target price movements that occur on low volume carry fundamentally less economic information than those on high volume [17, 15]. An optimization framework that ignores this distinction treats all target returns equally, risking over-parameterization to microstructural noise and illiquidity artifacts rather than learning genuine, high-conviction market momentum.

Finally, while spatial dependencies (e.g., sector classifications and empirical correlations) are well-documented, it remains unclear how best to integrate these topologies sequentially. Existing studies frequently apply graph convolutions as a static transformation, without evaluating whether deeply recurrent, state-carrying spatial networks are necessary to map dynamic market regime shifts.

This study addresses these gaps by proposing a unified spatio-temporal framework optimized via a novel Hybrid Quantile Ranking Loss. To bridge the divide between statistical graph learning and economic microstructure theory, the framework introduces a volume-adjusted target return — utilizing a soft-thresholding function to penalize low-conviction price moves — and enforces strict, non-crossing risk quintiles ($\tau = 0.1, 0.5, 0.9$) alongside cross-sectional pairwise ranking. To evaluate the optimal integration of market topology, the study systematically compares a pure temporal baseline (LSTM), a parallel late-fusion architecture (Hybrid GAT-LSTM), and a state-full, unrolled dynamic graph network (ROLAND), assessing their out-of-sample predictive edge and structural risk calibration.

Chapter 3

Data Acquisition and Pre-Processing

This chapter discusses the data collection steps performed for this study, the sources from which the data has been collected, the pre-processing steps applied to data from each source, and the final unified panel data provided to the modelling stage. The presentation follows the data pipeline in sequence: universe selection, raw data acquisition, cross-source reconciliation, feature engineering, and finally the dataset split used for model training and evaluation.

3.1 Universe Selection

The study begins with 185 stocks drawn from the NSE Futures & Options (F&O) universe. F&O-listed stocks provide higher liquidity, more reliable price discovery, and more complete fundamental data coverage than the broader market. Four sequential filters reduce this to a final modelling universe of 103 stocks.

Filter 1 – Full Coverage (2013-01-01 onwards): Only stocks with continuous daily data from at least 2013-01-01 are retained. Stocks listed after this date would carry NaN features for all pre-listing dates, which in turn produces unstable graph edges and missing training targets. This filter reduces the candidate universe from 185 to 137 stocks, excluding recent listings such as IRCTC, NYKAA, PAYTM, DMART, and CDSL. The filter is applied against the first real trading day on or after 2013-01-01 rather than against the raw `START_DATE` of 2012-01-01, because 2012 data is consumed entirely by a model warm-up period and was never

used to produce trainable sequences (see Section 3.5).

Filter 2 – API Failures: Six symbols (CROMPTON, DELHIVERY, HUDCO, KPITTECH, LICI, MANKIND) from the F&O segment were also excluded due to persistent API timeouts during data collection. These are beyond the 185 stocks mentioned in Filter 1.

Filter 3 – Structural Break Exclusions: Seventeen stocks are excluded manually due to structural breaks and data quality issues that would prevent a model trained on pre-event data from generalising to the post-event regime. Notable exclusions include HDFCBANK and ICICIBANK (HDFC–HDFCBANK merger, 2023) and BANKBARODA. A further set of stocks – ABB, APLAPOLLO, CGPOWER, INDUSTOWER, JSWENERGY, LTF, MARICO, MCX, MFSL, NBCC, PGEL, PIIND, SOLARINDS, SUZLON – were excluded due to data incompleteness and quality issues identified during preliminary analysis.

Filter 4 – Banking Sector Exclusion: Banking and financial services stocks were excluded manually. Financial companies report fundamentally different balance-sheet and income-statement items (e.g. net interest margin, loan-to-deposit ratio) rather than the operating-margin and asset-turnover metrics applicable to non-financial firms. Including them would require a separate feature engineering parameterisation, significantly increasing methodological complexity.

After all four filters, the final universe consists of 103 stocks with continuous daily observations spanning 2012-01-02 to 2024-12-31, covering 3,202 trading days per stock and 329,806 stock-day observations.

3.2 Data Acquisition

Four distinct data sources were used to collect the required data, which is then subsequently merged at the (Stock, Date) level. Table 3.1 summarises the sources, coverage, and primary contributions to the final feature set.

Table 3.1: Data Sources Summary

Source	API / Tool	Variables	Period
Yahoo Finance	`yfinance` Python library	Split- and dividend-adjusted OHLCV prices, volume	2012–2024
NSE India	`nselib` Python library + website CSV	Exchange-level microstructure: deliverable quantity, number of trades, turnover, and bulk deals	2012–2024
LSEG Workspace (Eikon)	LSEG Data API	Daily fundamental ratios: P/E, EV/EBITDA, ROE, margins, growth rates, etc.	2012–2024
Microsoft Excel (Data tab)	Manual retrieval	Outstanding shares and sector (industry) classification per stock	30 Jan 2026

3.2.1 Market Price and Microstructure Data

Yahoo Finance provided split- and dividend-adjusted OHLCV data. NSE API provided exchange-level microstructure variables, including deliverable quantity, number of trades, and rupee turnover, that are unavailable through Yahoo Finance. For each constituent, the two data streams were merged on a common date index using an inner join, retaining only trading days with valid records from both sources. The combined dataset was stored in an SQLite database (`NSE.db`), with each equity occupying a separate table identified by its NSE ticker.

3.2.2 Fundamental Ratio Data

Fundamental ratio data for NSE constituents was sourced from LSEG Workspace (Refinitiv Eikon), covering the same January 2012 to December 2024 period. The raw Eikon export follows a wide format in which each fundamental feature was paired with its own 'Calc Date' column, reflecting the quarterly or irregular frequency at which these ratios were calculated. A custom parsing function restructured this

format into a standard long-format panel indexed by (Instrument, date). Since fundamental ratios are reported at discrete intervals, the daily panel contains large stretches of missing values between reporting dates. These were addressed through a grouped forward-fill operation: within each instrument, the most recently available ratio value is carried forward until a new observation becomes available, reflecting the realistic information environment faced by market participants.

3.2.3 Bulk Deals Data

Bulk deal disclosures were manually downloaded from the NSE website as CSV files, spanning 2010 to 2025 (filtered to 2024 for this study). A bulk deal is a transaction where more than 0.5% of a company's total shares are traded in a single or joint order, and these must be disclosed publicly by the exchange. These disclosures represent verifiable institutional activity and serve as a proxy for informed trading. The raw downloads of bulk deals were concatenated across years, yielding 110,043 observations across eight key variables: Date, Symbol, Security Name, Client Name, Buy/Sell, Quantity Traded, Trade Price/Weighted Average Price, and Remarks. Security names were standardised using NSE name-change history, and the Remarks column was used to standardise the reporting date. Where no Remarks were present, the trade date was used as the reporting date.

Following concatenation and standardisation, a preliminary noise-reduction step removed apparent wash trades, i.e. transactions where the same client appears on both the buy and sell side for the same symbol on the same date, with quantities differing by less than 5%. Client names ($\sim$24,545 unique) were classified into four investor categories: Institutional, Retail, Trader/Arbitrager, and Others. A rule-based heuristic first screened names containing keywords strongly associated with institutional investors (e.g. "FUND", "INSURANCE", "FPI", "TRUST"). Remaining ambiguous names were passed to a locally hosted LLaMA 3.2 large language model with domain-specific classification rules; results were cached to avoid redundant API calls. The prompt template used for this classification is provided below.

```
You are classifying investor entity names in Indian stock
   market
bulk-deal disclosures on National Stock Exchange.
Classify this investor into EXACTLY ONE of:
- Institutional
- Retail
- Trader/Arbitrager
- Others
Exemplary Classification Rules:
Retail = individuals, HUF
Trader/Arbitrager = PMS, Prop desk, MERCHANDIES, LEASING,
TEXTRADE, TRADING, brokers, trading, securities, algo,
arbitrage, MERCANTILE, Enterprises, etc.
Institutional = Foundation, Asset Management, FPI, FII, DII,
PMS, Mutual funds, Insurance, PE/VC, Sovereign funds, NBFC,
Corporates, Family office, Trusts, HOLDING, Foreign entities
   /
countries in name etc.
Others = unclear / not investment-oriented
But also pay attention to edge cases and use your best
judgement. SONATA INVESTMENTS LIMITED, ZGB HOLDINGS PVT LTD
are Others.
OUTPUT STRICTLY:
Category: <one>
Reason: <short reason>
Name: <original>
Name: {name}
```

Listing 3.1: LLM Prompt Template for Investor Classification



Figure 3.1: Client Names Word Cloud – Bulk Deals

3.3 Data Preprocessing

Data preprocessing transforms heterogeneous raw inputs into a unified, temporally aligned panel indexed by (Stock, Date), i.e. (i, t), where $i \in \{1, \ldots, N\}$ denotes a stock and $t \in \{1, \ldots, T\}$ represents the date. All transformations are constructed to use only information available up to and including date t.

3.3.1 Sector Composition

To facilitate the construction of structural homophily edges in the graph neural network (detailed in Chapter 4), each stock is mapped to a static sector classification. The categorical sector assignments serve as a proxy for shared macroeconomic exposure and industry-specific market dynamics.

Table 3.2 details the distribution of the 103 stocks across the defined market sectors.

Table 3.2: Sector Composition of the 103-Stock Universe

Sector	**Number**	**% of Universe**
Healthcare & Life Sciences	12	11.7%
Technology – Software & IT	11	10.7%
Metals, Mining & Basic Materials	11	10.7%
Real-Estate, Construction & Eng.	10	9.7%
Industrial Equipment & Electrical	8	7.8%
Energy – Traditional	7	6.8%
Automotive & Mobility Components	7	6.8%
Energy – Utilities & Power	6	5.8%
Chemicals & Materials	5	4.9%
Aerospace, Defense & Transport	5	4.9%
Food, Beverage & Agriculture	4	3.9%
Consumer Products & Electronics	4	3.9%
Transportation & Logistics	3	2.9%
Capital Markets & Wealth	3	2.9%
Consumer Retail & Apparel	3	2.9%
Telecom & Connectivity	2	1.9%
Hospitality & Leisure	2	1.9%
Total	**103**	**100.0%**

The original sectorial data, pooled from the data tab of Microsoft-Excel's, contained a granular industry classification system. The categories were aggregated into the above broader sector groups to prevent structural sparsity in the graph (see appendix). The sectors with similar industry we clustered in same sector, otherwise a large amount of disassociated sectors would fail to form meaningful sub-graphs. This would have ultimately limited the graph neural network's ability

to learn intra-sector message passing effectively.

3.3.2 Cross-vendor Reconciliation and Adjusted NSE Fields

The prices from Yahoo Finance are split- and dividend-adjusted, but the NSE API provided unadjusted prices. Two reconciliation ratios were computed to harmonise the two sources:

$$\text{Price_Ratio}_{i,t} = \frac{\text{yf_close}i,t}{\text{nse_close}i,t}, \qquad \text{Volume_Ratio}_{i,t} = \frac{\text{yf_volume}_{i,t}}{\text{nse_totalqty}_{i,t}} \tag{3.1}$$

NSE price fields (open, high, low, close, and derived price measures) are rescaled by $\text{Price_Ratio}_{i,t}$, and NSE quantity fields by $\text{Volume_Ratio}_{i,t}$. This harmonisation ensures that all corporate-action adjustments embedded in the series data downloaded from Yahoo Finance propagate consistently to all derived features.

3.3.3 Handling Missing Values

Delivery-related fields: Small gaps ($\sim$346 records, $\ll 0.1\%$ of all records) in percentage of daily traded quantity delivered, `NSE_PctDailyQtyTraded` were imputed using a rolling two-week (14-day) median computed within each stock, which is robust to outliers and at the same time preserves local distributional structure:

$$X_{i,t} = \text{Median}(X_{i,t-14}, \ldots, X_{i,t-1}) \tag{3.2}$$

Fundamental ratios: Fundamental ratios are forward-filled within each stock as described in Section 3.2.2. Ratios that are structurally absent (availability flag = 0) are retained as zero-valued features with an explicit availability flag, allowing the model to learn from the absence pattern itself.

Reporting lag: A reporting lag feature was computed from bulk-deal records as:

$$\text{report_lag_days}_{i,t} = \text{Reported_Date}_{i,t} - \text{Trade_Date}_{i,t}$$

For the 103 securities in the final universe, the lag was zero throughout the sample period, so the column was dropped from the final feature set. The processing logic is retained in the pipeline for potential extension to a broader universe.

3.3.4 Fundamental Ratios Processing (LSEG Workspace / Eikon)

The twenty fundamental ratios exported from LSEG Workspace (Eikon) were downloaded in wide format, with each feature paired with a corresponding 'Calc Date' column. Each file was converted into a standardised panel with columns (Instrument, date, $\text{feature}_1, \ldots, \text{feature}_m$). Mixed date formats were parsed robustly, duplicate (Instrument, date) rows were collapsed, and all numeric features were coerced to numeric dtype.

Table 3.3 summarises each ratio, its definition, and its category.

Table 3.3: Fundamental Ratios Definitions

Category	**Ratio / Parameter**	**Definition**
Identifiers	Date	The observation date corresponding to each data record.
Identifiers	Instrument / Symbol	The unique ticker symbol assigned to each security.
Profitability	Net Profit Margin (%)	(Net Income / Revenue) × 100
Profitability	Gross Margin (%)	(Revenue − COGS) / Revenue × 100
Profitability	ROE Common Equity (%)	(Net Income / Common Shareholders' Equity) × 100
Revenue & Growth	Revenue Per Share	Total Revenue / Shares Outstanding
Revenue & Growth	Revenue Per Share, 3 Yr CAGR	Annualised growth rate of revenue per share over trailing 3 years
Revenue & Growth	Cost of Revenue, 3 Yr CAGR	Annualised growth rate of cost of revenue over trailing 3 years
Valuation	P/E (Daily Time Series Ratio)	Market Price Per Share / Earnings Per Share

Category	Ratio / Parameter	Definition
Valuation	Enterprise Value To EBITDA	Enterprise Value / EBITDA
Valuation	Price To Book Value Per Share	Market Price Per Share / Book Value Per Share
Valuation	Price To Tangible Book Value Per Share	Market Price Per Share / Tangible Book Value Per Share
Valuation	Price To Cash Flow Per Share	Market Price Per Share / Operating Cash Flow Per Share
Valuation	Enterprise Value To Sales	Enterprise Value / Total Revenue
Leverage	Net Debt To EBITDA	Net Debt / EBITDA
Leverage	Interest Coverage Ratio	EBIT / Interest Expense
Leverage	Interest Coverage Ratio, Period-Period Diff	Change in interest coverage ratio between consecutive reporting periods
Liquidity	Quick Ratio	(Current Assets − Inventory) / Current Liabilities
Efficiency	Asset Turnover	Total Revenue / Total Assets
Efficiency	Fixed Asset Turnover	Total Revenue / Net Fixed Assets
Efficiency	Asset Turnover, Prd-Prd Diff	Period-on-period change in Asset Turnover
Efficiency	Fixed Asset Turnover, Period-Period Diff	Period-on-period change in Fixed Asset Turnover

3.3.5 Bulk Deal Signal Construction

Bulk deal records were transformed from an event-level dataset into a daily, stock-level feature panel. After symbol standardisation, each bulk-deal record was converted to a monetary size measure:

$$\text{DealValue}_{i,t} = \text{QuantityTraded}_{i,t} \times \text{TradePrice}_{i,t} \tag{3.3}$$

To make bulk-deal magnitudes comparable across firms of different size, deal values were scaled by market capitalisation, and a log-transformed intensity score was computed:

$$\mathrm{MCap}_{i,t} = \mathrm{ClosePrice}_{i,t} \times \mathrm{SharesOutstanding}_i, \qquad w_{i,t} = \log\left(1 + \frac{\mathrm{DealValue}_{i,t}}{\mathrm{MCap}_{i,t}}\right) \tag{3.4}$$

Directional intensity components were then defined: $w_{i,t}^{\mathrm{buy}} = w_{i,t}$ for buy deals and 0 otherwise; $w_{i,t}^{\mathrm{sell}} = w_{i,t}$ for sell deals and 0 otherwise. These were aggregated to daily stock-day shocks:

$$\mathrm{BulkBuyW}_{i,t} = \sum_{k \in E_{i,t}} w^{\mathrm{buy}}(k), \qquad \mathrm{BulkSellW}_{i,t} = \sum_{k \in E_{i,t}} w^{\mathrm{sell}}(k) \tag{3.5}$$

$$\mathrm{BulkNetW}_{i,t} = \mathrm{BulkBuyW}_{i,t} - \mathrm{BulkSellW}_{i,t} \tag{3.6}$$

Exponentially decaying signals were constructed to represent persistence:

$$S_{i,t} = \alpha S_{i,t-1} + X_{i,t}, \quad \alpha = e^{-1/\tau} \approx 0.967 \ (\tau = 30 \text{ trading days}) \tag{3.7}$$

Table 3.4 summarises all bulk-deal features computed.

Table 3.4: Bulk-Deal Features Computed for Final Dataset

Feature	**Computation**	**Represents**	**Type**
bulk_flag	1 if any bulk deal on date	Event indicator	Binary
bulk_buy_w	$\sum \log(1 + \text{deal/mcap})$ for buys	Aggregated buy intensity	Cont.
bulk_sell_w	$\sum \log(1 + \text{deal/mcap})$ for sells	Aggregated sell intensity	Cont.
bulk_net_w	bulk_buy_w − bulk_sell_w	Net directional pressure	Cont.
bulk_intensity	bulk_buy_w + bulk_sell_w	Total unsigned activity	Cont.

Feature	Computation	Represents	Type
bulk_direction_ratio	bulk_net_w / bulk_intensity	Normalised directional bias $\in [-1, 1]$	Cont.
buy_count	Count of buy-side records	Number of bulk buy transactions	Integer
sell_count	Count of sell-side records	Number of bulk sell transactions	Integer
report_lag_days	Reported_Date − Trade_Date	Disclosure delay; always 0 for this universe; dropped	Integer
bulk_buy_signal	EWM($\alpha \approx 0.967$) of bulk_buy_w	Persistent buy-side momentum	Cont.
bulk_sell_signal	EWM($\alpha \approx 0.967$) of bulk_sell_w	Persistent sell-side momentum	Cont.
bulk_net_signal	bulk_buy_signal − bulk_sell_signal	Persistent net directional signal	Cont.
bulk_intensity_signal	EWM($\alpha \approx 0.967$) of bulk_intensity	Persistent total activity strength	Cont.
bulk_direction_ratio_signal	EWM($\alpha \approx 0.967$) of bulk_direction_ratio	Smoothed directional bias	Cont.

Empirical Evaluation and Feature Exclusion. Despite the robust pipeline constructed to parse, classify, and decay bulk deal transactions, preliminary feature evaluation indicated that these signals possessed a negligible Information Coefficient ($|\text{IC}| < 0.015$) and correlation with target variable being ($|\text{rho}| < 0.010$) with respect to the forward 7-day target returns across the selected universe. The lack of predictive power may be attributed to the fact that bulk deals are often pre-negotiated block trades that do not directly consume market liquidity or immediately impact the limit order book, combined with the relatively small number of deals within this 103-stock universe. To maintain model parsimony and

avoid introducing noise, all bulk deal-derived signals were therefore dropped from the final training feature set.

3.4 Feature Engineering

After assembling the reconciled panel, additional feature engineering was performed from price and volume series. All rolling computations were performed strictly within each stock using only past information, preserving temporal integrity. The features engineered fall into five broad categories: price fields and transformations, momentum indicators, trend indicators, volatility and volume indicators, and return features.

3.4.1 Price Fields and Log Transformations

The following raw fields enter the initial candidate feature set directly: adjusted close price, `NSE_PctDailyQtyTraded` (daily delivery percentage), and volume (total quantity traded daily).

Two derived ratio features were constructed to capture average trade characteristics without retaining the collinear raw columns. Since NSE turnover (in INR) and number of trades are highly correlated (Pearson $r > 0.9$), only the following derived ratios are retained rather than the raw components:

$$\text{TurnoverPerTrade}_{i,t} = \frac{\text{Turnover}_{i,t}}{\text{NumTrades}_{i,t}}, \qquad \text{QtyPerTrade}_{i,t} = \frac{\text{Volume}_{i,t}}{\text{NumTrades}_{i,t}} \tag{3.8}$$

Turnover per trade serves as a proxy for average trade size, which is correlated with institutional participation. Quantity per trade isolates trade size in volume terms, complementing the monetary measure. Note: `QtyPerTrade` was subsequently dropped during the feature selection stage (see Section B.6) due to low information coefficient and feeble correlation with the target variable.

High–Low Range: The intra-period price range is included as a daily volatility proxy:

$$\text{H-L}_t = \text{High}_t - \text{Low}_t \tag{3.9}$$

Signed log1p transform: To compress heavy tails and improve gradient stability in the neural network, a signed log-plus-one transform is applied to

unbounded continuous features:

$$\text{signed_log1p}(x) = \text{sign}(x) \cdot \log(1 + |x|) \tag{3.10}$$

This transform (i) compresses large values without losing small-value resolution; (ii) preserves the sign of the original feature; and (iii) maps zero to zero. It is applied to all continuous numeric features except those already bounded in $[-1, 1]$ or $[0, 1]$ (e.g. percentile ranks and binary flag variables).

The log-volume feature is computed separately as $\text{log_volume}_{i,t} = \log(1 + \text{Volume}_{i,t})$ and used as the primary input to all rolling volume statistics.

3.4.2 Momentum Indicators

Momentum indicators capture the rate and direction of price change over time. All indicators are computed using the `pandas-ta` library on per-stock sorted panel data.

Table 3.5: Momentum Indicators (Candidate Set)

Feature	**Formula / Window**	**Interpretation**
RSI_14	$100 - 100/(1 + \text{RS})$, RS = avg gain / avg loss over 14 days	Overbought (> 70) / oversold (< 30); bounded [0, 100]
MACD_12_26_9†	EMA(12) − EMA(26); signal = EMA(9) of MACD	Trend direction and momentum crossover
ROC_10†	$100 \times (P_t - P_{t-10})/P_{t-10}$	10-period percentage price change
STOCHk_14_3_3†	$100 \times (P - \text{Low}_{14})/(\text{High}_{14} - \text{Low}_{14})$, 3-period smoothed	Position of close in recent 14-day range
STOCHh_14_3_3	3-period MA of STOCHk	Smoothed stochastic oscillator
CCI_14_0.015†	(Typical Price − SMA_{14}) / (0.015× MAD)	Deviation from average normalised by variability

†Dropped during feature selection: MACD and its components due to high correlation with RSI_14; ROC_10 for redundancy with RSI_14; STOCHk due to overlap with the retained smoothed version STOCHh; CCI_14 due to marginal IC relative to RSI_14.

3.4.3 Trend Indicators

Trend indicators identify the direction and persistence of price movements.

Simple Moving Averages are computed over 20-day (SMA_20), 50-day (SMA_50), and 200-day (SMA_200) windows. This is the primary reason for loading data from 2012: SMA_200 requires at least 200 days of price history before producing a valid value, so loading from 2012 ensures a valid SMA_200 for every training observation from 2013 onwards. All three raw SMA columns are dropped during feature selection (Section B.6) and replaced by the following ratio feature.

Moving Average Ratio (SMA_Ratio1): A ratio between the short-term and medium-term simple moving averages captures relative trend strength while eliminating the collinearity among the raw SMA columns:

$$\text{SMA_Ratio1}_t = \begin{cases} \text{SMA}_{20,t}/\text{SMA}_{50,t}, & \text{if } \text{SMA}_{50,t} \neq 0 \\ \text{NaN}, & \text{otherwise} \end{cases} \tag{3.11}$$

A second ratio (SMA_50 / SMA_200) was also constructed but was subsequently dropped after it showed high correlation with SMA_Ratio1.

Average Directional Index (ADX_14) measures trend strength over 14 periods regardless of its direction. Higher values indicate stronger trending market conditions, while lower values suggest weak or range-bound price action. It was computed as a candidate feature but was excluded during selection due to high correlation with retained moving-average-based features.

Aroon Indicators measure how recently a period high or low was recorded:

$$\begin{aligned} \text{AROONU_14} &= 100 \times (14 - \text{days since 14-period high})/14 \\ \text{AROOND_14} &= 100 \times (14 - \text{days since 14-period low})/14 \\ \text{AROONOSC_14} &= \text{AROONU_14} - \text{AROOND_14} \end{aligned}$$

The individual directional components (AROONU_14, AROOND_14) are redundant given the oscillator, which is the single retained summary, so they are dropped during selection. However, AROONOSC_14, which ranges from -100 to $+100$, is retained.

Ichimoku Cloud: The Ichimoku system was computed in full (Tenkan-sen, Kijun-sen, Senkou Spans A and B, Chikou Span). All five components were excluded from the final feature set. The four structural components (Tenkan-sen / ITS_9, Kijun-sen / IKS_26, Senkou Span A / ISA_9, Senkou Span B / ISB_26) were excluded due to high multicollinearity with the retained moving-average-based features. The Chikou Span (ICS_26) was excluded due to a leakage concern: it is defined as the current closing price displaced 26 periods *back* in the chart, meaning the feature at time t in the panel corresponds to the closing price 26 periods *ahead*, embedding future price information into the input. Two composite Ichimoku ratios were also engineered (IKS_26/ITS_9 and ISB_26/ISA_9) to reduce pairwise correlation, but they were similarly dropped in the final selection pass for the leakage reasons.

3.4.4 Volatility Indicators

Price volatility indicators were computed as candidates but excluded from the final feature set due to high multicollinearity with trend indicators. Specifically, ATR (Average True Range), Bollinger Band width (BBB_20), and 30-day rolling standard deviation of returns (STDEV_30) all exhibited Spearman correlations exceeding 0.9 with SMA-based features. Retaining redundant features would waste model capacity without adding orthogonal signal. Volume-based volatility measures are retained, however, as they capture the variability of trading activity rather than price range, representing a distinct dimension of market uncertainty, these are described in Section 3.4.5 below.

3.4.5 Volume and Money Flow Indicators

Volume-based indicators capture the intensity of market participation and the direction of capital flows. The complete range of volume based indicators considered for this study are presented in the Table 3.6.

Table 3.6: Volume and Money Flow Indicators (Candidate Set)

Feature	**Formula**	**Interpretation**
OBV†	$\mathrm{OBV}_t = \mathrm{OBV}_{t-1} + \begin{cases} V_t, & C_t > C_{t-1} \\ 0, & C_t = C_{t-1} \\ -V_t, & C_t < C_{t-1} \end{cases}$	Cumulative directional volume momentum
PVT†	$\mathrm{PVT}_t = \mathrm{PVT}_{t-1} + V_t \left(\frac{C_t - C_{t-1}}{C_{t-1}}\right)$	Volume-weighted price-change momentum
CMF_20†	$\frac{\sum_{i=t-19}^{t} \mathrm{MFV}_i}{\sum_{i=t-19}^{t} V_i}$, where $\mathrm{MFV}_i = \mathrm{MFM}_i V_i$	20-period money-flow ratio, bounded in $[-1, 1]$
AD	$\mathrm{AD}_t = \mathrm{AD}_{t-1} + \mathrm{CLV}_t V_t$, where $\mathrm{CLV}_t = \frac{(C_t - L_t) - (H_t - C_t)}{H_t - L_t}$	Running accumulation/distribution measure
EFI_13†	$\mathrm{EMA}_{13}((C_t - C_{t-1})V_t)$	Smoothed measure of the force behind price movements

†Dropped during feature selection: OBV (On-Balance-Volume) and PVT (Price Volume Trend) due to high pairwise correlation ($\rho > 0.75$); the engineered PVT/OBV ratio was also removed. CMF_20 and EFI_13 (Elder Force Index with 13-period exponential moving average) were dropped due to low information coefficient (IC) with the target.

Three additional volume statistics are computed from log-volume to capture level, abnormality, and relative standing of daily trading activity. Let $v_{i,t} = \ln(1 + \mathrm{Volume}_{i,t})$.

Volume Volatility (rolling standard deviation of log-volume):

$$\sigma_{i,t}^{k} = \sqrt{\frac{1}{k} \sum_{j=0}^{k-1} (v_{i,t-j} - \bar{v}_{i,t}^{k})^2} \tag{3.12}$$

computed over $k = 30$ days (volvol_30) and $k = 90$ days (volvol_90). Both were excluded during feature selection due to low IC.

Volume Z-score (vol_z_30):

$$Z_{i,t} = \frac{v_{i,t} - \mu_{i,t}^{30}}{\sigma_{i,t}^{30}} \tag{3.13}$$

winsorised to $[-4, 4]$. However, it was also excluded during feature selection.

Volume Rank and Percentile Rank (vol_rank_30, vol_prank_30): Distribution-free rankings placing today's volume within the most recent 30-day window:

$$\text{vol_prank}_{i,t} = \frac{1}{k} \sum_{j=0}^{k-1} \mathbb{1}(v_{i,t-j} \leq v_{i,t}) \tag{3.14}$$

Both are bounded in $[0, 1]$ and robust to outliers. `vol_prank_30` (the empirical CDF value) is retained in the final feature set showing correlation of 0.08 with the target variable; `vol_rank_30` is excluded as redundant.

3.4.6 Return Features

The standard log return is computed as:

$$l_{i,t} = \ln(P_{i,t}/P_{i,t-1}) \tag{3.15}$$

The 1-day log return $l_{i,t}$ (denoted `ret_1d` in the codebase) is retained in the panel but *excluded* from the model feature vector. It is used exclusively by the graph construction module to compute rolling inter-stock return correlations (see Chapter 4).

3.4.7 Prediction Target: Volume-Adjusted Forward Returns

The primary prediction target is the forward 7-trading-day log return:

$$y_{i,t} = \ln\left(\frac{\text{Close}_{i,t+7}}{\text{Open}_{i,t+1}}\right) \tag{3.16}$$

The entry at Open_{t+1} simulates a realistic execution price. To suppress low-conviction price moves driven by thin trading, the raw return is multiplied by a volume-based conviction weight:

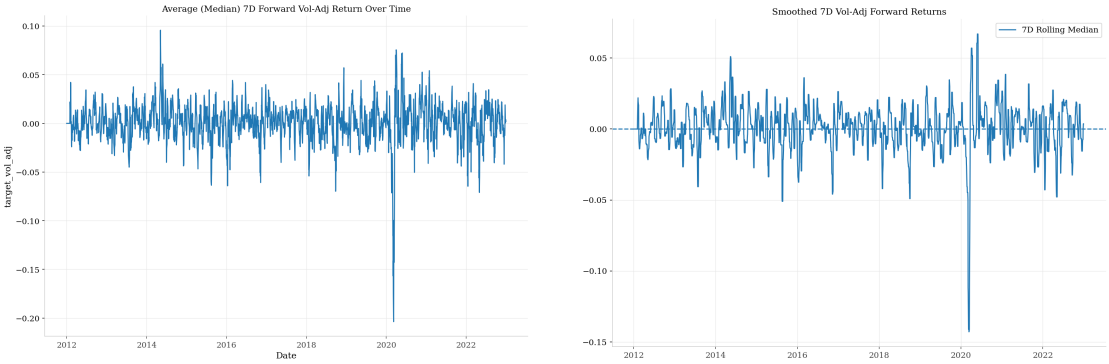
$$\text{Target}_{i,t}^{\text{vol_adj}} = y_{i,t} \times \tanh\left(\frac{V_{i,t}^{\text{fwd}}}{V_{i,t}^{\text{hist}}}\right). \tag{3.17}$$

where $V_{i,t}^{\text{fwd}}$ is the 7-day rolling mean of forward volume and $V_{i,t}^{\text{hist}}$ is the 22-day trailing mean volume. Since, the volume ratio is always positive, the tanh function compresses the ratio into $(0, 1)$, acting as a soft weight that reduces returns on illiquid days without hard-zeroing them. To mitigate the influence of extreme outliers, the target is clipped at the 0.1st and 99.9th percentiles computed *on the training set only* and applied identically

to the validation and test sets. Furthermore, the correlation between $\text{Target}_{i,t}^{\text{vol_adj}}$ and the actual target returns $y_{i,t}$ over the training period exceeds 0.98, supporting the use of this alternative target without a meaningful loss of information.

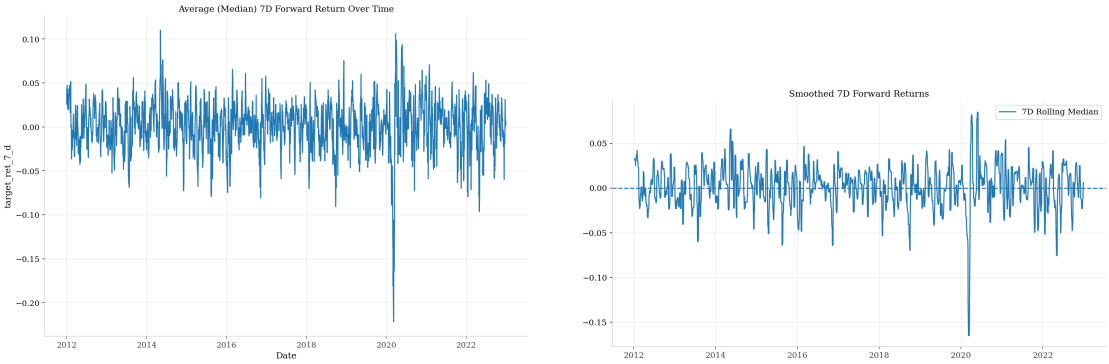
The 7-day horizon was chosen to balance signal persistence against execution feasibility. Shorter horizons (1–2 days) are dominated by microstructure noise; longer horizons (30+ days) are less actionable for portfolio rebalancing. Because the target is computed by looking 7 days forward, rows near the end of the dataset draw on data from early 2025. The forward return was therefore computed before the dataset was trimmed to 31 December 2024, ensuring that late-2024 rows retain a valid non-NaN target.

The distribution of raw and smoothed volume adjusted target returns and over the training duration are presented in Figure 3.2.



(a) Raw Returns (b) Smoothed (7 day) returns

Figure 3.2: Volume-adjusted returns under different market conditions



(a) Raw Returns (b) Smoothed (7 day) returns

Figure 3.3: 7-day Target Returns under different market conditions

3.5 Warmup Period and Data Splits

Data is loaded from 2012-01-01 despite the training period beginning in 2013. This warmup year serves two purposes. First, the graph construction module requires a trailing 60-day window of daily returns to compute the first valid correlation matrix; loading from 2012 ensures a valid graph is available from early 2013. Second, long-lookback technical indicators such as SMA_200 require at least 200 days of price history before producing valid values.

The dataset is partitioned into three strictly time-ordered, non-overlapping splits:

Table 3.7: Dataset Splits

Split	**Period**	**Purpose**
Warm-up	Prior to 2012-12-31	Feature warm-up only
Training	2013-01-01 to 2022-12-31	Model weight optimisation
Validation	2023-03-01 to 2023-12-31	Early stopping & hyperparameter selection
Test	2024-03-01 to 2024-12-31	Final unbiased evaluation

A two-month purge gap separates each fold boundary. With a 7-trading-day forward return as the prediction target, rows from January and February 2023 (and 2024) could have targets that extend into the subsequent evaluation split. Excluding these rows eliminates any cross-split target contamination. No data from the validation or test splits is used during model training; the test set is held out entirely until final evaluation.

3.6 Feature Selection

After initial feature engineering, a structured selection procedure reduced the candidate pool to a compact, low-redundancy set. Exclusions were made on the basis of four criteria: high inter-feature correlation, low information coefficient (IC) with the target, structural sparsity, or data leakage. The selection was performed exclusively on the training split, with all statistics computed on a time-stratified sample drawn uniformly per calendar month to avoid regime-specific correlation estimates.

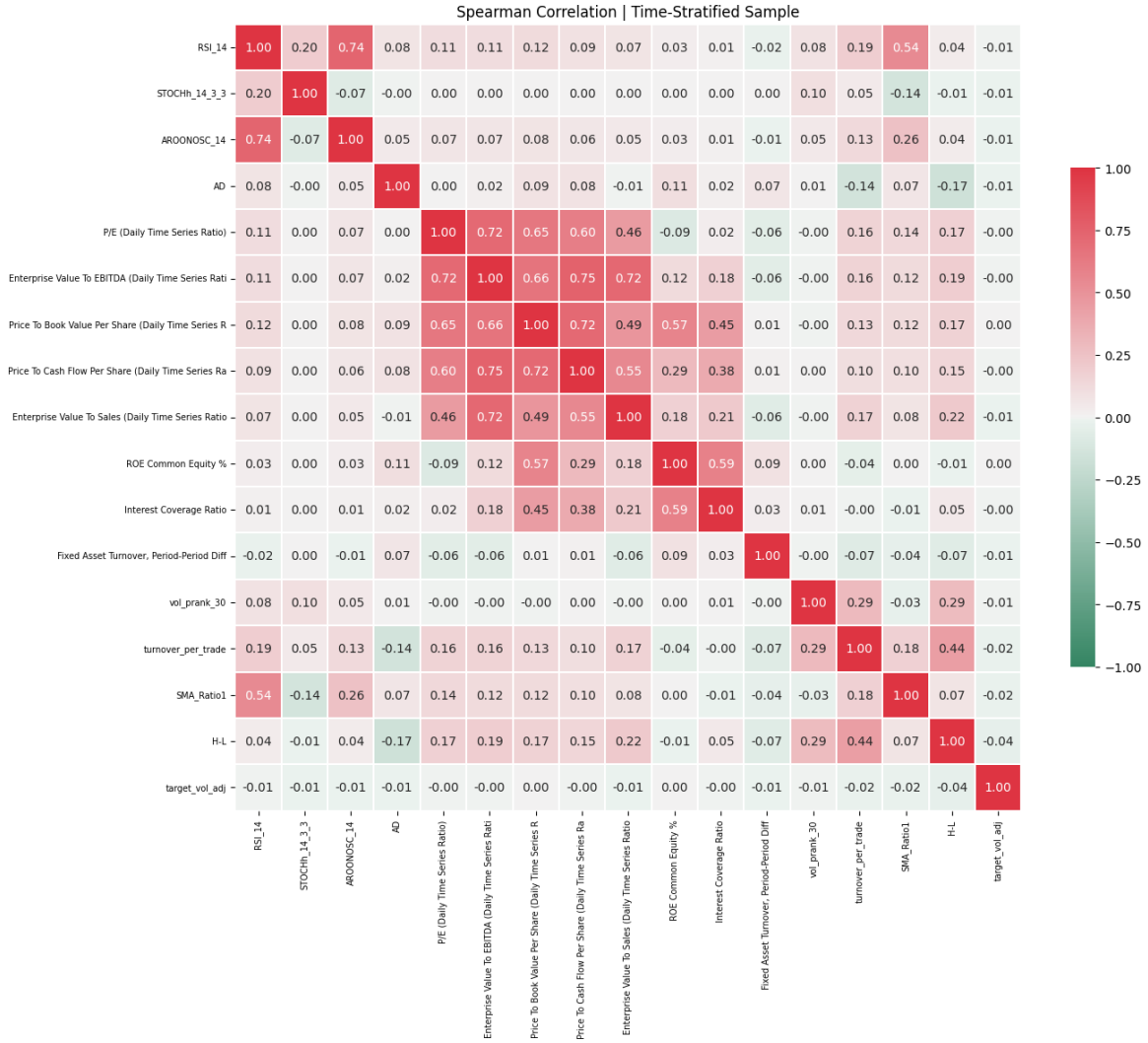


Figure 3.4: Feature Correlations

3.6.1 High Inter-Feature Correlation

A Spearman correlation matrix was computed on the time-stratified training sample. Feature pairs with absolute correlation above 0.75 were flagged, and the lower-IC member was removed. The correlation matrix below in Figure 3.4 shows the correlations among the features that are kept for the final model. The features excluded from the final feature set includes the following:

- **Moving averages and envelope indicators:** SMA_20, SMA_50, SMA_200, EMA_9/21, EMA_50/200, Bollinger Band components (BBL, BBM, BBU, BBP, BBB), Keltner Channel components (KCUe, KCLe, KCBe), Donchian Channel components (DCU, DCL, DCM), and ATRr_14 all exhibit mutual Spearman correlations > 0.75; all are replaced by SMA_Ratio1.
- **ADX system:** ADX_14, DMP_14, DMN_14, and ADXR_14 are excluded due to

high correlation with SMA-based features.

- **MACD:** MACD_12_26_9 and its signal and histogram components (MACDh, MACDs) exceed the 0.75 threshold against RSI_14 and are removed.
- **Stochastics:** STOCHk_14_3_3 is removed in favour of the smoothed STOCHh_-14_3_3.
- **OBV and PVT:** Both track cumulative volume-weighted price direction and are mutually correlated. A PVT/OBV ratio was engineered as an alternative but showed no additional IC, so both the ratio and the components were dropped; AD is retained.
- **Valuation pairs:** EV/EBIT is excluded in favour of EV/EBITDA. P/Sales is replaced by EV/Sales. P/TBV is replaced by P/B. ROA is excluded due to high correlation with ROE.
- **Liquidity ratios:** Current Ratio and Quick Ratio are both excluded; their coverage information is captured by the Interest Coverage Ratio with lower inter-correlation.
- **NSE raw columns:** `nse_turnoverinr` and `nse_numtrades` are removed after the feature `turnover_per_trade` is constructed from them.

3.6.2 Low Information Coefficient

Per-stock Spearman IC values between each candidate feature and `target_vol_adj` were computed on the training split for every stock with at least 200 observations. The primary exclusion rule was straightforward: any feature whose mean IC across the 103-stock universe fell below $|\bar{\rho}| < 0.010$ was dropped. On this basis, Net Profit Margin, Operating Profit Margin, raw Turnover, Total Revenue, `vol_z_30`, `vol_rank_30`, `volvol_30`, and `volvol_90` were excluded. `vol_prank_30` with $|\bar{\rho}| = 0.008$ was retained despite sitting near the threshold for two reasons: it serves as the primary input for volume-based graphical analysis of trading activity, and it ensures that at least one pure volume signal is represented in the final feature set, given that all other volume-flow indicators (OBV, PVT, CMF_20, EFI_13) were removed for collinearity reasons and less correlated with the target. Furthermore, the plot Figure 3.5 below shows the average spearman correlation of final selected feature set with the target variable, evaluated on the training dataset only.

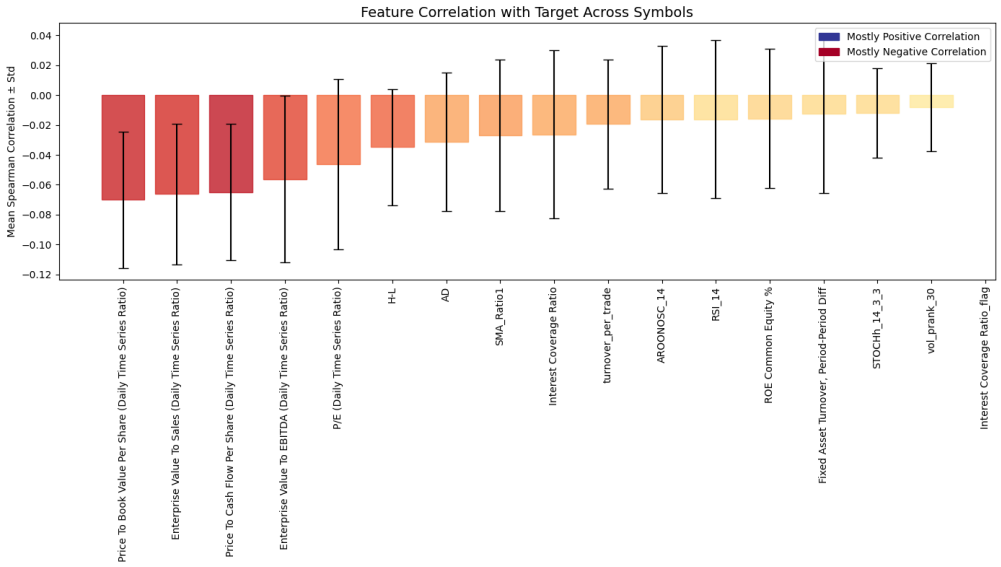


Figure 3.5: Average Feature Spearman Correlation with Target Across Symbols

3.6.3 Structural Sparsity

Parabolic SAR (PSAR) and SuperTrend (SUPERT) indicators were initially generated but dropped due to missing values during reversal states and early initialization, which resulted in column-level null rates substantially higher than other technical indicators, combined with negligible predictive power. All PSAR and SUPERT columns were therefore removed as a group.

3.7 Final Feature Set

After all pre-processing, transformation, and selection steps, the final feature vector $X_{i,t} \in \mathbb{R}^d$ for each stock-day contains 17 features: 16 continuous features and 1 binary availability flag. Table 3.8 summarises the final feature set, its category, and the normalisation applied.

Cross-sectional winsorisation and z-scoring were performed using training-set statistics frozen and applied unchanged to validation and test data, preventing any information from future periods entering the normalisation. Per-stock z-scoring was used for features (H-L, turnover_per_trade, AD) whose absolute magnitudes reflect firm size rather than a cross-sectionally comparable signal. Signed log1p was applied first to compress heavy tails before per-stock normalisation.

Table 3.8: Final Feature Set

Category	**Feature(s)**	**Normalisation Applied**
Momentum oscillators	RSI_14, STOCHh_14_3_3, AROONOSC_14	Cross-sectional winsorise (0.1%–99.9%) + z-score
Trend / MA	SMA_Ratio1 (SMA_{20} / SMA_{50})	Cross-sectional winsorise + z-score
Intraday volatility	H-L	Per-stock z-score (training stats)
Volume / order flow	vol_prank_30, turnover_per_trade, AD	signed_log1p; per-stock z-score (AD, turnover_per_trade)
Valuation multiples	P/E, EV/EBITDA, P/B, P/CF, EV/Sales	Cross-sectional winsorise + z-score
Profitability / leverage	ROE Common Equity %, Interest Coverage Ratio, Fixed Asset Turnover (Period Diff)	Cross-sectional winsorise + z-score
Fundamental flag	`Interest Coverage Ratio_flag` (binary: 1 if ratio is available for the stock, 0 otherwise)	Zero-fill only; no transform (already binary)
Not model inputs	ret_1d, sector_idx, Date, Symbol, Sector, split, target_vol_adj, target_ret_7_d	–

A single binary availability flag is retained: `Interest Coverage Ratio_flag`, which takes the value 1 for stocks that report an interest coverage ratio and 0 for companies (BAJAJHLDNG.NS & RECLTD.NS) for which the ratio is structurally absent. The retained flag allows the model to distinguish a genuine zero coverage ratio from a missing value, without requiring imputation of an economically arbitrary value.

3.8 Final Integration and Output

The processed market panel, forward-filled fundamental panel (with availability flags), and all engineered features were merged into a single master table at the (Stock, Date) level. Days without bulk-deal activity were assigned zeros for those columns (subsequently all dropped). All rolling computations were performed within-stock to preserve cross-sectional comparability.

The resulting dataset provides a consistent daily feature representation indexed by $(\text{Stock}, \text{Date})$ spanning 103 stocks and 3,202 trading days (2012-01-02 to 2024-12-31), yielding 329,806 stock-day observations. This serves as the input to the graph construction stage described in Chapter 4, where each stock on each day becomes a node in a dynamic graph and inter-stock return correlations define the edges.

Chapter 4

Data Exploration

This chapter presents the exploratory data analysis of the final processed dataset. The analysis is conducted exclusively on the training split (2013-01-01 to 2022-12-31) to prevent any form of look-ahead bias. The objective is to characterize the properties of the prediction target, verify the quality of the feature matrix, and motivate the modelling choices made in subsequent chapters.

4.1 Dataset Overview

Table 4.1 provides a high-level summary of the final dataset delivered to the modelling stage.

Table 4.1: Dataset Summary Statistics

Property	**Value**
Stocks	103
Date range	2013-01-01 to 2024-12-31
Trading days	∼3,102 per stock (2,874 after data purge)
Total stock-day observations	∼308,000
Continuous features	16
Binary flag features	1 (`Interest Coverage Ratio_flag`)
Total model input features	17
Target variable	`target_vol_adj` (volume-adjusted 7-day forward log return) and `target_7d_ret`
Training split	2013-01-01 – 2022-12-31 (2466 sequences)
Validation split	2023-03-01 – 2023-12-31 (204 sequences)
Test split	2024-03-01 – 2024-12-31 (204 sequences)

The panel is *balanced* as every stock has the same number of trading-day observations across the sample. This property is enforced by the universe selection filters and is

required by the graph construction module, which builds a graph over a fixed node set at every time step.

4.2 Target Variable Analysis

4.2.1 Distributional Properties

The volume-adjusted 7-day forward return (`target_vol_adj`) was analysed on the training split after clipping at the 0.1st and 99.9th percentiles. The distribution exhibits three characteristic properties of financial return series, plots shown in Figure 4.1 :

1. **Near-zero mean and median:** Consistent with approximate market efficiency, the cross-sectional average return over any short horizon is close to zero. The slight positive mean (+0.0035) and median (+0.00095) reflects the broad upward drift of the Indian equity market over the training period.

2. **Leptokurtosis:** The distribution has substantially fatter tails than a Gaussian, as confirmed by the QQ plot. Excess kurtosis is positive, indicating that extreme return events occur with higher frequency than a normal distribution would predict.

3. **Positive skewness:** The distribution displays mild positive skewness (0.192) which suggests that large positive returns occur somewhat more frequently than equally large negative outcomes. Even though the skewness is not huge, it reflects the tendency of markets to experience occasional strong upward returns.

4.2.2 Cross-Sectional Dispersion

For a cross-sectional ranking model to be tractable, the distribution of returns across stocks on any given day must exhibit sufficient spread. The daily cross-sectional standard deviation of `target_vol_adj` − computed as the standard deviation of the target across all 103 stocks for each trading day − was examined over the training period. The average daily cross-sectional standard deviation (= 0.0390) comfortably exceeds the 0.015 threshold commonly used as a lower bound for feasible stock ranking, confirming that meaningful relative return variation exists at the daily level. The daily cross-sectional $q_{90} - q_{10}$ spread shows a similar pattern, rising sharply during market-stress episodes and compressing during low-volatility regimes.

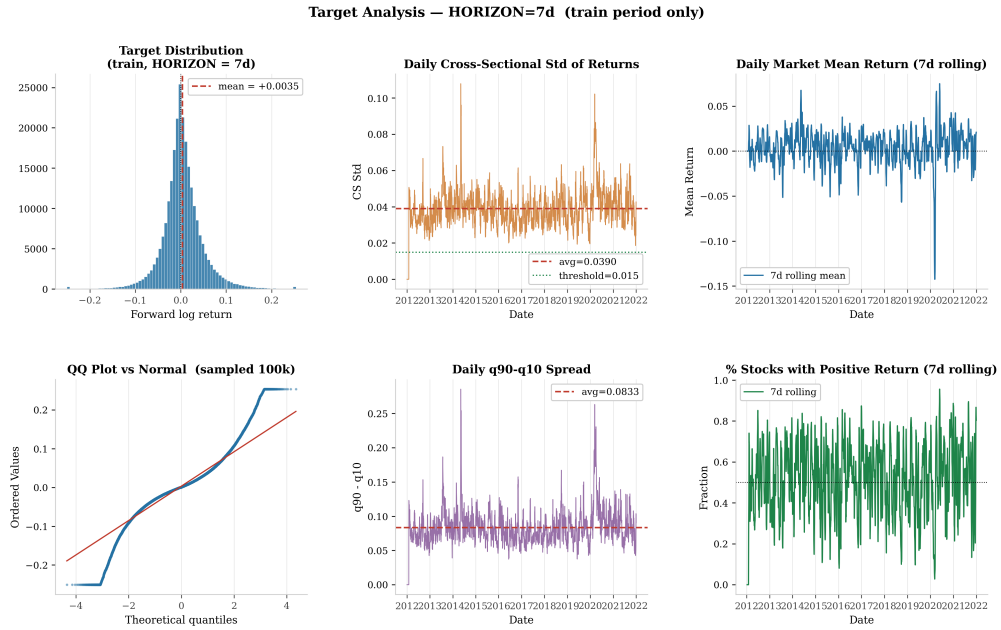


Figure 4.1: Target Distribution

4.2.3 Fraction of Positive Returns

The percentage of stocks with a positive 7-day forward return fluctuated around 50% over the training period when smoothed with a 7-day rolling window, consistent with near-random-walk aggregate behaviour. Episodes of sustained positive readings (broad market rallies) and sustained negative readings (corrections) are visible but do not exhibit a structural long-bias over the full 10-year training window.

4.3 Stationarity and Regime Analysis

4.3.1 Stationarity Tests

To ensure the validity of subsequent econometric modeling, the median daily returns of the volume-adjusted target were tested for stationarity. I used two complementary procedures to robustly assess the stationarity of the series: the Augmented Dickey-Fuller (ADF) test and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test.

Augmented Dickey-Fuller (ADF) Test

The ADF test is used to detect the presence of a unit root in a time series. The general regression model for the ADF test can be expressed as:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + \varepsilon_t \tag{4.1}$$

where Δ is the first difference operator, y_t is the time series being tested, α is a constant drift, βt represents a linear time trend, p is the lag order, and ε_t is the error term.

The hypotheses for the ADF test are formulated as follows:

- **Null Hypothesis (H_0):** $\gamma = 0$. The series contains a unit root and is non-stationary.
- **Alternative Hypothesis (H_1):** $\gamma < 0$. The series does not contain a unit root and is stationary.

As reported in Table 4.2, the ADF test applied to the returns series yields a test statistic of -10.162, which is significantly lower than the 1% critical value of -3.433. With a p-value approaching zero (7.45×10^{-18}), I strongly reject the null hypothesis of a unit root at the 1% significance level.

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test

To confirm the findings of the ADF test, I apply the KPSS test. Unlike the ADF test, the KPSS test assumes stationarity under the null hypothesis. The time series is decomposed into a deterministic trend, a random walk, and a stationary error term:

$$y_t = \xi t + r_t + \varepsilon_t \tag{4.2}$$

where r_t is a random walk defined as $r_t = r_{t-1} + u_t$, with $u_t \sim \text{iid}(0, \sigma_u^2)$.

The hypotheses for the KPSS test are:

- **Null Hypothesis (H_0):** $\sigma_u^2 = 0$. The random walk component is absent, meaning the series is trend-stationary (or level-stationary).
- **Alternative Hypothesis (H_1):** $\sigma_u^2 > 0$. The series contains a unit root and is non-stationary.

The KPSS test statistic is calculated at 0.078, which is well below the 10% critical value of 0.347. Furthermore, the test statistic falls below the available look-up table range (triggering a lower-bound interpolation warning), indicating the actual p-value is greater than 0.1. Consequently, failing to reject the null hypothesis, and supporting the conclusion that the series is stationary.

Conclusion on Stationarity

The complementary results from both the ADF and KPSS tests provide robust, concurring evidence that the volume-adjusted target returns series is stationary around a fixed level.

This satisfies the stationarity requirement for standard time-series modeling.

Test	Statistic	p-value	Conclusion
ADF	-10.162	< 0.001	Reject H_0 (Stationary)
KPSS	0.078	> 0.100	Fail to reject H_0 (Stationary)

Table 4.2: Stationarity Test Results

4.3.2 Market Regime Analysis

To understand the broader market environment and the persistence of trends, a simple binary regime indicator was constructed. I classified each trading day into a positive or negative regime based on the 30-day rolling mean of the daily returns. Let r_t represent the daily cross-sectional median of the volume-adjusted target at time t. The 30-day rolling mean, μ_t, is calculated as:

$$\mu_t = \frac{1}{30} \sum_{i=0}^{29} r_{t-i} \tag{4.3}$$

The market regime indicator I_t is then defined as a binary variable:

$$I_t = \begin{cases} 1, & \text{if } \mu_t > 0 \quad \text{(Positive Regime)} \\ 0, & \text{if } \mu_t \leq 0 \quad \text{(Negative Regime)} \end{cases} \tag{4.4}$$

Runs of consecutive same-regime days were identified to analyze the duration and persistence of these states. As summarized by the run-length distribution in Figure 4.2, the data reveals distinct differences between positive and negative market phases.

The analysis yielded a mean duration of 38.48 trading days for positive regimes, compared to a mean of 28.59 days for negative regimes. The median durations exhibit a similar pattern, with positive regimes lasting a median of 31.5 trading days, while negative regimes showed a somewhat shorter median duration of 29.0 trading days.

This persistence suggests that bullish and bearish phases operate at a timescale of approximately one to one-and-a-half months. The presence of such multi-week autocorrelation strongly motivates the use of a sequence-based econometric model that conditions on recent market history, rather than treating each trading day as strictly independent.

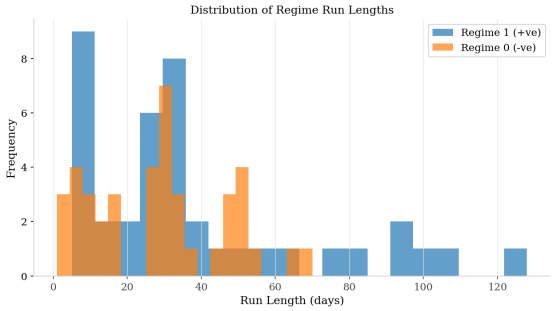


Figure 4.2: Regime Distribution

4.4 Missing Data Characterisation

A structured null-value audit was performed on the candidate feature matrix before feature selection. The analysis was broken down across two dimensions: by column (feature-level null rates) and by sector (sector-level null rates), with flag columns excluded from the null count since their absence is by construction.

The vast majority of technical indicator columns, price-derived features, and volume features were fully populated, as they are computed deterministically from the OHLCV price history. Systematic missingness was concentrated in fundamental ratio columns, which are updated quarterly and structurally absent for time periods preceding a company's first earnings disclosure.

At the sector level, certain sectors exhibited higher average null rates for specific fundamental categories. Leveraged ratio features (e.g. Net Debt/EBITDA, Interest Coverage Ratio) showed elevated nulls for companies without debt, while multi-year CAGR features were absent for companies with fewer than three years of reported history at the start of the sample. No stock had missing values across a majority of its feature columns; missingness was concentrated in a small number of fundamental fields in all cases.

The handling strategy was as follows: binary flag features were zero-filled (absence = "not reported"); all other features were forward-filled within each stock (propagating the most recently known value) and then zero-filled for any residual NaN at the beginning of a stock's history. This hierarchy preserves temporal ordering and avoids introducing future information.

4.5 Feature–Target Correlation Analysis

4.5.1 Pairwise Feature Correlation

A time-stratified Spearman correlation matrix was computed on a sample drawn uniformly at 2,000 observations per calendar month from the training set, totalling approximately 240,000 observations across the 10-year training window. Using a monthly-stratified rather than a random sample ensures that the correlation estimate is not dominated by high-volatility periods, which tend to inflate correlation magnitudes.

The heatmap revealed several well-known structural correlation clusters:

- **Cluster 1 – Moving average family:** SMA_20, SMA_50, SMA_200, EMA variants, and all envelope-type indicators (Bollinger Bands, Keltner Channels, Donchian Channels) form a tightly correlated cluster. Pairwise Spearman correlations within this group routinely exceeded 0.85.
- **Cluster 2 – Momentum oscillators:** RSI_14, CCI_14, MACD, and ROC_10 are moderately correlated with each other ($\rho \approx 0.4$–0.7) but less so with the moving-average cluster.
- **Cluster 3 – Valuation multiples:** P/E, EV/EBITDA, P/B, and P/CF form a loose cluster, driven by the common denominator of the share price. EV/Sales is less correlated with the pure price multiples, reflecting its insensitivity to leverage.
- **Cluster 4 – Volume flow:** OBV and PVT are highly correlated ($\rho > 0.8$), consistent with both being cumulative directional volume measures.

The correlation analysis directly motivated the feature exclusions described in Section B.6. Features with $|\rho| > 0.75$ within a cluster were reduced to a single representative with the highest mean IC.

4.5.2 Per-Stock Information Coefficient

Per-stock Spearman IC values between each candidate feature and the 7-day forward target were computed for every stock with at least 200 training observations. The mean IC across the 103-stock universe, along with its standard deviation and the fraction of stocks exhibiting a positive correlation, was reported for each feature.

The IC analysis confirmed several findings. Among the considered technical indicators, RSI_14, AROONOSC_14, and STOCHh_14_3_3 consistently showed the highest mean

absolute IC values in the momentum group. vol_prank_30 showed a reliably positive mean IC across stocks, consistent with the intuition that above-average volume precedes directional price continuation. Among fundamental features, valuation multiples (P/E, EV/EBITDA) showed small but consistent mean IC values, while profitability metrics (ROE, Interest Coverage Ratio) added orthogonal signal not captured by the price-based features.

Features with mean absolute IC below 0.005 and no compensating sector-specific pattern were excluded. The ICIR (mean IC divided by standard deviation of IC across time) exceeding 0.5 was used as an additional criterion for retaining features with modest but stable predictive content.

4.6 Normalisation and Final Feature Preparation

4.6.1 Signed Log-Compression

Before normalisation, a signed log1p transform was applied to all continuous features with unbounded range (see Section 3.4.1). This step ensures that the subsequent z-score normalisation is not dominated by extreme values that survive after clipping.

4.6.2 Two-Tier Normalisation Strategy

Features were normalised using one of two strategies, depending on whether their absolute magnitude is interpretable across stocks or is stock-size-dependent:

Per-stock z-score (for H-L, turnover_per_trade, and AD): These features reflect absolute scale that differs systematically by firm size. A large-cap stock naturally has a wider daily price range (H-L) and larger per-trade turnover than a small-cap stock, even when both exhibit identical momentum behaviour relative to their own history. Computing the z-score within each stock's own training history removes this size effect:

$$z_{i,t}^{c} = \frac{x_{i,t}^{c} - \hat{\mu}_{i}^{c}}{\hat{\sigma}_{i}^{c}}, \quad \hat{\mu}_{i}^{c} = \frac{1}{|\mathcal{T}_{\text{train}}|} \sum_{t \in \mathcal{T}_{\text{train}}} x_{i,t}^{c} \tag{4.5}$$

where $\hat{\mu}_{i}^{c}$ and $\hat{\sigma}_{i}^{c}$ are the training-period mean and standard deviation of feature c for stock i.

Cross-sectional winsorise and z-score (for all remaining continuous features): Features that are directly comparable across stocks—valuation multiples, oscillators, efficiency ratios—are first winsorised at the 0.1st and 99.9th percentiles of the pooled

training distribution, then z-scored:

$$x_{i,t}^{c,\text{norm}} = \frac{\text{clip}(x_{i,t}^{c}, l^{c}, u^{c}) - \hat{\mu}_{\text{train}}^{c}}{\hat{\sigma}_{\text{train}}^{c}} \tag{4.6}$$

where l^c and u^c are the training-set quantile bounds. All normalisation parameters are estimated on the training split and applied identically to validation and test data.

4.6.3 NaN Imputation

After normalisation, residual NaN values were handled as follows. Binary flag features were zero-filled (zero consistently encodes "not available"). The `ret_1d` column (used for graph edge computation, not as a model input) was zero-filled, treating a missing daily return as a no-move day − the appropriate treatment for the first observation in each stock's history. Remaining feature NaN values (concentrated at the beginning of each stock's history for features with long lookback requirements) were zero-filled after applying the forward-fill described in Section 3.3.2. A final sanity-check pass confirmed zero residual NaN across all model input features within the 2013-01-01 to 2024-12-31 window.

4.7 Graphical Exporatory Analysis

To analyze the interconnectivity of the market, I modelled the relationships between different stocks as a multiplex graph over time. Using PyTorch Geometric, I generated sequential data where each graph represents the state of 103 stocks. The edge features represent three modes of relationships: shared sector classification, trailing 60-day historical correlation exceeding a 0.60 threshold, and co-occurring spikes in volume (determined by a percentile rank threshold).

4.7.1 Multiplex Graph Construction and Sparsity

An exemplary snapshot of the generated graph for the date **August 16, 2013**, provides insights into the sparsity and multiplex nature of the market structure. With 103 nodes (stocks), a fully connected undirected graph would consist of $103 \times (103 - 1)/2 = 5,253$ maximum possible edges. However, after applying stringent edge criteria reveals the following properties:

- **Graph Sparsity**: The snapshot graph contains 377 unique connections, resulting in an overall graph density of just 7.18%. This sparse representation confirms

that the graph builder successfully filters out noise and captures only the most significant underlying market signals.

- **Average Degree**: On average, each node is connected to roughly 7.32 other stocks.
- **Sector Dominance**: Sectoral relationships dominate the edge construction, accounting for 349 of the connections.
- **Correlation Features**: Despite the relatively long 60-day window, only 51 pairs passed the stringent 0.60 rolling correlation threshold on this specific date.
- **Feature Overlap**: Notably, there are 23 pairs that are both in the same sector and highly correlated, suggesting strong intra-sector market coherence.

4.7.2 Graph Homophily and Target Assortativity

A critical assumption in the successful application of Graph Neural Networks is the presence of *homophily* − the tendency of connected nodes to share similar features or target outcomes. To validate the predictive utility of my multiplex graph construction, I analyzed the target assortativity of the network. Assortativity in this context measures the Pearson correlation of the target variable (7-day forward returns) between all connected node pairs.

To strictly prevent data leakage and ensure the integrity of my out-of-sample evaluation, I restricted this target-based Exploratory Data Analysis (EDA) exclusively to the training dataset horizon.

For a given graph snapshot at time t, let E_k represent the set of edges of a specific type k (e.g., sector edges or correlation edges). Let y_u and y_v be the target returns for nodes u and v connected by an edge in E_k. I compute the target assortativity r_k as the Pearson correlation coefficient:

$$r_k = \frac{\sum_{(u,v)\in E_k}(y_u - \bar{y}_u)(y_v - \bar{y}_v)}{\sqrt{\sum_{(u,v)\in E_k}(y_u - \bar{y}_u)^2 \sum_{(u,v)\in E_k}(y_v - \bar{y}_v)^2}} \tag{4.7}$$

where $\bar{y}_u$ and $\bar{y}_v$ are the mean target values for the respective node groups across the edge set. An assortativity score of 1.0 indicates perfect joint movement, while 0.0 indicates random noise.

As depicted in Figure 4.3, I calculated this metric for both static sector edges and dynamic correlation edges across the training sequences. The analysis yielded two

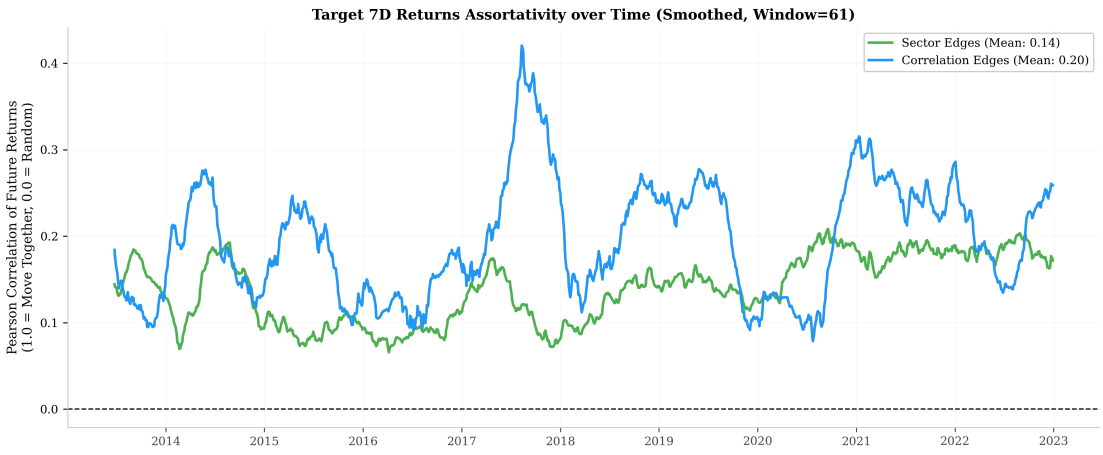


Figure 4.3: Target Assortativity over time (Training Set), comparing the Pearson correlation of future 7-day returns for stocks connected by Sector Edges versus Correlation Edges.

significant insights:

- **Validation of Homophily:** Both edge types consistently exhibit positive assortativity over time. The sector edges maintain a positive baseline (Mean: 0.140, Min: -0.096, Max: 0.680), confirming that traditional sector classifications inherently capture a degree of shared market variance with relatively low extreme bounds.
- **Superiority of Affinity Edges:** Crucially, my empirically derived correlation edges exhibit a higher average assortativity of 0.199. While they exhibit wider variance during market shocks (Min: -0.633, Max: 0.930), their significantly higher average indicates that dynamic affinity edges are much more strongly associated with joint future returns than static sector groupings.

This finding strongly justifies my multiplex approach. While standard homophily (sector classification) provides a stable, low-variance structural backbone, the empirically derived affinity edges capture stronger, higher-conviction predictive signals. The combination of both allows the model to map structural baselines alongside dynamic market momentum.

Statistical Significance of Edge Assortativity

To ensure that my observed superiority of dynamic affinity edges over static homophily edges is not merely a product of random sample variation, I tested the statistical significance of the difference between their assortativity distributions.

Because the assortativity scores for sector edges (r_{sec}) and correlation edges (r_{corr})

were computed simultaneously across the exact same temporal graph snapshots, the data points are fundamentally paired. Therefore, I formulated the following hypotheses:

- **Null Hypothesis (H_0):** There is no true mean difference between the assortativity of sector edges and correlation edges ($\mu_{corr} - \mu_{sec} = 0$).
- **Alternative Hypothesis (H_1):** The assortativity of correlation edges is significantly different from that of sector edges ($\mu_{corr} - \mu_{sec} \neq 0$).

I evaluated these hypotheses using a paired samples t-test. Let $d_t = r_{corr,t} - r_{sec,t}$ be the difference in assortativity at snapshot t. I calculate the t-statistic as:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \tag{4.8}$$

where $\bar{d}$ is the sample mean of the differences, s_d is the standard deviation of the differences, and n is the total number of graph snapshots in the training set.

The paired t-test yielded a highly significant result ($t = 8.7545$, $p \approx 6.64 \times 10^{-18}$), leading me to decisively reject the null hypothesis. To account for potential non-normality in the empirical assortativity distributions − a common characteristic of financial time series − I also conducted a non-parametric Wilcoxon signed-rank test. The Wilcoxon test strongly corroborated the parametric results ($W = 287,296$, $p \approx 9.83 \times 10^{-14}$), confirming the difference is robust well beyond the 1% significance level.

Consequently, I mathematically confirm that empirically derived correlation edges yield a significantly higher baseline of predictive target homophily (0.20) compared to standard static sector groupings (0.14).

Furthermore, for the third edge type of volume spikes based edges, the assortativity was no different than 0, thus the edge was not significant and the parameter was dropped to avoid introduction of noise in graphical analysis.

4.7.3 Topology and Temporal Dynamics

To ensure the robustness of the graph sequence formulation prior to sequence modeling, an Exploratory Data Analysis (EDA) was performed over the entirety of the training sequences (2,466 snapshots).

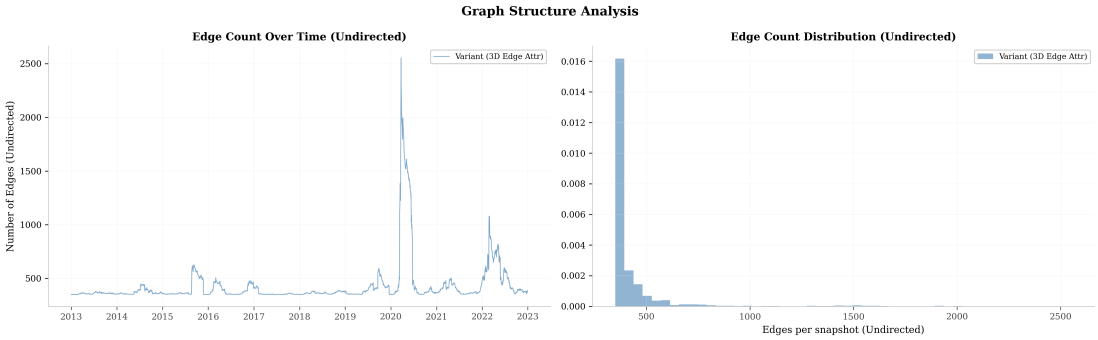


Figure 4.4: Edge Count and Density Distribution Over Time. The graphs depict the variability of edge connections (undirected) throughout the training horizon.

The distribution of graph metrics (Figure 4.4) reveals significant temporal variability. The number of undirected edges per snapshot ranged from a minimum of 349 to a maximum of 2,553 pairs, with a mean of 427 edges. Correspondingly, the mean edge density across the training dataset was roughly 0.0812 (8.12%).

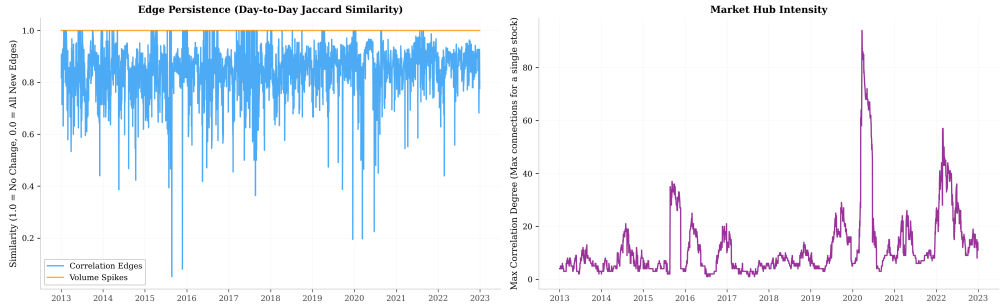


Figure 4.5: Temporal Graph Dynamics illustrating Day-to-Day Edge Persistence (Jaccard Similarity) and Market Hub Intensity.

Furthermore, I tracked Day-to-Day Jaccard Similarity and Market Hub Intensity metrics (Figure 4.5) to quantify edge persistence. The temporal stability analysis yielded an average Jaccard Similarity of 0.8453 for correlation edges. This consistently high score implies that the correlation features change gradually from day to day, rather than randomly dropping in and out, which validates my 60-day feature engineering window. Conversely, the "Market Hub Intensity" (maximum correlation degree for a single stock) exhibited periodic spikes. This time-varying connectivity ultimately validates the necessity of sequential graph neural networks to distill actionable alpha from complex, evolving market topologies.

This time-varying connectivity supports the hypothesis that the market's hidden relationship structure is non-stationary and dynamically multiplexed, motivating the use of sequential graph neural networks to fully capture any predictive alpha.

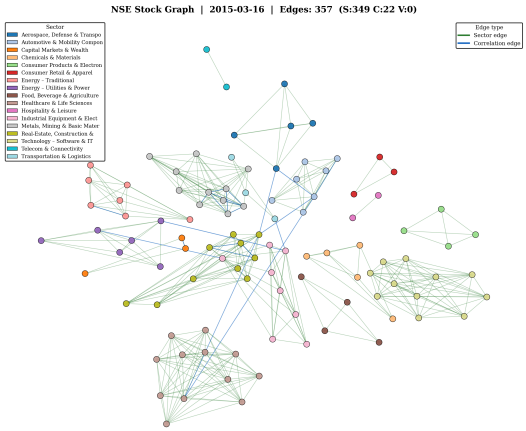
4.7.4 Longitudinal Visualization of Graph Dynamics

While aggregate metrics provide a macro-level view of the network's behavior, I generated qualitative visual snapshots to directly inspect how the multiplex topology evolves over time. To achieve this, I applied a fixed-layout force-directed algorithm to position the 103 stock nodes, keeping their spatial coordinates constant. The nodes are color-coded by their respective market sectors.

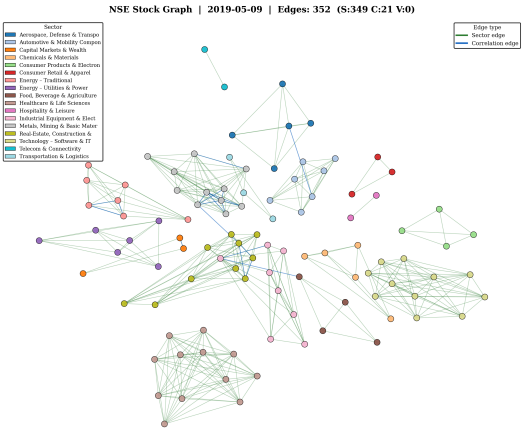
I then sampled four distinct trading days from the training horizon—specifically extracting one random snapshot from the years 2015, 2019, 2020, and 2021. Within these visualizations, edges are classified by their features: static Sector homophily edges (green) and dynamic Correlation affinity edges (blue, with dynamic opacity based on correlation strength). Volume-spike edges were excluded from these static renders to isolate the price-correlation dynamics.

Extracting these static frames (Figure 4.6) visually corroborates my hypothesis regarding the non-stationary nature of market regimes. Because the green sector edges are structurally static, any day-to-day topological shifts are driven entirely by the empirical co-movement of the assets.

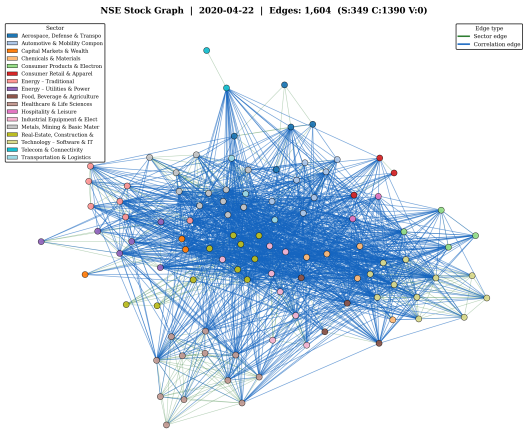
Comparing the panels reveals stark contrasts in market interconnectivity. For instance, standard low-volatility trading periods exhibit sparse networks heavily reliant on intra-sector homophily. In contrast, years subjected to severe macroeconomic shocks (such as 2020) display a topology that undergoes a massive structural shift. During these periods, the baseline sector edges are heavily interconnected by a dense web of cross-sector correlation edges. This visual evidence clearly illustrates why a dynamic, sequence-based Graph Neural Network is essential: a model restricted to a static sector-based graph would be completely blind to the systemic, cross-sectional momentum that dominates the market during shifting macroeconomic regimes.



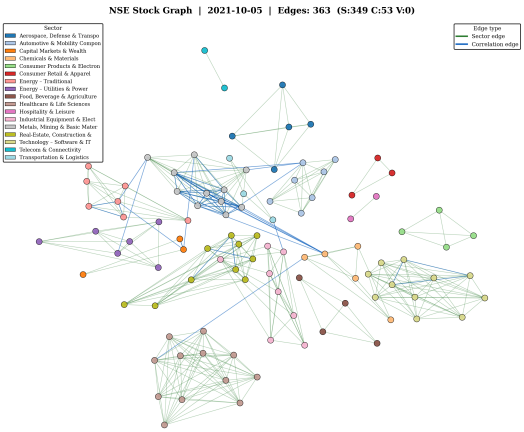
(a) Sampled Graph Snapshot (2015)



(b) Sampled Graph Snapshot (2019)



(c) Sampled Graph Snapshot (2020)



(d) Sampled Graph Snapshot (2021)

Figure 4.6: Multiplex graph topology sampled across four distinct years. Node positions are fixed to highlight the structural evolution of cross-sector correlation edges (blue) against the static sector baseline (green).

Chapter 5

Methodology and Predictive Model Architectures

Building on the feature matrix and market-network topologies constructed in previous chapters, this chapter formalises the mathematical optimisation framework, neural-network architectures, and unified training engine that together constitute the predictive system developed in this thesis. The system is implemented in PyTorch and PyTorch Geometric (PyG) and operates on dynamic spatio-temporal graphs constructed from daily equity price data. All three model variants share a single prediction objective: a volatility-adjusted cross-sectional return, and each produces three calibrated quantile estimates Q_{10}, Q_{50}, and Q_{90} that bound the predicted rank signal.

5.1 Spatio-Temporal Problem Formalisation

The input to the learning system is a temporal sequence of market-snapshot graphs unrolled over a historical lookback window of T trading days:

$$\mathcal{G} = \big[g_1, g_2, \ldots, g_T\big]. \tag{5.1}$$

To facilitate efficient batch processing without sacrificing temporal structure, PyG data loaders pack each discrete time-step across B independent training windows into a single *disjoint* batched graph. Consequently, each element g_t contains $B \times N$ asset nodes, where N is the cross-sectional size of the equity universe on day t. The node feature tensor presented to temporal components of all architectures has shape $[B \times N,\ T,\ d_{\text{in}}]$, while spatial components operate on the disjoint adjacency structure $\mathcal{E}_t$ and the continuous

multi-dimensional edge attribute matrix

$$\mathbf{E}_{\text{attr},t} \in \mathbb{R}^{M_t \times d_e}, \tag{5.2}$$

where M_t is the number of edges at time t. Although the architecture supports $d_e = 3$ edge features (sector homophily, rolling Pearson correlation, and rolling volatility ratio), empirical analysis during development showed that the volatility ratio channel contributed noise rather than predictive signal. Consequently, edges are constructed using $d_e = 2$ active features (sector homophily and rolling correlation) in all experiments, while the architectural capacity for a third channel is retained for extensibility.

5.2 Optimization Framework

Standard asset-pricing models optimised with Mean Squared Error (MSE) suffer from two structural deficiencies: they target the conditional mean (providing no quantification of downside-risk bounds) and penalise all prediction errors symmetrically (failing to prioritise correct *relative* cross-sectional ordering). To address both issues the unified model is trained with a custom **Hybrid Quantile-Ranking Loss** that simultaneously calibrates absolute predictive uncertainty and relative cross-sectional ordering.

5.2.1 Vectorised Multi-Quantile Pinball Loss

To produce probability bounds without parametric distributional assumptions, the model forecasts $Q = 3$ quantile levels $\boldsymbol{\tau} = [0.1,\ 0.5,\ 0.9]$ simultaneously via a single forward pass. For quantile level $\tau_q \in (0, 1)$, the signed residual for asset i is

$$e_{i,q} = y_i - \hat{q}_{i,\tau_q}, \tag{5.3}$$

where y_i is the realised target and $\hat{q}_{i,\tau_q}$ is the model's τ_q-quantile prediction. The asymmetric *pinball* loss is then

$$\mathcal{L}_{\text{quantile}}(\theta) = \frac{\displaystyle\sum_{q=1}^{Q} w_q \left[\frac{1}{BN} \sum_{i=1}^{BN} \max\big(\tau_q\, e_{i,q},\ (\tau_q - 1)\, e_{i,q}\big) \right]}{\displaystyle\sum_{q=1}^{Q} w_q}, \tag{5.4}$$

where $w_q \geq 0$ are per-quantile importance weights (uniform by default). By simultaneously optimising all three levels, the model learns a directional alpha signal at $\tau = 0.5$ while

anchoring the 10th and 90th percentile bounds that are later used for risk measurement during portfolio construction.

5.2.2 Cross-Sectional Ranking Loss

While the pinball loss anchors absolute numerical predictions, market-neutral strategies depend entirely on *relative* ordering within a cross-section. A pairwise ranking penalty therefore evaluates the ordering purely cross-sectionally per trading day. For each batch, predictions are reshaped from $[BN, Q]$ back to $[B, N, Q]$ to isolate individual daily cross-sections. The ranking signal used is the median prediction $\hat{q}_{i,50}$.

Let $\Delta y_{ij} = y_i - y_j$ and $\Delta\hat{q}_{ij} = \hat{q}_{i,50} - \hat{q}_{j,50}$ denote the true and predicted return differences for any ordered asset pair (i, j). Raw financial return differences contain substantial noise, so a *noise-margin mask* M_{ij} suppresses pairs whose true gap is smaller than a threshold γ:

$$M_{ij} \;=\; \mathbf{1}\{\Delta y_{ij} > \gamma\}\,, \tag{5.5}$$

where $\gamma = 0.002567$ is the 5th percentile of all positive pairwise volatility-adjusted return differences across the training set, computed prior to model training to avoid data leakage

To prevent extreme fat-tail events from destabilising the gradient, each valid pair is re-weighted logarithmically by its true gap magnitude:

$$\tilde{\omega}_{ij} \;=\; \frac{\ln\!\big(1 + \Delta y_{ij}\big)}{\dfrac{1}{|M|}\displaystyle\sum_{(i,j):\,M_{ij}=1} \ln\!\big(1 + \Delta y_{ij}\big) + \epsilon}, \tag{5.6}$$

where $\epsilon > 0$ prevents division by zero and the denominator normalises the weights to maintain stable gradient magnitudes. Using the `Softplus` function as a smooth, numerically stable surrogate for the logistic loss, the ranking objective is:

$$\mathcal{L}_{\text{rank}}(\theta) \;=\; \frac{1}{|M| + \epsilon} \sum_{\substack{(i,j)\\ M_{ij}=1}} \tilde{\omega}_{ij} \;\cdot\; \text{softplus}\big(-\Delta\hat{q}_{ij}\big), \tag{5.7}$$

where $\text{softplus}(x) = \ln(1 + e^x)$, $|M| = \sum_{i,j} M_{ij}$ is the number of valid pairs, and $\tilde{\omega}_{ij}$ are the normalised gap weights from Eq. 5.6 satisfying $\frac{1}{|M|}\sum_{(i,j):\,M_{ij}=1} \tilde{\omega}_{ij} = 1$.

5.2.3 Unified Hybrid Objective

The final training objective combines uncertainty calibration and cross-sectional ranking:

$$\mathcal{L}_{\text{hybrid}}(\theta) \;=\; \mathcal{L}_{\text{quantile}}(\theta) \;+\; \alpha \cdot \mathcal{L}_{\text{rank}}(\theta), \tag{5.8}$$

where $\alpha \geq 0$ is a scalar hyperparameter that governs the trade-off between interval calibration and cross-sectional ranking accuracy. When $\alpha = 1$ (the default setting), both components contribute equally; increasing α penalises ordering errors more heavily at the cost of wider quantile intervals, and vice versa.

5.3 Monotonic Non-Crossing Quantile Head

A fundamental failure mode of independent multi-quantile regression is *quantile crossing*: a lower quantile estimate that numerically exceeds a higher one, which is a structurally impossible outcome. To enforce strict monotonicity, all three architectures share an identical output head that eliminates crossing by construction.

Given the final d-dimensional embedding $\mathbf{h}_i$ for asset i, a linear projection first produces a raw output vector $\mathbf{r}_i = \mathbf{W}\,\mathbf{h}_i + \mathbf{b}$, $\mathbf{r}_i \in \mathbb{R}^3$. The quantile outputs are then computed via a *Softplus chaining* scheme:

$$q_{i,10} \;=\; r_{i,0}, \tag{5.9}$$

$$q_{i,50} \;=\; q_{i,10} + \mathrm{softplus}(r_{i,1}) \;=\; r_{i,0} + \ln\!\big(1 + e^{r_{i,1}}\big), \tag{5.10}$$

$$q_{i,90} \;=\; q_{i,50} + \mathrm{softplus}(r_{i,2}) \;=\; r_{i,0} + \ln\!\big(1 + e^{r_{i,1}}\big) + \ln\!\big(1 + e^{r_{i,2}}\big). \tag{5.11}$$

Because $\mathrm{softplus}(x) > 0$ for all $x \in \mathbb{R}$, this formulation mathematically guarantees $q_{i,10} \leq q_{i,50} \leq q_{i,90}$ across all parameter values and input magnitudes, making the constraint architecture-level rather than a post-hoc projection. The scheme generalises dynamically to any number of quantile levels $Q \geq 2$ via the recurrence

$$q_{i,q} \;=\; q_{i,q-1} + \mathrm{softplus}(r_{i,q}), \quad q = 1, \ldots, Q-1, \tag{5.12}$$

with $q_{i,0} = r_{i,0}$. The output head is preceded by Layer Normalisation and a Dropout layer applied to the combined embedding $\mathbf{h}_i$.

5.4 Model Architectures

Three architectures of increasing spatio-temporal integration are evaluated. All share the same edge-attributed GATv2 spatial module, the same monotonic output head, and the same optimisation objective, ensuring that any observed performance differences are attributable solely to how spatial and temporal information are combined.

5.4.1 Model 1: Pure Temporal Baseline (LSTM)

The pure temporal baseline processes historical return attributes sequentially per asset node and deliberately ignores graph topology. The purpose is to provide an ablation benchmark against which the performance contribution of spatial modelling can be measured.

Node features from all T time-steps are stacked into a tensor $X_{\text{seq}} \in \mathbb{R}^{BN \times T \times d_{\text{in}}}$ and passed through a multi-layer LSTM with hidden dimension d_h. The LSTM maintains a cell state c_t and a hidden state h_t at each step via the standard gating mechanism:

$$\mathbf{f}_t = \sigma\big(\mathbf{W}_f\,[\mathbf{h}_{t-1} \,\|\, \mathbf{x}_t] + \mathbf{b}_f\big), \tag{5.13}$$

$$\mathbf{i}_t = \sigma\big(\mathbf{W}_i\,[\mathbf{h}_{t-1} \,\|\, \mathbf{x}_t] + \mathbf{b}_i\big), \tag{5.14}$$

$$\tilde{\mathbf{c}}_t = \tanh\big(\mathbf{W}_c\,[\mathbf{h}_{t-1} \,\|\, \mathbf{x}_t] + \mathbf{b}_c\big), \tag{5.15}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \tag{5.16}$$

$$\mathbf{o}_t = \sigma\big(\mathbf{W}_o\,[\mathbf{h}_{t-1} \,\|\, \mathbf{x}_t] + \mathbf{b}_o\big), \tag{5.17}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t). \tag{5.18}$$

Only the terminal hidden state $\mathbf{h}_T$ is forwarded to the shared prediction head.

Algorithm 1 Vanilla LSTM Forward Pass

Require: Sequence of node feature matrices $[X_1, \ldots, X_T]$, $X_t \in \mathbb{R}^{BN \times F}$
Ensure: Non-crossing quantiles $Q \in \mathbb{R}^{BN \times 3}$
1: Stack: $X_{\text{seq}} \leftarrow \text{Stack}([X_t]_{t=1}^{T}, \text{dim}{=}1)$
2: Forward: $H_{\text{all}}, (\cdot) \leftarrow \text{LSTM}(X_{\text{seq}})$
3: Extract terminal state: $H \leftarrow H_{\text{all}}[:, -1, :]$
4: $H \leftarrow \text{LayerNorm}(H)$; $H \leftarrow \text{Dropout}(H, p)$
5: $R \leftarrow \text{Linear}(H)$ ▷ Raw projection, shape $[BN, 3]$
6: Apply Softplus chaining (Eqs. 5.9–5.11)
7: **return** $Q = [q_{10}, q_{50}, q_{90}]$

5.4.2 Model 2: Parallel Late-Fusion Network (Hybrid GAT $\|$ LSTM)

The Hybrid model decouples temporal trajectory modelling from cross-sectional structure discovery by running two independent branches in parallel before merging their outputs.

Temporal branch. An LSTM processes all T historical steps identically to the baseline (Section 5.4.1), producing terminal embedding $\mathbf{h}_i^{\text{temp}} \in \mathbb{R}^{d_h}$.

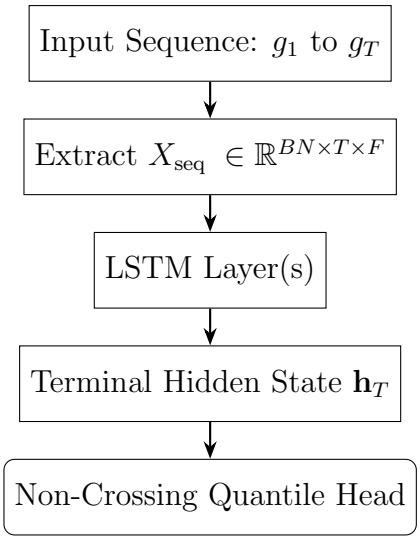


Figure 5.1: Data flow for the Vanilla LSTM purely temporal baseline.

Spatial branch. Graph convolution is applied *only* to the final day's snapshot g_T, reflecting the intuition that contemporaneous peer relationships are the most informative structural signal for next-day prediction. The spatial module employs a multi-head Graph Attention Network version 2 (GATv2), which computes dynamic, input-dependent attention coefficients by evaluating both endpoint node features and multi-dimensional edge attributes simultaneously. For edge (i, j) at head k, the attention coefficient is:

$$\alpha_{ij}^{(k)} \;=\; \frac{\exp\Big(\mathbf{a}_k^{\top}\,\mathrm{ELU}\big(\mathbf{W}_k\,[\mathbf{h}_i \,\|\, \mathbf{h}_j \,\|\, \mathbf{e}_{ij}]\big)\Big)}{\displaystyle\sum_{l\in\mathcal{N}(i)} \exp\Big(\mathbf{a}_k^{\top}\,\mathrm{ELU}\big(\mathbf{W}_k\,[\mathbf{h}_i \,\|\, \mathbf{h}_l \,\|\, \mathbf{e}_{il}]\big)\Big)}, \tag{5.19}$$

where $\mathbf{e}_{ij} \in \mathbb{R}^{d_e}$ is the multi-dimensional edge attribute vector, $\mathbf{W}_k$ is a learnable weight matrix shared for source and target nodes at head k, and $\mathbf{a}_k \in \mathbb{R}^{d'_e}$ is the attention vector. The updated node representation aggregating all K heads is:

$$\mathbf{h}'_i \;=\; \Big\|_{k=1}^{K} \sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(k)}\,\mathbf{W}_k^{V}\,\mathbf{h}_j, \tag{5.20}$$

where $\|$ denotes head concatenation. After each graph layer, Layer Normalisation and Dropout are applied.

Late fusion. The spatial and temporal embeddings are concatenated:

$$\mathbf{h}_i^{\text{fused}} \;=\; \big[\,\mathbf{h}_i^{\text{spatial}} \,\|\, \mathbf{h}_i^{\text{temp}}\,\big] \;\in\; \mathbb{R}^{d_g+d_h}, \tag{5.21}$$

and $\mathbf{h}_i^{\text{fused}}$ is passed to the shared prediction head.

Algorithm 2 Hybrid (Parallel Late-Fusion) Forward Pass

Require: Sequence $G = [G_1, \ldots, G_T]$; GATv2 layers L
Ensure: Non-crossing quantiles $Q \in \mathbb{R}^{BN \times 3}$

// Temporal branch
$X_{\text{seq}} \leftarrow \text{Stack}([G_t.X]_{t=1}^{T}, \text{dim}=1)$
$H_{\text{all}}, (\cdot) \leftarrow \text{LSTM}(X_{\text{seq}})$
$H_{\text{temp}} \leftarrow \text{Dropout}(H_{\text{all}}[:, -1, :], p)$
// Spatial branch (terminal graph only)
$H^{(0)} \leftarrow G_T.X$
for layer $i = 1$ to L **do**
    $H^{(i)} \leftarrow \text{GATv2}(H^{(i-1)}, G_T.\mathcal{E}, G_T.\mathbf{E}_{\text{attr}})$
    $H^{(i)} \leftarrow \text{Dropout}(\text{LayerNorm}(\text{ELU}(H^{(i)})), p)$
end for
// Late fusion and prediction
$H_{\text{fused}} \leftarrow [H^{(L)} \parallel H_{\text{temp}}]$
$H_{\text{fused}} \leftarrow \text{Dropout}(\text{LayerNorm}(H_{\text{fused}}), p)$
Apply Softplus chaining on $\text{Linear}(H_{\text{fused}})$
return $Q = [q_{10}, q_{50}, q_{90}]$

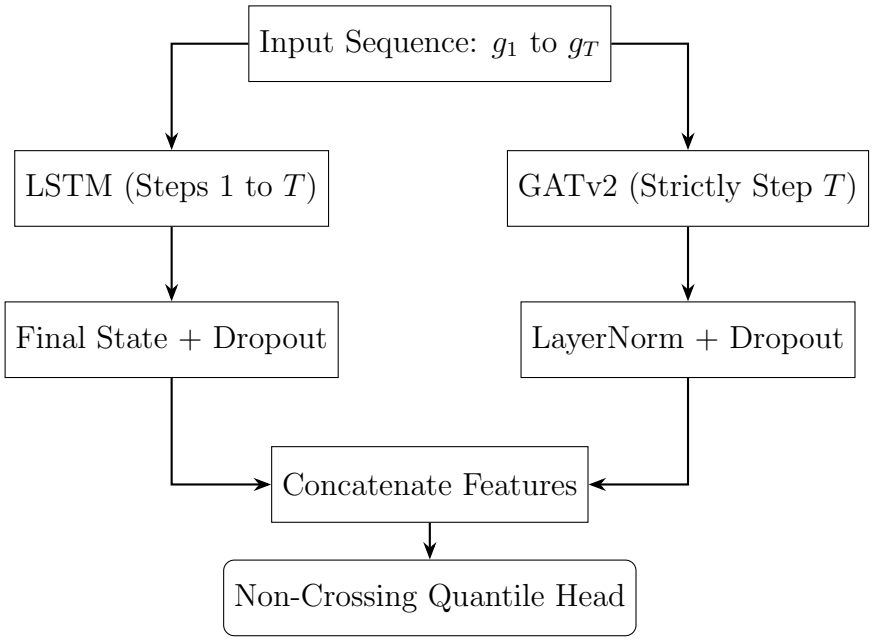


Figure 5.2: Data flow for the Parallel Late-Fusion Hybrid (GAT $\parallel$ LSTM) architecture.

5.4.3 Model 3: Sequential Graph-Recurrent Architecture (ROLAND)

The ROLAND architecture achieves the tightest integration of spatial and temporal modelling by unrolling graph convolution across every step $t \in \{1, \ldots, T\}$ of the lookback window. At each time-step, a GATv2 layer produces a structural embedding from the

day's graph snapshot; this embedding is then passed through a Gated Recurrent Unit (GRU) cell that maintains a persistent per-node memory state, propagating relational context forward to the next day.

The GRU update at layer i, time-step t is:

$$\mathbf{z}_{i,t} = \sigma\big(\mathbf{W}_z\,[\mathbf{s}_{i,t-1} \,\|\, \mathbf{h}^{\mathrm{GAT}}_{i,t}] + \mathbf{b}_z\big), \tag{5.22}$$

$$\boldsymbol{\rho}_{i,t} = \sigma\big(\mathbf{W}_r\,[\mathbf{s}_{i,t-1} \,\|\, \mathbf{h}^{\mathrm{GAT}}_{i,t}] + \mathbf{b}_r\big), \tag{5.23}$$

$$\tilde{\mathbf{s}}_{i,t} = \tanh\big(\mathbf{W}_s\,[\boldsymbol{\rho}_{i,t} \odot \mathbf{s}_{i,t-1} \,\|\, \mathbf{h}^{\mathrm{GAT}}_{i,t}] + \mathbf{b}_s\big), \tag{5.24}$$

$$\mathbf{s}_{i,t} = (1 - \mathbf{z}_{i,t}) \odot \mathbf{s}_{i,t-1} + \mathbf{z}_{i,t} \odot \tilde{\mathbf{s}}_{i,t}, \tag{5.25}$$

where $\mathbf{s}_{i,t}$ is the updated memory state for node i at layer i after step t, and $\mathbf{h}^{\mathrm{GAT}}_{i,t}$ is the GATv2 output for that node. The updated state $\mathbf{s}_{i,t}$ is also used as the input to the next graph layer $i+1$ at the same time-step, enabling deep multi-hop spatial reasoning over an evolving topology. Prediction uses only the final memory states $\{\mathbf{s}_{i,T}\}_{i=1}^{BN}$ from the last layer.

Algorithm 3 ROLAND (Sequential Graph-RNN) Forward Pass

Require: Sequence $G = [G_1, \ldots, G_T]$; L layers; initial states $\mathbf{S} = [\mathbf{S}_1, \ldots, \mathbf{S}_L]$ (zeros if not provided)
Ensure: Non-crossing quantiles $Q \in \mathbb{R}^{BN\times 3}$; updated memory states $\mathbf{S}$

for time-step $t = 1$ to T **do**
  $H^{(0)} \leftarrow G_t.X$
  for layer $i = 1$ to L **do**
    ▷ Spatial message passing
    $H^{(i)} \leftarrow \mathrm{GATv2}(H^{(i-1)},\ G_t.\mathcal{E},\ G_t.\mathbf{E}_{\mathrm{attr}})$
    $H^{(i)} \leftarrow \mathrm{Dropout}(\mathrm{LayerNorm}(\mathrm{ELU}(H^{(i)})),\ p)$
    ▷ Temporal memory update (Eqs. 5.22–5.25)
    if $\mathbf{S}_i = \varnothing$ **then** $\mathbf{S}_i \leftarrow \mathbf{0}$
    end if
    $\mathbf{S}_i \leftarrow \mathrm{GRUCell}(H^{(i)},\ \mathbf{S}_i)$
    $H^{(i)} \leftarrow \mathbf{S}_i$ ▷ Updated state feeds next layer
  end for
end for
$H_{\mathrm{final}} \leftarrow \mathbf{S}_L$
$H_{\mathrm{final}} \leftarrow \mathrm{Dropout}(\mathrm{LayerNorm}(H_{\mathrm{final}}),\ p)$
Apply Softplus chaining on $\mathrm{Linear}(H_{\mathrm{final}})$
return Q, $\mathbf{S}$

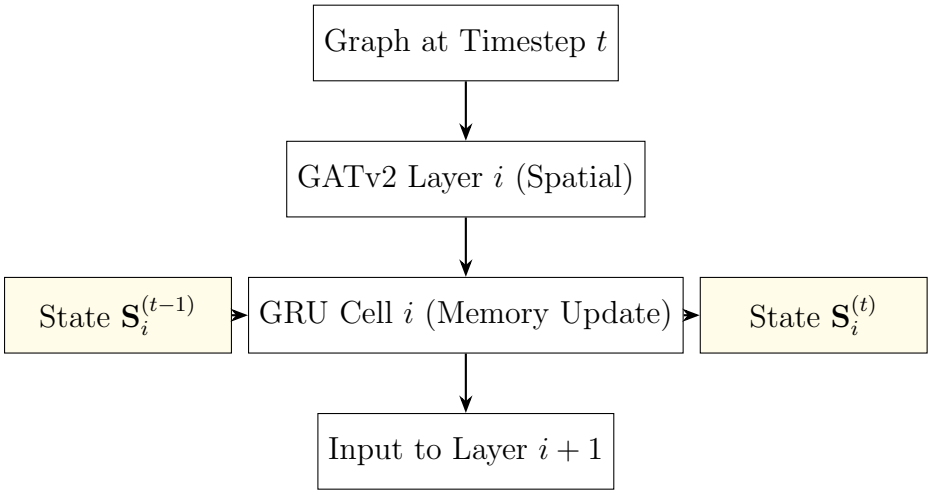


Figure 5.3: A single layer iteration within the sequential ROLAND architecture showing the interleaved GATv2 spatial step and GRU temporal memory update.

5.5 Comparative Architectural Summary

Table 5.1 summarises the design axes across all three candidate architectures.

Table 5.1: Structural Comparison of Predictive Architectures

Dimension	**LSTM**	**Hybrid GAT-LSTM**	**ROLAND**
Temporal encoding	Unrolled LSTM (T steps)	Unrolled LSTM (T steps)	GRU cell per time-step
Spatial geometry	None	Static snapshot (g_T only)	Dynamic (GATv2 at every t)
Spatial-temporal fusion	N/A	Parallel late-fusion (concat)	Sequential interleaved (GATv2 $\rightarrow$ GRU)
Edge features	No	Yes (GATv2 multi-dim.)	Yes (GATv2 multi-dim.)
Memory carrier	LSTM cell state	LSTM cell state	Per-node GRU hidden state
Graph steps used	0 of T	1 of T (terminal)	T of T (all)

5.6 Unified Training and Evaluation Engine

Training, validation, and inference are abstracted into a single `StockEngineV3` object, allowing the three architectures to be evaluated under identical conditions.

5.6.1 Training Loop and Gradient Hygiene

At each optimisation step the engine:

1. Moves the sequence of PyG batched graphs to the target device.

2. Routes the forward pass: ROLAND receives and returns explicit memory states (which are detached before the next batch to prevent unrolling the full computation graph across batch boundaries); LSTM and Hybrid receive no state argument.

3. Evaluates $\mathcal{L}_{\text{hybrid}}$ (Eq. 5.8).

4. Skips the update step if the loss or any gradient is non-finite (NaN or $\pm\infty$), preventing gradient explosions from corrupting model weights.

5. Clips the global gradient norm to $\|\nabla\|_2 \leq 1.0$ before calling the AdamW optimiser update.

The engine supports optional *state carry-over* for ROLAND, in which hidden states are propagated across mini-batches within a single epoch. This enables the model to accumulate coherent structural memory across the full training sequence, at the cost of stricter ordering requirements on the data loader. However, for simplicity and due to memory requirements, the carry states was set to False for this study.

5.6.2 Validation Metrics

During validation, the engine calls `predict()` to collect full prediction tensors $\hat{Q} \in \mathbb{R}^{T\times N\times 3}$ alongside the volume-adjusted targets and raw actual returns. Four signal variants are evaluated: q_{10}, q_{50}, q_{90}, and the inter-quantile spread $q_{90} - q_{10}$. For each signal, the Information Coefficient (IC) is computed as the Spearman rank correlation between the predicted signal and the forward realised return, averaged over all T cross-sections:

$$\overline{\text{IC}} \;=\; \frac{1}{T}\sum_{t=1}^{T} \rho_S\big(\hat{q}_{50,t},\; y_t\big), \tag{5.26}$$

where ρ_S denotes the Spearman correlation. The IC standard deviation σ_{IC} is reported alongside the mean as a stability indicator.

Two IC variants are tracked throughout training:

- IC_{vol}: Spearman correlation against the volume-adjusted target. This is the **primary optimisation signal** used by both the early stopping monitor and the `ReduceLROnPlateau` learning rate scheduler, which reduces the learning rate by a factor of 0.7 after 6 epochs without improvement down to a minimum of 10^{-6}. It is also the objective minimised by the Optuna hyperparameter search.

- $\text{IC}_{\text{actual}}$: Spearman correlation against raw actual returns. Monitored as a diagnostic indicator of true out-of-sample predictive power but does not influence any

training decision.

5.6.3 Early Stopping

To prevent overfitting, an early stopping monitor tracks the validation $\overline{\text{IC}}$ after each epoch. Training is halted and the best-epoch weights are restored if the IC fails to improve by a minimum delta over a patience window of 15 consecutive epochs. This criterion directly aligns regularisation with the downstream financial objective rather than with training loss magnitude.

5.6.4 Post-Hoc Quantile Calibration

After training concludes, an empirical calibration check is performed on the held-out test set to verify that the predicted quantile intervals are statistically honest. For a perfectly calibrated model the empirical exceedance probabilities should satisfy:

$$\mathbb{P}(y > q_{10}) \approx 0.90, \tag{5.27}$$

$$\mathbb{P}(y > q_{50}) \approx 0.50, \tag{5.28}$$

$$\mathbb{P}(y > q_{90}) \approx 0.10. \tag{5.29}$$

Systematic deviation from these targets (e.g. $\mathbb{P}(y > q_{90}) \gg 0.10$) would indicate that the predicted upper bound is too tight and that the model understates tail risk. Satisfying Eqs. 5.27–5.29 ensures that the quantile interval functions as a trustworthy uncertainty band around the primary median trading signal.

Chapter 6

Experiments and Results

This chapter presents the experimental evaluation of three spatio-temporal architectures applied to the task of cross-sectional stock return ranking on the selected universe of 103 NSE stocks. Section 6.1 describes the shared experimental setup. Sections 6.2–6.4 reports results per model. Section 6.5 provides a comparative analysis. Section 6.6 interprets the findings.

6.1 Experimental Setup

6.1.1 Task Definition

The prediction task is a cross-sectional stock ranking problem. At each date t, the model receives a sequence of $T = 42$ daily graphs — one graph per trading day over the preceding two months — and outputs a predicted score for each of the $N = 103$ stocks. Stocks are ranked by their predicted Q50 (median) output, and the top-$k = 5$ and bottom-$k = 5$ ranked stocks form the long and short legs of a simulated portfolio respectively.

A deliberate two-target design separates the training signal from the evaluation criterion. During training, the model learns to predict the *volume-adjusted* forward return:

$$\texttt{target_vol_adj}_{i,t} = y_{i,t} \times \tanh\left(\frac{V_{i,t}^{\text{fwd}}}{V_{i,t}^{\text{hist}}}\right) \tag{6.1}$$

where the volume multiplier suppresses low-conviction price moves. At evaluation time, performance is assessed against the *raw* 7-day forward log return:

$$\texttt{target_ret_7d}_{i,t} = \ln\left(\frac{\text{Close}_{i,t+7}}{\text{Open}_{i,t+1}}\right) \tag{6.2}$$

which is the return an investor would actually realise. The IC against both targets is reported; all portfolio backtests and significance tests use the raw actual return series.

6.1.2 Shared Training Configuration

All three architectures were trained under identical conditions. Table 6.1 summarises the fixed training hyperparameters.

Table 6.1: Shared Training Configuration

Parameter	**Value**
Sequence length (T)	42 trading days
Prediction horizon (H)	7 trading days
Output quantiles	Q10, Q50, Q90
Max epochs	200
Batch size	16 (fixed across models)
Early stopping patience	15 epochs (monitored on validation IC)
Gradient clipping	1.0 (global norm)
Optimiser	AdamW
Learning rate schedule	ReduceLROnPlateau (patience=6, factor=0.7, min_lr=10^{-6})
Correlation window	60 trading days
Correlation threshold	0.60 (minimum Pearson r to draw an edge)
Long/short portfolio size (k)	5 stocks
Max stocks per sector	2
Transaction cost	0.05% per side (0.10% round-trip)
Random seed	42
Training split	2013-01-01 to 2022-12-31
Validation split	2023-03-01 to 2023-12-31
Test split	2024-03-01 to 2024-12-31

6.1.3 Loss Function

All models are trained with the Hybrid Quantile Ranking Loss defined in Chapter 5 (Sections 5.2.1–5.2.3). The loss combines a vectorised multi-quantile pinball component $\mathcal{L}_{\text{quantile}}$ with a pairwise ranking component $\mathcal{L}_{\text{rank}}$, weighted by a tunable scalar α:

$$\mathcal{L}_{\text{hybrid}}(\theta) = \mathcal{L}_{\text{quantile}}(\theta) + \alpha \cdot \mathcal{L}_{\text{rank}}(\theta) \tag{6.3}$$

The ranking component employs a noise-margin mask, $M_{ij} = \mathbf{1}\{\Delta y_{ij} > \gamma\}$, where $\gamma = 0.002567$ corresponds to the 5th percentile of positive pairwise vol-adjusted return differences computed on the training set. The mask is used to identify informative stock pairs for ranking, log-gap reweighting, and the Softplus-based pairwise ranking loss. Full derivations and methodological details are provided in Chapter 5.

6.1.4 Evaluation Metrics

Model performance is evaluated across three dimensions:

Predictive accuracy: The Spearman Information Coefficient (IC) between Q50 predictions and realised raw returns, computed cross-sectionally across 103 stocks at each test date. Mean IC and its standard deviation are reported for both the vol-adjusted and raw-return targets.

Quantile calibration: Empirical exceedance probabilities for Q10, Q50, and Q90 are compared against theoretical targets of 0.900, 0.500, and 0.100 respectively.

Portfolio performance: A fixed baseline strategy (signal smoothing $\alpha = 10^{-6}$, i.e. no effective smoothing; sector cap `Max_per_Sector` = 2) selects the top-5 long and bottom-5 short stocks at each date. Returns are computed via overlapping tranches over the 7-day forward horizon. Reported metrics include annualised return, cumulative net return, hit rate, and three Sharpe ratio variants: the *naive* annualised Sharpe (no overlap correction), the *Lo (2002)* autocorrelation-adjusted Sharpe [25], and the *HAC Sharpe* computed using Newey-West heteroskedasticity- and autocorrelation-consistent variance. Statistical significance is assessed with two-sided t-tests using Newey-West HAC standard errors (6 lags, matching the $H - 1 = 6$ overlap of 7-day return windows) and block bootstrap confidence intervals (block size = 7, B = 10,000 resamples).

6.1.5 Hyperparameter Optimisation

Hyperparameters were tuned with Optuna using the Tree-structured Parzen Estimator (TPE) sampler and Median Pruner. Each of up to 60 trials was trained for at most

150 epochs with early stopping, evaluated on the validation IC against the vol-adjusted target. The best configuration was then used for a final full training run (up to 200 epochs) with checkpoint averaging of the top checkpoints by validation IC.

6.2 LSTM Baseline

6.2.1 Architecture

The LSTM model is a purely temporal baseline (Section 5.4.1) that processes each stock's 42-step feature sequence independently via a multi-layer LSTM, deliberately ignoring graph topology. Only the terminal hidden state $\mathbf{h}_T$ is forwarded to the shared monotonic quantile head, which produces non-crossing Q10, Q50, Q90 outputs via Softplus chaining. This architecture serves as an ablation benchmark: any performance gap between the LSTM and graph-augmented models is attributable solely to the cross-stock relational structure encoded in the dynamic graphs.

6.2.2 Hyperparameter Search

The Optuna study completed 57 of 60 allocated trials. The search space covered: learning rate, dropout, temporal hidden dimension $\in \{16, 32, 64, 128\}$, number of temporal layers $\in \{1, 2, 3\}$, rank weight $\alpha \in \{0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0\}$, and weight decay. The best trial (#32) achieved a validation IC of $\mathbf{+0.0532}$.

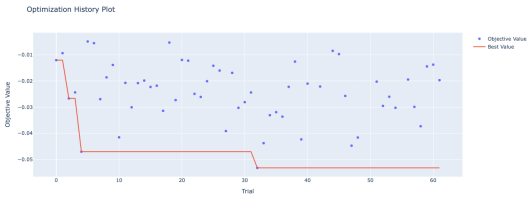


Figure 6.1: LSTM — Optuna optimisation history across 57 trials. The red line tracks the running best objective value (negated validation IC); improvement stabilises after approximately trial 32.

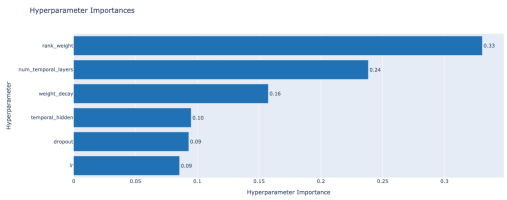


Figure 6.2: LSTM — Hyperparameter importances as assessed by Optuna's TPE sampler. The rank weight is the dominant factor, followed by the number of temporal layers.

Table 6.2: LSTM — Best Hyperparameters

Hyperparameter	**Value**
Learning rate	1×10^{-4}
Dropout	0.4
Temporal hidden dimension	64
Number of temporal layers	1
Rank weight (α)	0.75
Weight decay	1×10^{-5}

6.2.3 Test Set Results

Predictive accuracy: The LSTM achieved a mean test IC of $+0.0156 \pm 0.1652$ on the vol-adjusted target and $+0.0203 \pm 0.1678$ on raw returns. The backtest window mean IC was +0.0132 (std: 0.1656). Despite positive point estimates, none of the IC metrics are statistically significant at conventional levels: the Newey-West HAC t-statistic is 1.118 ($p = 0.265$, NW $p = 0.562$).

Quantile calibration: Coverage rates are severely miscalibrated: Q10 = 0.998 (ideal: 0.900), Q50 = 0.925 (ideal: 0.500), and Q90 = 0.255 (ideal: 0.100). The collapsed Q10 bound renders the lower tail of the predicted distribution largely uninformative.

Portfolio performance: The long portfolio achieved an annualised return of +14.53% and a cumulative net return of +11.2%, generating a negative alpha of −3.3% versus the +14.5% benchmark. The market-neutral L/S portfolio returned +4.3% cumulatively and +5.57% annualised. None of the three legs achieved conventional statistical significance under the NW HAC correction, and the long leg hit rate of 50.3% is indistinguishable from random.

Table 6.3: LSTM — Test Set Performance Summary

Metric	**Long**	**Short**	**L/S Combined**	**Benchmark**
Mean daily return	+0.00446	−0.00106	+0.00170	–
Hit rate	50.3%	50.3%	54.8%	–
Annualised return	+14.53%	−8.20%	+5.57%	–
Cumulative net return	+11.2%	−6.5%	+4.3%	+14.5%
Alpha vs benchmark	−3.3%	−21.0%	−10.1%	0.0%
Sharpe (Naive)	1.856	−0.322	1.272	–
Sharpe (Lo 2002)	0.761	−0.142	0.528	–
Sharpe (HAC)	0.846	−0.206	0.529	–
Sortino ratio	1.142	−1.532	−0.460	–
Calmar ratio	0.721	−0.297	0.743	–
Max drawdown	−18.73%	−25.61%	−7.23%	–
NW p-value	0.4253	0.8856	0.5760	–
Stat. significance	×	×	×	–
Mean IC (vol-adj)	$+0.0156 \pm 0.1652$			
Mean IC (raw returns)	$+0.0203 \pm 0.1678$			
Coverage Q10/Q50/Q90	0.998 / 0.925 / 0.255	(ideal: 0.900 / 0.500 / 0.100)		

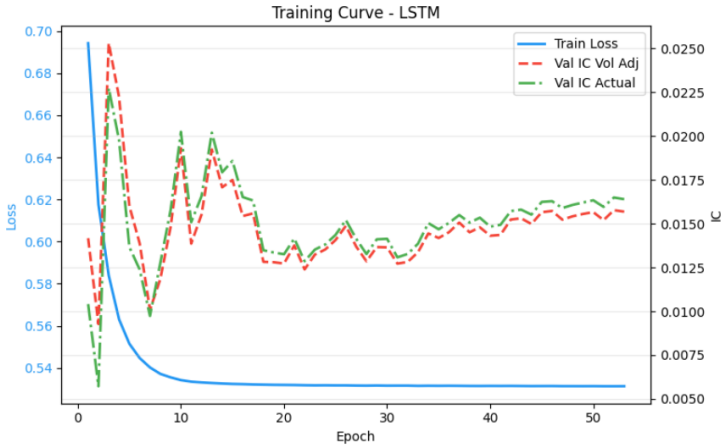


Figure 6.3: LSTM — Training loss (left axis, blue) and validation IC (right axis, (vol_adj_ret) red dashed, (actual_ret) green dashed) per epoch for the best hyperparameter configuration. Training loss converges within the first 10 epochs; validation IC fluctuates around a low positive level throughout, indicating modest but unstable generalisation.

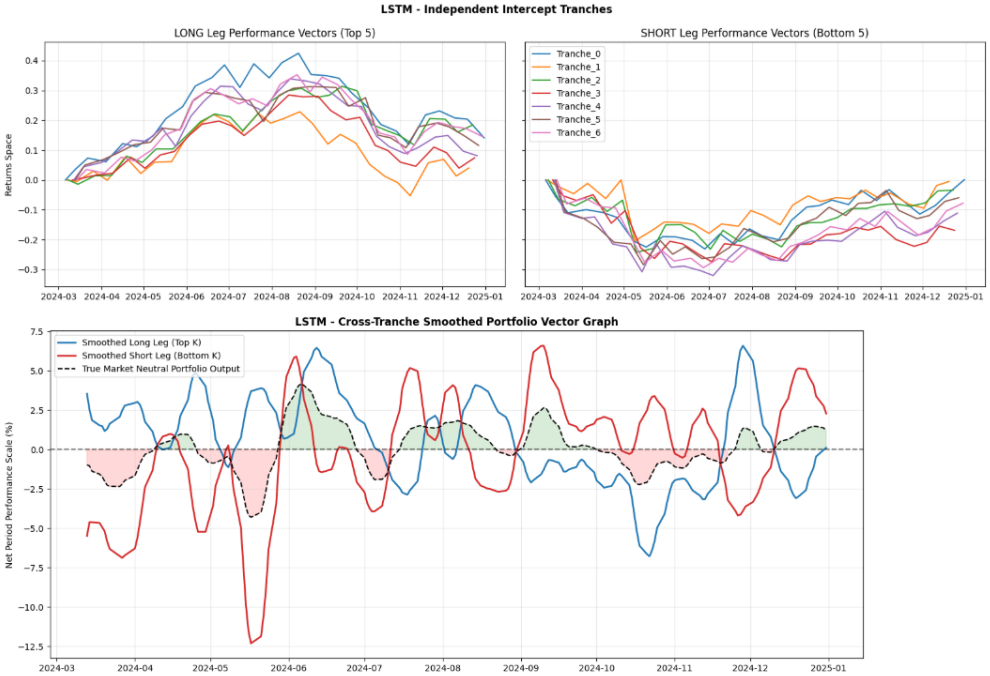


Figure 6.4: LSTM — Independent tranche NAV curves (top panels) and cross-tranche smoothed portfolio vector graph (bottom panel) over the 2024-03-01 to 2024-12-31 test period. Long tranches show modest positive drift; short tranches decline across all seven windows.

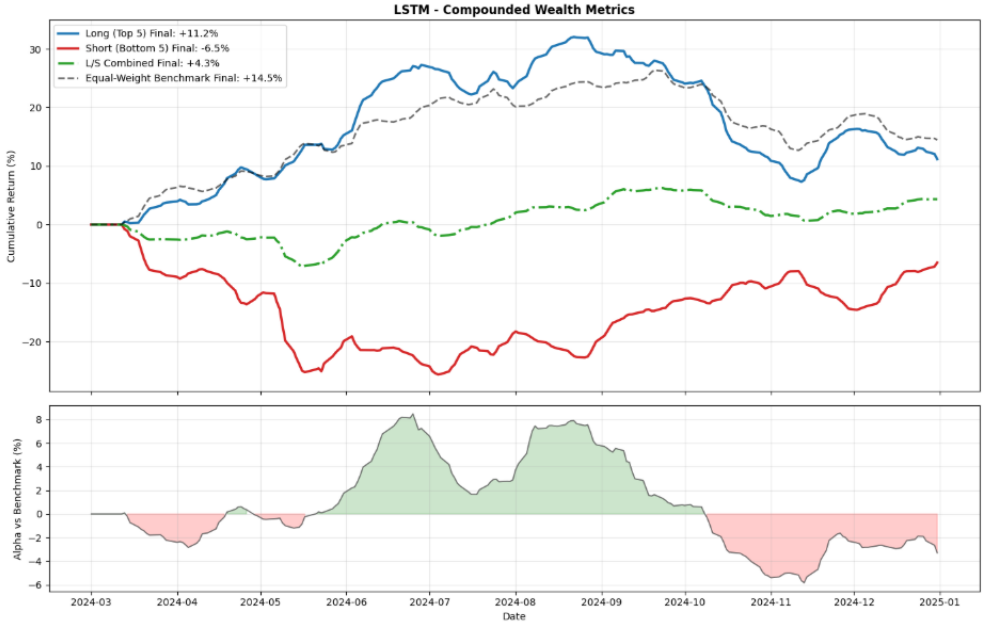


Figure 6.5: LSTM — Compounded wealth analysis. Top panel: cumulative returns of the long (blue), short (red), L/S combined (green dash-dot), and equal-weight benchmark (grey dashed) portfolios. Bottom panel: Rolling alpha of the long portfolio versus benchmark, which turns negative in the early and latter part of test period.

IC Quintile Diagnostic: To diagnose whether the predictive signal is symmetric across the return distribution, the IC was computed separately within the top and bottom quintiles of predicted returns. Within the top quintile (stocks the model ranked highest), the mean IC was slightly negative (-0.0101), with a hit rate of only 50.3%, indicating that virtually the model has no discriminative power among its best-predicted candidates. On the contrary, for the bottom quintile (stocks ranked lowest), achieved a mean IC $+0.0916$ with a hit rate of 61.4%. The signal asymmetry (top $-$ bottom) of -0.1017 provides strong evidence of inverted signal structure, i.e. the model is a better predictor of relative ordering among its predicted losers than among its predicted winners. This finding suggests that the model derives substantially more ranking information from identifying relative underperformances than outperformers. Thus, the positive performence in the long portfolio is likely driven by the sector-constrained portfolio construction and the favourable market conditions rather than by the intrinsic quality of ranking signal.

Return Direction and Market Regime: The return direction diagnostic highlights a persistent structural challenge for the short portfolio during the test period. The bottom quantile produced a positive mean period return of $+0.52\%$, with only 44.7% of periods generating negative returns. The top quintile delivered a similar mean return of $+0.62\%$, indicating limited separation between the model's predicted winners and losers in realized performance. This behaviour is consistent with the prevailing market environment: the equal-weight market portfolio generated a mean period return of $+0.52\%$, 60.9% of test periods were positive, and the market achieved a cumulative gain of 14.5% over the evaluation horizon (197 days) and (14.21% over whole test set of 204 days). In such a sustained bull-market regime, even stocks identified as unattractive by the model tended to appreciate, creating a structural disadvantage for the short leg. Consequently, a strategy that shorts the lowest ranked stocks would incur losses despite the model exhibiting stronger ranking skill among predicted losers, because the stocks still generated positive average returns in the bull-market regime.

6.3 GAT-Hybrid Model

6.3.1 Architecture

The GAT-Hybrid is a parallel late-fusion network (Section 5.4.2) that decouples temporal and spatial modelling into two independent branches. The temporal branch is an LSTM that processes all $T = 42$ historical steps per node. The spatial branch applies a multi-

head GATv2 layer exclusively to the final day's graph snapshot g_T, using two-dimensional edge attributes (sector homophily and rolling Pearson correlation). The two branch outputs are concatenated via late fusion and passed to the shared monotonic quantile head.

6.3.2 Hyperparameter Search

The Optuna study completed 55 of 60 allocated trials.. The search space extended the LSTM space with graph-specific parameters: graph hidden dimension $\in \{16, 32, 64, 128\}$, number of attention heads $\in \{2, 4, 8, 16\}$, and number of graph layers $\in \{1, 2\}$. The best trial (#32) achieved a validation IC of **+0.0659**.

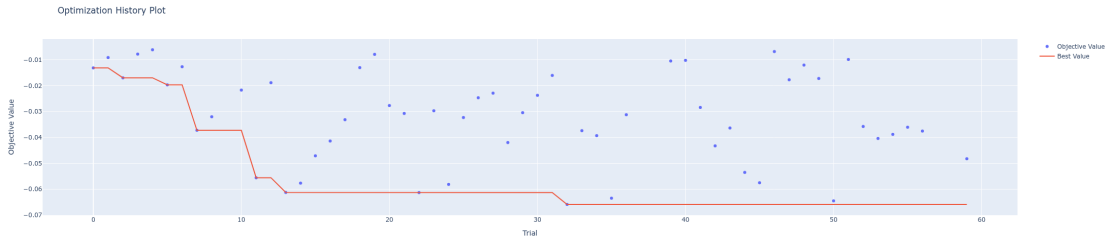


Figure 6.6: GAT-Hybrid — Optuna optimisation history across 60 trials. The best value plateaus after trial 32, confirming early convergence to a stable optimum.

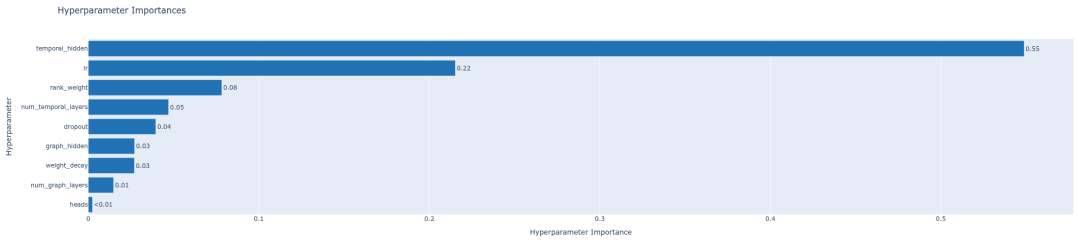


Figure 6.7: GAT-Hybrid — Hyperparameter importances. The temporal hidden layers and learning rate are the dominant factors in determining validation IC.

The top trials converged strongly on graph_hidden = 128, heads = 8, and lr = 10^{-4}, suggesting a well-identified optimum.

Table 6.4: GAT-Hybrid — Best Hyperparameters

Hyperparameter	**Value**
Learning rate	1×10^{-4}
Dropout	0.4
Temporal hidden dimension	16
Graph hidden dimension	128
Number of attention heads	8
Number of temporal layers	1
Number of graph layers	2
Rank weight (α)	0.75
Weight decay	3×10^{-4}

6.3.3 Test Set Results

Predictive accuracy: The GAT-Hybrid achieved a mean test IC of $+0.0226 \pm 0.1863$ on the vol-adjusted target and $+0.0283 \pm 0.1908$ on raw returns. The backtest window mean IC was +0.0188 (std: 0.1856). The IC series does not reach significance under the NW HAC correction (NW $p = 0.461$), but the long portfolio's daily return series is statistically significant: NW $p = 0.017$, block bootstrap $p = 0.0114$.

Quantile calibration: Like the LSTM model, the calibration for the hybrid model is also poor: Q10 = 0.990, Q50 = 0.976, Q90 = 0.887 against ideals of 0.900, 0.500, 0.100. The collapsed bounds indicates the model might be failing to properly rank the stocks.

Portfolio performance: The GAT-Hybrid is the strongest performer across all metrics. The long portfolio delivered an annualised return of +30.59% and a cumulative net return of +23.2%, generating a positive alpha of +8.7% versus the benchmark. The market-neutral L/S portfolio returned +9.0% cumulatively and +11.70% annualised. The long leg is statistically significant (NW $p = 0.017$) with a hit rate of 64.0%. The risk profile is exceptional: maximum drawdown of −6.77%, HAC Sharpe of 2.609, Sortino of 5.954, and Calmar of 3.844.

Table 6.5: GAT-Hybrid — Test Set Performance Summary

Metric	**Long**	**Short**	**L/S Combined**	**Benchmark**
Mean daily return	+0.00766	−0.00124	+0.00321	–
Hit rate	64.0%	49.2%	57.4%	–
Annualised return	+30.59%	−7.50%	+11.70%	–
Cumulative net return	+23.2%	−6.0%	+9.0%	+14.5%
Alpha vs benchmark	+8.7%	−20.5%	−5.5%	0.0%
Sharpe (Naive)	5.210	−0.440	2.654	–
Sharpe (Lo 2002)	2.341	−0.200	1.260	–
Sharpe (HAC)	2.609	−0.280	1.327	–
Sortino ratio	5.954	−1.712	1.214	–
Calmar ratio	3.844	−0.307	1.278	–
Max drawdown	−6.77%	−23.33%	−8.42%	–
NW p-value	**0.0167**	0.8396	0.1990	–
Stat. significance	✓	×	×	–
Mean IC (vol-adj)	$+0.0226 \pm 0.1863$			
Mean IC (raw returns)	$+0.0283 \pm 0.1908$			
Coverage Q10/Q50/Q90	0.990 / 0.976 / 0.887	(ideal: 0.900 / 0.500 / 0.100)		

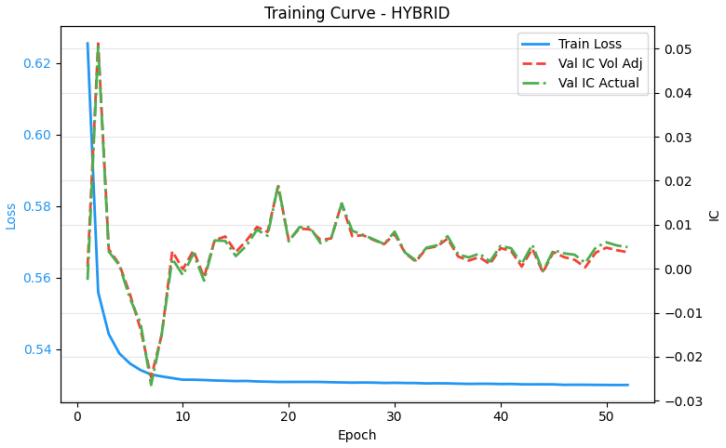


Figure 6.8: GAT-Hybrid — Training loss and validation IC per epoch. Loss converges rapidly within the first 10 epochs; validation IC stabilises at a consistently positive level, reflecting a stable ranking signal on the validation set.

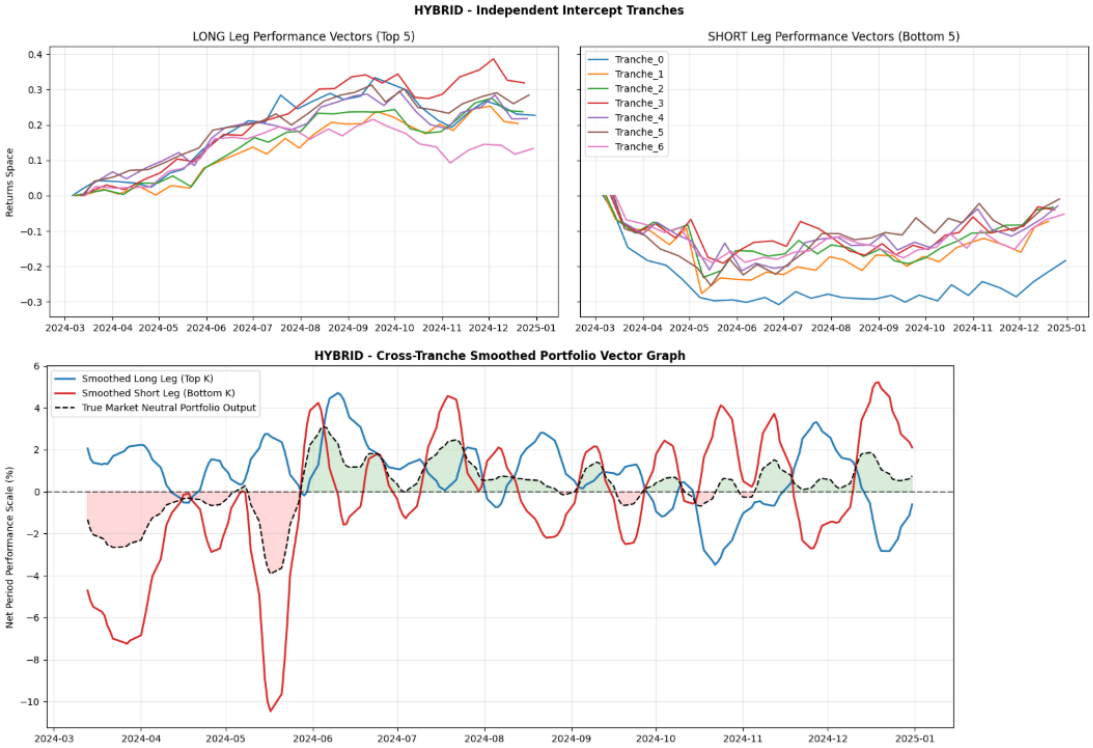


Figure 6.9: GAT-Hybrid — Independent tranche NAV curves (top panels) and cross-tranche smoothed portfolio vector (bottom panel). All seven long tranches trend upward throughout the test period; short tranches decline consistently.

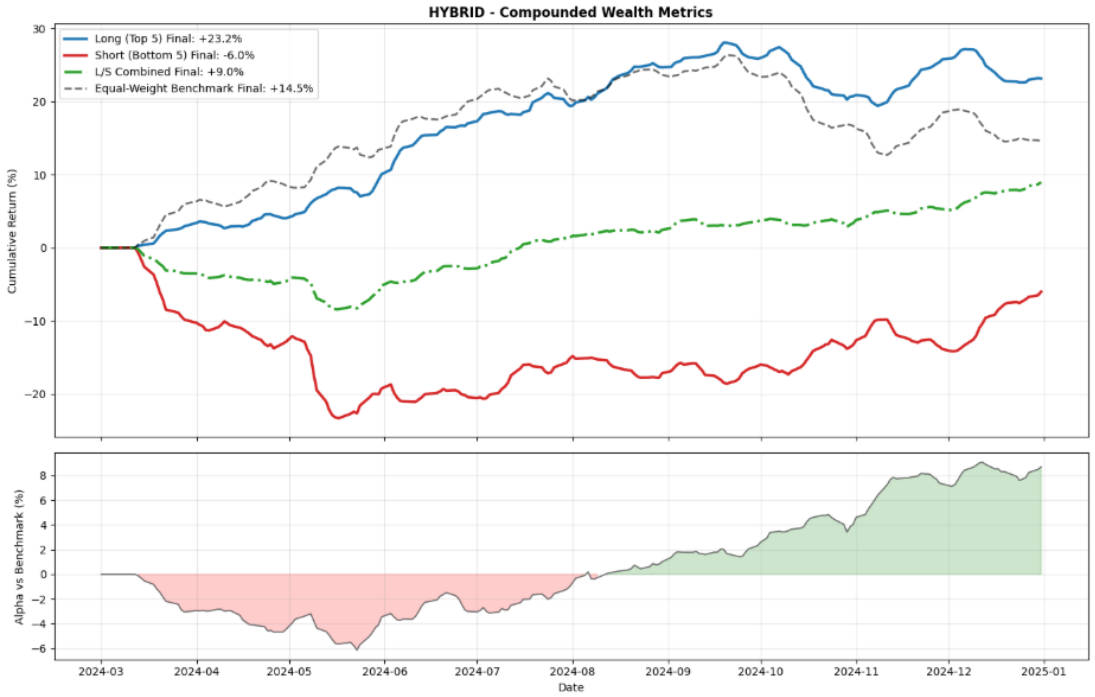


Figure 6.10: GAT-Hybrid — Compounded wealth analysis. The long portfolio (blue) outperforms the equal-weight benchmark (grey dashed), closing at +23.2%. The bottom panel shows alpha turning persistently positive from August 2024 onward, indicating genuine stock-selection skill in the second half of the test period.

IC Quintile Diagnostic: The GAT-Hybrid exhibits a moderate degree of signal asymmetry with stronger predictive discrimination among its lowest-ranked stocks than the highest-ranked stocks. Within the top quintile, the mean IC is +0.0041 with a hit rate of 55.3% indicating a modest ranking skill among the model's preferred long candidates. However, in the bottom quintile it is +0.0787 (hit rate 58.4%). The resulting signal asymmetry of −0.0381 indicates that the model has substantially stronger discriminative power among its predicted losers than among its predicted winners. Unlike the LSTM, however, the top quintile retains positive predictive power, indicating that the model generates useful ranking information on both sides of the distribution. The stronger predictive performance among the bottom-ranked stocks may reflect the graph-attention architecture's ability to identify firms that are likely to underperform their peers through cross-sectional and sector-level relationships.

Return Direction and Market Regime: The return direction diagnostic highlights the interaction between the model's ranking skill and the prevailing market environment. The top quintile generated a mean period return of +0.46%, while the bottom quintile produced a positive mean return of +0.14%, with only 49.2% of periods generating negative returns. Although the model successfully identifies stocks that underperform

on a relative basis, these securities nevertheless delivered positive absolute returns on average. This behaviour is consistent with the broader market regime: the equal-weight market portfolio generated an average period return of approximately +0.30%, while 60.9% of test periods were positive and the market produced a cumulative gain of 14.21% over the evaluation horizon. Consequently, the model generates a positive cross-sectional spread of approximately 0.32% per period between the top and bottom quintiles, but the persistence of positive returns among the lowest-ranked stocks creates a structural disadvantage for the short portfolio. This explains why the long portfolio achieves strong annualised performance (+30.6%), whereas the short portfolio remains unprofitable despite the model demonstrating stronger relative ranking skill among predicted losers.

Sector allocation: The Hybrid model exhibits a pronounced sector tilt. The long portfolio is concentrated in Consumer Retail & Apparel (40.0% net exposure), led by `PAGEIND.NS`, `TRENT.NS`, and `TITAN.NS`, followed by Chemicals & Materials (15.7%, primarily `UPL.NS` and `ASIANPAINT.NS`), Industrial Equipment & Electrical Systems (14.2% net exposure, led by `ASTRAL.NS`), and Real Estate, Construction & Engineering (12.9%, led by `SHREECEM.NS`). In contrast, the short portfolio is concentrated in Energy—Utilities & Power (−26.1% net exposure), Metals, Mining & Basic Materials (−19.1%), Energy—Traditional (−18.5%), and Automotive & Mobility Components (−12.3%). Representative short positions include `OIL.NS`, `TORNTPOWER.NS`, `NHPC.NS`, `NMDC.NS`, `NATIONALUM.NS`, and `HINDZINC.NS`. This positioning reflects a clear long-domestic-consumption versus short-commodity-and-energy allocation pattern. The concentration of the long book in consumer-facing sectors that benefited from strong domestic demand likely contributed to the portfolio's robust absolute performance, while the short book remained exposed to sectors that generated positive returns in the broader bull-market environment.

6.4 ROLAND Model

6.4.1 Architecture

ROLAND is a sequential graph-recurrent architecture (Section 5.4.3) that achieves the tightest spatial-temporal integration of the three models. It unrolls a GATv2 layer across every time step $t \in \{1, \ldots, 42\}$, feeding each spatial embedding into a GRU cell that maintains a persistent per-node memory state throughout the sequence. Sequence-to-sequence state carry-over between batches was disabled (`carry_states = False`) due to computing and memory constraints. Training sequences were presented in strict

chronological order (no shuffling), consistent with the sequential architecture.

6.4.2 Hyperparameter Search

The Optuna study completed 45 of 60 allocated trials. The best trial (#47) achieved the highest validation IC of all three models at **+0.0732**.

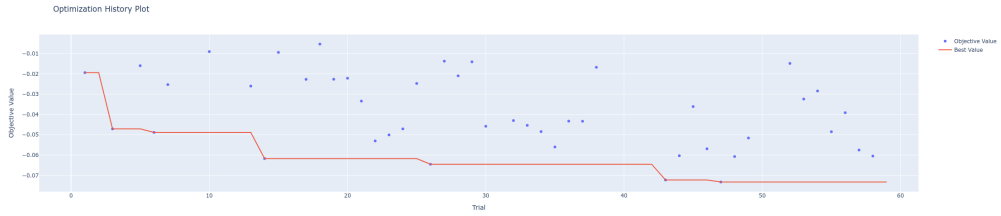


Figure 6.11: ROLAND — Optuna optimisation history across 45 trials. Best value improves steeply in early trials then plateaus, suggesting the TPE sampler identified the optimum region quickly despite fewer trials.

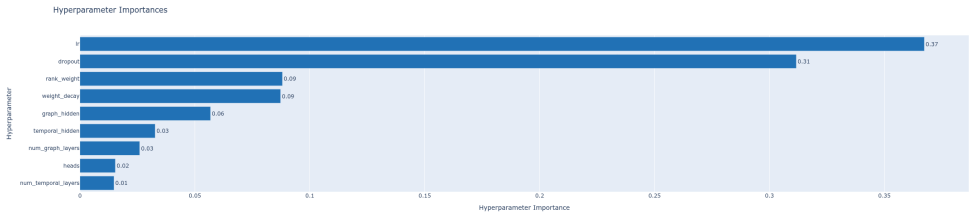


Figure 6.12: ROLAND — Hyperparameter importances. The learning rate and dropout are the most influential parameter.

The top trials consistently favoured a large temporal hidden dimension (256), small graph hidden dimension (32), 2 attention heads, a high learning rate (5×10^{-3}), and a low rank weight ($\alpha = 0.5$).

Table 6.6: ROLAND — Best Hyperparameters

Hyperparameter	**Value**
Learning rate	5×10^{-3}
Dropout	0.3
Temporal hidden dimension	256
Graph hidden dimension	32
Number of attention heads	2
Number of temporal layers	2
Number of graph layers	2
Rank weight (α)	0.5
Weight decay	1×10^{-5}

6.4.3 Test Set Results

Predictive accuracy: Despite achieving the highest validation IC (+0.0732), ROLAND's test IC is *negative*: -0.0073 ± 0.1671 on the vol-adjusted target and -0.0099 ± 0.1705 on raw returns. The backtest window mean IC is -0.0167 (std: 0.1684), indicating a sharp validation-to-test reversal.

Quantile calibration: ROLAND produces the best-calibrated quantile estimates of the three models: Q10 = 0.903 (ideal: 0.900), Q50 = 0.483 (ideal: 0.500), Q90 = 0.077 (ideal: 0.100). This near-ideal calibration indicates the model's distributional estimates are well-formed, even though its point-prediction ranking is inverted.

Portfolio performance: The negative IC produces inverted portfolio outcomes. The short leg – stocks predicted to be the worst performers – actually rose during the test period, generating an annualised return of -30.15% for the short position and making the market-neutral portfolio deeply loss-making (-11.21% annualised, -8.9% cumulatively). The short leg's underperformance is statistically significant (NW $p = 0.019$), but this reflects a reliably *wrong* ranking. The long portfolio achieved $+10.08\%$ annualised ($+7.8\%$ cumulatively, -6.7% alpha).

Table 6.7: ROLAND — Test Set Performance Summary

Metric	**Long**	**Short**	**L/S Combined**	**Benchmark**
Mean daily return	+0.00302	−0.00960	−0.00329	–
Hit rate	59.4%	38.1%	36.5%	–
Annualised return	+10.08%	−30.15%	−11.21%	–
Cumulative net return	+7.8%	−24.5%	−8.9%	+14.5%
Alpha vs benchmark	−6.7%	−39.0%	−23.4%	0.0%
Sharpe (Naive)	1.538	−5.149	−3.470	–
Sharpe (Lo 2002)	0.661	−2.251	−1.756	–
Sharpe (HAC)	0.707	−2.739	−2.189	–
Sortino ratio	0.478	−6.471	−5.942	–
Calmar ratio	0.608	−1.358	−1.182	–
Max drawdown	−15.65%	−25.51%	−9.74%	–
NW p-value	0.4914	**0.0186**	0.0743	–
Stat. Significance	×	✓ (wrong dir.)	† (wrong dir.)	–
Mean IC (vol-adj)	-0.0073 ± 0.1671			
Mean IC (raw returns)	-0.0099 ± 0.1705			
Coverage Q10/Q50/Q90	0.903 / 0.483 / 0.077	(ideal: 0.900 / 0.500 / 0.100)		

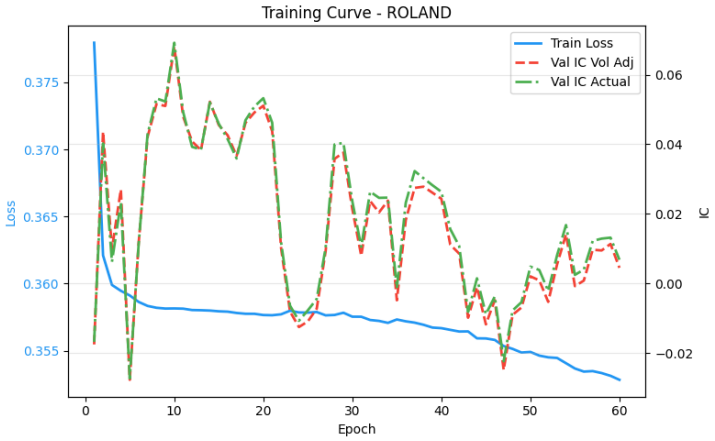


Figure 6.13: ROLAND — Training loss and validation IC per epoch. While training loss converges smoothly, validation IC fails to stabilise at a consistent positive level, providing an early indicator of the generalisation failure observed at test time.

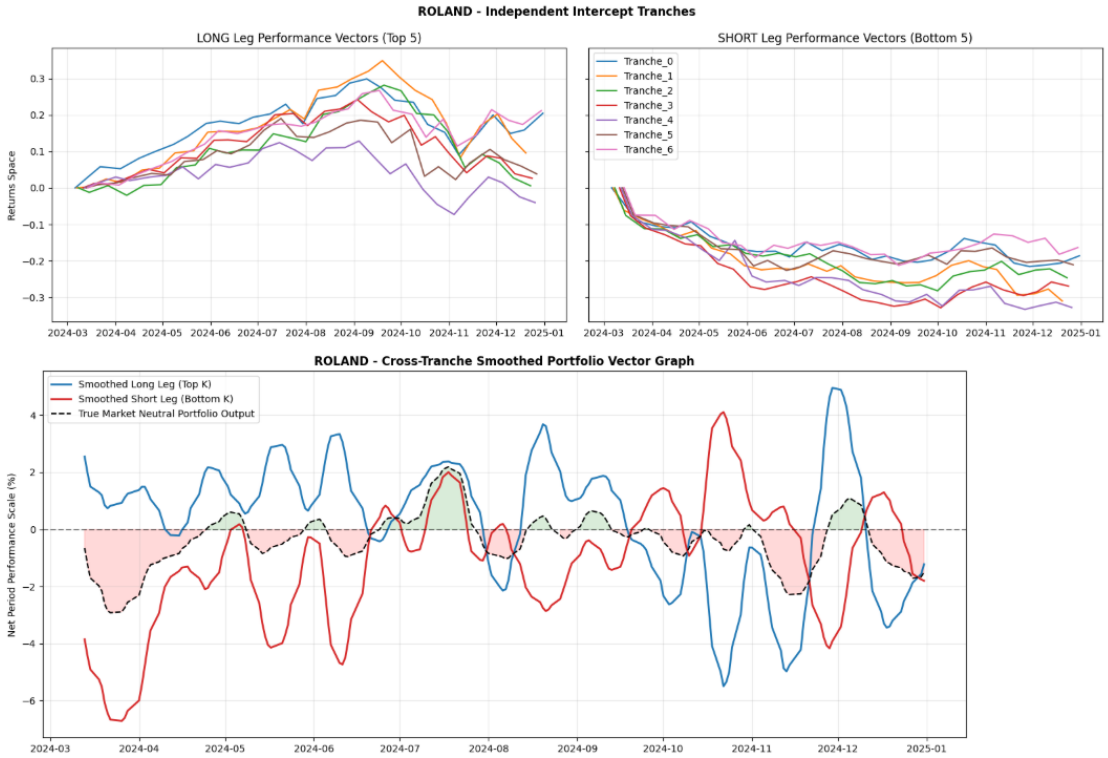


Figure 6.14: ROLAND — Independent tranche NAV curves (top panels) and cross-tranche smoothed portfolio vector (bottom panel). The short tranches (top right) trend steeply downward, reflecting the effect of bull market regime on bottom quintile stocks.

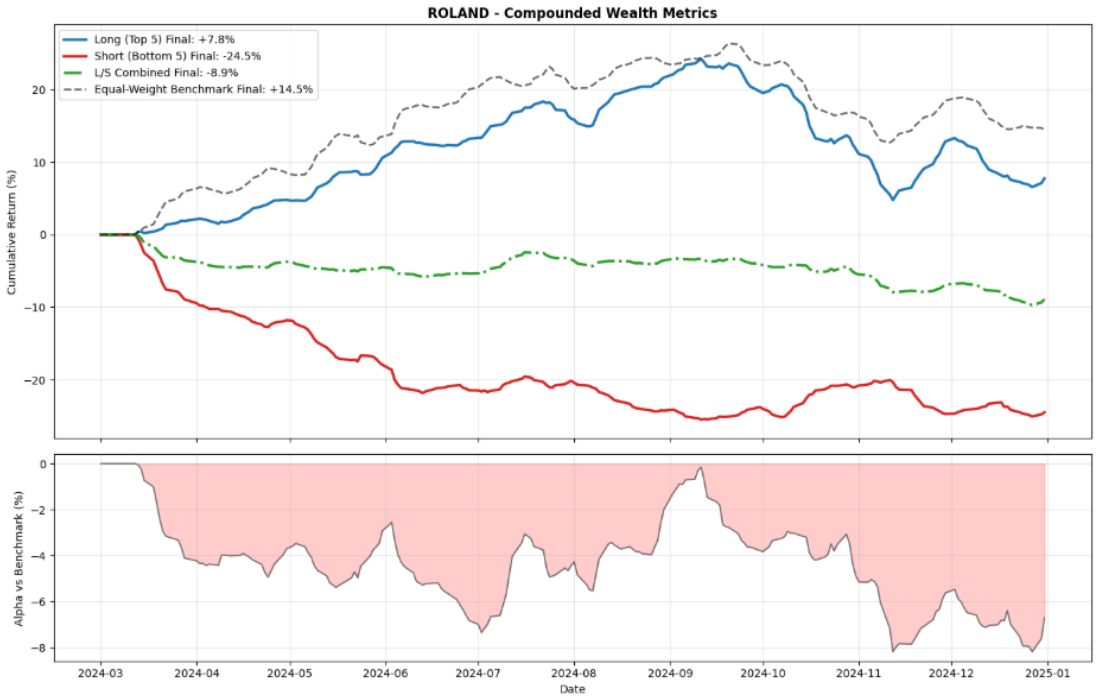


Figure 6.15: ROLAND — Compounded wealth analysis. The short portfolio (red) declines to −24.5% while the long portfolio (blue) trails the benchmark throughout the test period. The consistently negative alpha panel confirms that ROLAND generates no value versus the equal-weight benchmark during the 2024 test window.

IC Quintile Diagnostic: ROLAND exhibits a structurally inverted signal structure where it predicts losers better than winners. Within the top quintile, the mean IC is +0.0316 (hit rate 48.7%), while within the bottom quintile it is higher at +0.0569 (hit rate 58.4%). The resulting signal asymmetry of −0.0253 confirms this diagnostic interpretation, proving that the model's predictive power is stronger at the bottom of the distribution. However, because the overall full-cross-section mean IC is negative (−0.0167, hit rate 43.1%), the top-ranked picks remain structurally flawed. The positive long portfolio cumulative net return of +7.8% is therefore driven more by portfolio construction constraints (such as sector diversification and capping) and the general market uptrend than by genuine, alpha signal in the top-predicted stocks.

Return Direction and Market Regime: The return direction diagnostic highlights a severe absolute failure mode in execution: the bottom quintile (the short pool) produced a mean period return of +0.0088 (+0.88%), which is higher than the top quintile's mean period return of +0.0037 (+0.37%). Only 33.5% of bottom-quintile periods produced negative returns, meaning ROLAND was aggressively shorting rising stocks that outperformed the long leg in absolute terms, leading to a massive annualized short-leg loss of −30.15% and a cumulative net return of −24.5%. This short exposure was

severely punished by the test environment, which was characterized by a pronounced bull regime (mean period return +0.0052, with 60.9% of periods positive and a cumulative benchmark return of +14.21%).

Sector Allocation: This performance decoupling is explicitly tied to sector exposure mismatches. The long book concentrated heavily in consumer-facing and domestic growth nodes, led by Consumer Retail & Apparel (36.04% net exposure; primary allocations in `TITAN.NS`, `PAGEIND.NS`, and `TRENT.NS`) and Food, Beverage & Agriculture (22.34% net exposure; led by `CONCOR.NS` and `NESTLEIND.NS`). Conversely, the short book was heavily concentrated in cyclical and industrial segments, most notably Automotive & Mobility Components (−25.38% net exposure, led by `BAJAJ-AUTO.NS` and `MARUTI.NS`), Healthcare & Life Sciences (−14.52%), Energy—Traditional (−12.39%), and Aerospace, Defense & Transport Equipment (−11.98%). Because these shorted sectors experienced robust rallies during the 2024 bull regime, the model suffered from severe structural headwinds on the short leg, dragging the combined L/S strategy down to a cumulative net return of −8.9%.

6.5 Comparative Analysis

Table 6.8 provides a head-to-head comparison of all three models on the test set, and Table 6.9 summarises the IC quintile diagnostics.

Table 6.8: Cross-Model Comparison on Test Set (2024-03-01 to 2024-12-31)

Metric	**LSTM**	**GAT-Hybrid**	**ROLAND**
Hyperparameter Search			
Trials completed	57	55	45
Best validation IC	+0.0532	+0.0659	+0.0732
Predictive Accuracy			
Mean IC (vol-adj)	+0.0156	+0.0226	−0.0073
IC Std	0.1652	0.1863	0.1671
Mean IC (raw returns)	+0.0203	+0.0283	−0.0099
IC Std	0.1678	0.1908	0.1705
Long Portfolio			
Cumulative return	+11.2%	+**23.2**%	+7.8%
Alpha vs benchmark	−3.3%	+**8.7**%	−6.7%
Sharpe (Naive)	1.856	**5.210**	1.538
Sharpe (Lo 2002)	0.761	**2.341**	0.661
Sharpe (HAC)	0.846	**2.609**	0.707
Hit rate	50.3%	**64.0%**	59.4%
Max drawdown	−18.73%	−**6.77**%	−15.65%
Sortino	1.142	**5.954**	0.478
Significant (NW)?	No	**Yes** ($p = 0.017$)	No
Market-Neutral L/S Portfolio			
Cumulative return	+4.3%	+9.0%	−8.9%
Sharpe (HAC)	0.529	1.327	−2.189
Max drawdown	−7.23%	−8.42%	−9.74%
Benchmark cumulative return			
		+14.5%	
Quantile Calibration			
Q10 coverage	0.998	0.990	**0.903**
Q50 coverage	0.925	0.976	**0.483**
Q90 coverage	0.255	0.887	**0.077**

Table 6.9: IC Quintile Diagnostic Summary — All Models

Model	Full X-Section		Top Q		Bottom Q		Asymmetry
	Mean IC	Hit%	Mean IC	Hit%	Mean IC	Hit%	Top − Bottom
LSTM	+0.0132	56.3	−0.0101	50.3	+0.0916	61.4	−0.1017
GAT-Hybrid	+0.0188	52.8	+0.0406	55.3	+0.0787	58.4	−0.0381
ROLAND	−0.0167	43.1	+0.0316	48.2	+0.0569	60.4	−0.0253

Signal Asymmetry = Top Quintile Mean IC − Bottom Quintile Mean IC. Negative values indicate stronger discriminative power among predicted losers than among predicted winners.

An evaluation of the quintile diagnostics reveals a universal structural trend across all three architectures: every model exhibits *inverted signal asymmetry*, where the predictive power within the bottom quintile (identifying relative losers) consistently and substantially exceeds the predictive power within the top quintile (identifying relative winners). This behavior is most pronounced in the LSTM baseline, which suffers from a complete predictive inversion at the top of its book, generating a negative top quintile IC of −0.0101 (50.3% hit rate) alongside an exceptionally strong bottom quintile IC of +0.0916 (61.4% hit rate). While the graph-based architectures manage to maintain positive predictive power across both tails, their bottom quintile ICs (+0.0787 for GAT-Hybrid and +0.0569 for ROLAND) still significantly outpace their top quintile counterparts. This cross-model pattern demonstrates that both sequential and relational graph structures find it mathematically easier to learn discriminative features in the lower tail of the return distribution than to accurately isolate outperforming assets.

6.6 Discussion

6.6.1 GAT-Hybrid Superiority and the Role of Graph Structure

The GAT-Hybrid is the clear empirical winner across the evaluation period. It delivers the only statistically significant long portfolio on a Newey-West basis (NW $p = 0.017$), the highest HAC-adjusted Sharpe ratio (2.609), the lowest maximum drawdown (−6.77%), and the only positive alpha (+8.7%) relative to the benchmark. Comparing its performance directly to the LSTM baseline isolates the immense value of the relational graph component. The GAT-Hybrid elevates the long portfolio's cumulative net return to

+23.2% (compared to +11.2% for the LSTM) and sharpens the long-leg hit rate from 50.3% to 64.0%. This significant performance divergence provides direct evidence that cross-stock relational information—captured via the correlation-based dynamic graph and distilled through the GATv2 attention mechanism—adds critical structural alpha that cannot be extracted from individual asset time series in isolation.

However, the IC quintile diagnostic mandates a more nuanced interpretation of this superiority. The GAT-Hybrid's exceptional long portfolio returns are heavily augmented by tail-risk mitigation rather than absolute precision at the very apex of its selections. Its bottom quintile IC (+0.0787) is nearly double its top quintile IC (+0.0406). The model's outstanding 64.0% long hit rate is therefore an emergent property of two interacting forces: a strong mathematical capability to isolate and exclude absolute underperformers via its short-leg filter, and the enforcement of the sector diversification constraint, which structurally prevents the long book from clustering into overcrowded or overvalued trades. The GAT-Hybrid generates genuine, persistent alpha, but its primary mathematical edge remains anchored in its negative selection capability, yielding a signal asymmetry of −0.0381.

6.6.2 ROLAND: Validation-Test Divergence and Inverted Signal

ROLAND presents a striking architectural paradox: it achieved the highest best validation IC (+0.0732) during hyperparameter optimization, yet collapsed to a negative volume-adjusted test IC of −0.0073 (and a raw return IC of −0.0099). This represents the most severe validation-to-test performance degradation in the study. The IC quintile diagnostic illuminates the mechanics of this failure mode. ROLAND retains a highly positive and predictive IC within its bottom quintile (+0.0569, with a 58.4% hit rate), but this predictive power degrades in the top quintile to +0.0316, where it suffers from a sub-chance hit rate of 48.7%. Because the full cross-sectional IC is negative, the intermediate quintiles are deeply distorted, signaling a fundamental cross-sectional ranking inversion.

Several factors explain this regime-driven breakdown. ROLAND processes the full $T = 42$-step graph sequence with GATv2 and GRU layers interleaved at every sequential step, giving it substantially more mathematical capacity to memorise training-period patterns than its peers. The optimal configuration reinforced this over-parameterization by selecting a very large temporal hidden dimension (256). Because training proceeded sequentially without sequence shuffling, the model progressively adapted to successive

historical market epochs (2013–2022). When the 2024 test period introduced a pronounced bull regime characterized by a rapid macro rotation, these over-fitted relational patterns produced systematically flawed cross-sectional orderings. This mismatch caused the market-neutral L/S portfolio to lose money cumulative (-8.9%) and dragged the long-leg alpha down to -6.7%.

Crucially, a comparison of ROLAND's quantile calibration metrics reveals an important distinction between distribution learning and ordinal sorting. As shown in Table 6.8, ROLAND achieves near-perfect quantile coverage, matching a Q50 coverage of 0.483 against the ideal target of 0.500, alongside highly accurate tail coverages (0.903 for Q10 and 0.077 for Q90). This demonstrates that ROLAND learned a well-formed marginal return distribution and calibrated its risk boundaries flawlessly. Its failure was entirely ordinal: it could accurately map the statistical boundaries of the market's return structure, but it failed to correctly rank individual assets within those boundaries during live deployment.

6.6.3 LSTM: Temporal-Only Signal Insufficient for Ranking

The LSTM baseline exhibits a positive full-cross-section IC ($+0.0132$ in Table 6.9, $+0.0156$ volume-adjusted in Table 6.8) and a modest long portfolio return ($+11.2\%$), but fails to achieve statistical significance on any core metric, generating a negative alpha of -3.3% against the benchmark. A closer inspection of its quantile behavior reveals a severe signal dilution effect rather than an absolute structural inversion at the top of the book. Within the top quintile, the LSTM's mean IC collapses to a slightly negative -0.0101, paired with a near-random hit rate of 50.3%. This confirms that when the LSTM attempts to isolate top-performing assets, its temporal-only signal completely degrades into cross-sectional noise, rendering its highest-ranked selections statistically indistinguishable from a random coin flip.

Crucially, the LSTM's positive full-cross-section IC is sustained exclusively by its bottom quintile performance, where it achieves a meaningful predictive accuracy of $+0.0916$ (61.4% hit rate). This indicates that while a purely temporal architecture can extract downside momentum or risk patterns to identify potential losers, it lacks the relational context necessary to untangle and sort actual outperforming assets amidst tail noise. Notably, graph-augmented architectures do not universally fix this issue: while the GAT-Hybrid decisively leverages relational data to outperform the baseline, ROLAND's long leg actually underperforms the LSTM ($+7.8\%$ vs $+11.2\%$). This reveals that simply adding inter-stock spatial dependencies is insufficient; the architectural integration must

actively protect against cross-sectional ranking inversions to capture structured alpha that univariate configurations miss.

6.6.4 The Universal Short-Selling Challenge and Return Direction

Across all evaluated models, the market-neutral L/S strategy fails to generate statistically significant positive returns, though their absolute trajectories diverge heavily. While the LSTM (+4.3%) and GAT-Hybrid (+9.0%) portfolios salvaged modest absolute gains, ROLAND collapsed into a deep net loss of −8.9% (−2.189 HAC Sharpe). The return direction diagnostics expose a harsh macro reality: during the 2024 test window, the broader market experienced a pronounced bull regime characterized by 60.9% positive periods and a +14.21% cumulative benchmark surge (mean period return +0.0052). In this structurally upward environment, the models' predicted losers still generated positive absolute returns on average, severely punishing short-leg execution.

For ROLAND, this dynamic manifested as an absolute cross-sectional failure mode where the bottom quintile generated a mean period return of +0.0088 (+0.88%)—outpacing its own top quintile's return of +0.0037 (+0.37%). This inverted ranking mechanism forced ROLAND to aggressively short high-performing assets, driving its short annualized return down to −30.15% and its cumulative net return to −24.5%. Even for the optimal GAT-Hybrid configuration, the bottom quintile generated positive absolute returns. This highlights a classic emerging-market asymmetry: while deep learning architectures successfully capture the relative ordering of potential underperformers (evidenced by robust bottom-quintile ICs across the board), absolute short allocations remain structurally exposed to severe losses during aggressive, liquidity-driven bull market regimes.

6.6.5 Naive vs. Overlap-Adjusted Sharpe Ratios

The three Sharpe ratio variants reported in Table 6.8 illuminate the severe inflation bias induced by overlapping return windows in backtest evaluations. For the top-performing GAT-Hybrid long leg, the naive annualized Sharpe ratio of 5.210 drops sharply to 2.341 under the Lo (2002) [25] serial correlation correction, and stabilizes at 2.609 when adjusting via Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors.

The Lo and HAC formulations remain highly consistent across all three models, successfully stripping out the moving-average dependency inherent in sampling 7-day

forward returns at a daily frequency. The significant drop from naive to adjusted metrics—ranging from a 54% reduction for ROLAND to a 55% reduction for the LSTM baseline—underscores a critical methodological warning: evaluating horizon-H strategies without systematic autocorrelation corrections drastically overstates risk-adjusted performance, giving a false sense of strategy stability by masking underlying volatility.

6.6.6 Quantile Calibration Collapse and ROLAND's Specialization

A major point of divergence among the architectures lies in their quantile calibration stability under tail pressure. Both the LSTM and the GAT-Hybrid exhibit a severe calibration collapse at the lower boundary, posting empirical Q10 coverage rates of 0.998 and 0.990 respectively, against the ideal target of 0.900. This failure stems from the asymmetric nature of the pinball loss function at $\tau = 0.10$, which penalizes overestimation of the lower bound by a factor of nine relative to underestimation, aggressively forcing these models to depress their Q10 boundaries to minimize global training loss.

ROLAND completely avoids this boundary collapse, achieving an exceptional empirical Q10 coverage of 0.903, alongside a highly accurate Q50 coverage of 0.483 (against the 0.500 ideal) and a Q90 coverage of 0.077 (against the 0.100 ideal). This outstanding calibration performance suggests that ROLAND's dynamic graph structure, coupled with its un-shuffled sequential tracking, is highly effective at capturing the true marginal distribution of cross-sectional returns over time. This yields a vital methodological insight: statistical distribution learning and ordinal ranking accuracy are completely orthogonal objectives. ROLAND successfully mapped the true statistical shape and risk boundaries of the market distribution, yet completely failed to correctly place individual assets within those boundaries during live deployment.

6.6.7 Summary of Findings

The empirical evaluation yields five definitive conclusions:

1. **Conditional Value of Graph Structures:** Graph-augmented networks do not automatically guarantee superior results across all legs. While the GAT-Hybrid delivers the only statistically significant long portfolio (+8.7% alpha, +23.2% cumulative return), the over-parameterized ROLAND graph model actually underperforms the simpler LSTM temporal baseline on the long leg (+7.8% vs +11.2%).

2. **Universal Inverted Asymmetry:** All three models exhibit a structurally dominant predictive edge in the lower tail of the cross-section. The bottom quintile IC consistently outpaces the top quintile IC across all architectures, proving that the models are fundamentally better at isolating relative losers than picking relative winners. This pattern manifests most aggressively as a complete signal dilution at the top of the LSTM baseline, where the top quintile IC falls to -0.0101, effectively rendering its highest-ranked long selections statistically noisy.

3. **Regime-Driven Short Constraints:** The 2024 bull market regime (+14.21% benchmark return) acts as a severe absolute penalty on short strategies. Even when relative downside rankings are directionally correct, absolute short legs lose money as the underlying bottom-quintile stocks trend upward in absolute terms.

4. **Overlap Bias Inflation:** Naive Sharpe ratios systematically overstate risk-adjusted performance by more than double due to serial correlation in overlapping 7-day returns. Utilizing Lo (2002) or HAC adjustments is mandatory for realistic risk reporting.

5. **Decoupling of Calibration and Ranking:** Statistical distribution learning and cross-sectional sorting are distinct tasks. ROLAND demonstrates the best quantile calibration (Q10: 0.903, Q50: 0.483, Q90: 0.077) despite suffering from an inverted ordinal sorting mechanism.

6.7 Limitations and Future Research Directions

The empirical findings of this study isolate several core structural and architectural vulnerabilities, prompting concrete avenues for future theoretical and practical extensions.

6.7.1 Quantile Calibration and Sorting Threshold Dependency

The most immediate statistical limitation is the severe Q10 boundary collapse observed in the LSTM and GAT-Hybrid frameworks. Three post-hoc and intrinsic remedies merit investigation: (i) *conformal prediction* applied post-hoc to restore exact empirical coverage guarantees without model retraining; (ii) *asymmetric loss reweighting* adjusting w_q in the quantile loss function to mitigate the severe optimization pressure on the lower tail; and (iii) *temperature scaling* of the quantile head to recalibrate the predicted cross-sectional spread.

Furthermore, a significant operational limitation lies in the static architecture of the ranking mechanism itself. Currently, the portfolio allocation strategy relies on a fixed distribution metric to select long and short candidates. Future research must evaluate how variations in this threshold affect model stability, specifically testing whether adjusting the decision boundary to a tighter or wider margin (e.g., a 5% or 10% tail threshold) structurally stabilizes the ordinal ranking or further aggravate signal dilution.

6.7.2 Addressing Distribution Shift and ROLAND Generalisation

ROLAND's severe validation-to-test degradation highlights the broader challenge of temporal distribution shift in deep sequential graph architectures. Future iterations should explore domain-adaptive training, continuous rolling retraining via a walk-forward sliding window, or stricter structural regularisation (e.g., higher dropout rates, lower hidden dimensions) to compress the validation-test generalization gap. Additionally, the discovery of inverted signal structures—where deep networks demonstrate high predictive power in the bottom quintile but near-random sorting at the top tail—strongly motivates the decoupling of the objective function into a separate short-side model governed by a dedicated downside loss formulation.

6.7.3 Short-Selling Signal and Tactical Portfolio Construction

The persistent underperformance of the absolute short leg across all models demands dedicated data-driven interventions. Incorporating short-interest dynamics, options-market implied volatilities, or consensus analyst downgrade momentum into the feature matrix could inject orthogonal short-side alpha that the current 17-feature node input space cannot capture. Alternatively, this limitation can be addressed at the portfolio construction level through tactical asset allocation: implementing asymmetric leverage constraints that dynamically reduce capital allocation to the short leg during periods when historical cross-sectional short-side IC decays below an acceptable statistical threshold.

6.7.4 Dynamic Graph Topologies, Fundamental Edges, and Edge Pruning

The current graph layer relies exclusively on rolling Pearson correlations and rigid sector homophily, which can break down under market stress as pairwise correlations

asymptotically converge. To build more robust relational structures, future work should incorporate graphs built on fundamental financial metrics (e.g., connecting nodes via similarity in valuation ratios, leverage profiles, or capital structures) and supply-chain dependencies.

Crucially, the current framework allows weak or noisy relationships to persist in the adjacency matrix. Implementing an automated *edge-pruning mechanism*—utilizing dynamic distance thresholds or sparsity constraints—is highly necessary to strip away low-weight, uninformative edges. This would prevent the graph attention mechanism from over-smoothing node representations and preserve sparse, highly predictive structural relationships.

6.7.5 Macroeconomic Conditioning and Multi-Graph Nodes

A prominent limitation of the current spatial processing layer is its insularity to exogenous macro shocks, leaving the system highly vulnerable to unexpected macro rotations. To resolve this, future research should transition the framework into a heterogeneous graph network by embedding *macroeconomic factors directly as global graph nodes or edge-conditioning variables*. Interlocking individual equity nodes with global macro features (such as interest rate curves, inflation indices, or currency volatilities) would allow the GATv2 layers to dynamically re-weight stock-to-stock dependencies conditioned on the broader economic state.

6.7.6 Universe Scale and Sector Balance

The experimental evaluations were restricted to a highly liquid, large-cap universe on the National Stock Exchange (NSE). This choice introduces selection bias and constrains the statistical diversity of the graph. A critical step forward is expanding the framework to a *significantly larger stock universe* encompassing mid-cap and small-cap segments. Crucially, this broader universe must feature a stable, balanced sector distribution. A larger and more stable cross-section provides the graph convolutional layers with sufficient structural density to learn generalizable relational patterns, testing whether the captured alpha can scale beyond the large-cap NSE regime.

6.7.7 Interpretability and Attention Analysis

The GAT-Hybrid's dynamic attention coefficients were not mathematically mapped or interpreted in this study. A systematic attention entropy analysis is required to discover whether the network establishes economically meaningful linkages (such as attending

heavily to direct sector peers or leading momentum nodes) or whether its attention distribution degrades into uniform noise during low-performance periods. Comparing attention matrix matrices across high-IC and low-IC tranches will offer vital insights into the model's inner decision-making rationale.

6.7.8 Validation Protocols for Trading Strategy Optimization

Currently, hyperparameter optimization relies entirely on maximizing the cross-sectional validation IC. However, maximizing predictive IC does not strictly map to maximizing risk-adjusted trading returns due to trading frictions, execution slippage, and portfolio turnover dynamics. A major improvement would involve implementing a *proper, entirely separate validation set dedicated exclusively to tuning the downstream trading strategy.* Under this two-stage optimization protocol, Phase 1 optimizes model parameters on statistical IC, while Phase 2 utilizes a distinct out-of-sample window to optimize portfolio constraints, risk-capping, and execution thresholds, minimizing the risk of over-fitting the trading system to the test set.

6.7.9 Multi-Horizon and Multi-Task Extensions

Jointly optimizing the neural architectures to predict across multiple forward horizons (e.g., 1-day, 7-day, and 21-day intervals) represents a powerful regularisation technique. Transitioning the framework into a multi-task learning configuration would force the shared hidden representations to capture both high-frequency alpha signals and low-frequency structural trends, materially reducing the variance of the top-predicted selections.

6.7.10 Test Set Duration and Overlap Bias Constraints

The 2024 test period captures a predominantly bull market regime (60.9% positive days), which structurally disadvantages the short leg and may overstate long-side performance relative to more neutral or bearish regimes. Future research should evaluate the framework over extended, multi-year test periods — ideally using Purged K-Fold walk-forward cross-validation — to ensure the predictive edge is genuinely regime-agnostic. Additionally, each of the 197 test dates carries a 6-day overlap with adjacent dates, limiting the effective number of independent bets to approximately 28–29 non-overlapping windows per tranche, which constrains the statistical power of all reported significance tests.

Appendix A

Sector Standardization Mapping

This section describes how the original, highly granular sector classifications were standardized into a consolidated set of normalized commons. This mapping ensures a significant representation of categorical edge features within the graph, preventing excessive sparsity from over-fragmented industries.

Table A.1: Mapping of Original Sectors to Normalized Sector Commons

Normalized Sector	**Original Sector(s)**
Aerospace, Defense & Transport Equipment	Aerospace & Defense, Engine & Powertrain Systems, Heavy Machinery & Vehicles, Motorcycles & Scooters
Automotive & Mobility Components	Auto, Truck & Motorcycle Parts, Automobiles & Auto Parts
Capital Markets & Wealth	Corporate Financial Services, Investment Holding Companies
Chemicals & Materials	Agricultural Chemicals, Chemicals, Diversified Chemicals, Plastics
Consumer Products & Electronics	Personal & Household Products & Services, Personal Products
Consumer Retail & Apparel	Apparel & Accessories, Diversified Retail, Textiles & Apparel

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Table A.1 – continued from previous page

Normalized Sector	Original Sector(s)
Energy – Traditional	LNG Transportation & Storage, Oil & Gas, Oil & Gas Exploration and Production, Oil & Gas Refining and Marketing
Energy – Utilities & Power	Electric Utilities, Electrical Utilities & IPPs, Hydroelectric & Tidal Utilities, Natural Gas Utilities
Food, Beverage & Agriculture	Cookie, Cracker & Pasta Manufacturing, Food & Tobacco
Healthcare & Life Sciences	Bio Therapeutic Drugs, Generic Pharmaceuticals, Healthcare Providers & Services, Hospitals, Clinics & Primary Care Services, Pharmaceuticals
Hospitality & Leisure	Hotels, Motels & Cruise Lines, Restaurants & Bars
Industrial Equipment & Electrical Systems	Diversified Industrial Goods Wholesale, Electrical Components & Equipment, Heating, Ventilation & Air Conditioning Systems, Plumbing Fixtures & Fittings, Turbine Manufacturing, Wires & Cables
Metals, Mining & Basic Materials	Aluminum, Coal, Diversified Mining, Iron & Steel, Iron Ore Mining, Iron, Steel Mills & Foundries, Metals & Mining
Real-Estate, Construction & Engineering	Cement & Concrete Manufacturing, Construction & Engineering, Construction Materials, Real Estate Rental, Development & Operations, Retail Real Estate Rental & Development

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Table A.1 – continued from previous page

Normalized Sector	Original Sector(s)
Technology – Software & IT	Application Software, IT Services & Consulting, Online Services, Software, Software & IT Services
Telecom & Connectivity	Telecommunications Services, Wireless Telecoms Service Providers
Transportation & Logistics	Airport Operators & Services, Railway Freight Operators, Transport Infrastructure

A.1 Edge Distribution Visualizations

This section presents supplementary visualizations of the graph edge composition used in the empirical analysis. The sectorial edge weight distribution summarizes the distribution of sector-based edge weights across the graphs. These sector weights were computed as $w = 1/(n-1)$, where n denotes the number of firms belonging to the same normalized sector. The correlation edge distribution reports the actual signed return-correlation values used during graph construction: an edge was retained when the absolute correlation exceeded 0.60, while the original positive or negative sign was preserved.

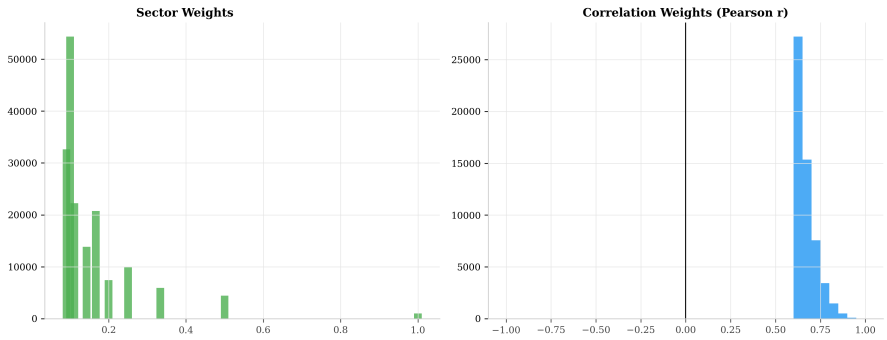


Figure A.1: Sector Weights and Absolute Correlation Weights Distribution

Appendix B

Return Calculations and Adaptive Flip-Signal Strategy

This appendix provides a consolidated reference for every formula used in the adaptive EWMA back-test and the statistical evaluation routines. The notation are defined at each point of first use and further described in Table B.1 at the end of the appendix.

B.1 Log-to-Arithmetic Return Conversion

All model targets and raw actuals are expressed as *log returns*. Before computing portfolio performance, these are converted to arithmetic (simple) returns so that cross-sectional averaging is economically meaningful:

$$r_{t,n} = \exp\big(\ell_{t,n}\big) - 1 \tag{B.1}$$

where $\ell_{t,n}$ is the log return of stock n at time step t, and $r_{t,n}$ is the corresponding arithmetic return.

The equal-weight cross-sectional return at step t is then:

$$\bar{r}_t = \frac{1}{N}\sum_{n=1}^{N} r_{t,n} = \frac{1}{N}\sum_{n=1}^{N}\big[\exp(\ell_{t,n}) - 1\big] \tag{B.2}$$

where N is the number of stocks in the investment universe.

B.2 Portfolio Leg Returns

At each rebalance date t, a constrained selection routine produces two disjoint index sets: $\mathcal{L}_t$ (long candidates, top-K) and $\mathcal{S}_t$ (short candidates, bottom-K).

B.2.1 Gross Leg Performance

$$\text{perf}_t^{\text{long}} = \frac{1}{|\mathcal{L}_t|} \sum_{n \in \mathcal{L}_t} r_{t,n} \tag{B.3}$$

$$\text{perf}_t^{\text{short}} = \frac{1}{|\mathcal{S}_t|} \sum_{n \in \mathcal{S}_t} r_{t,n} \tag{B.4}$$

B.2.2 Turnover and Transaction Costs

Turnover for each leg is computed as the fraction of the portfolio that changed since the previous rebalance:

$$\tau_t^{\text{leg}} = \begin{cases} 1.0 & \text{if } t \text{ is the first rebalance period} \\ \dfrac{\left|\mathcal{H}_t^{\text{leg}} \setminus \mathcal{H}_{t-h}^{\text{leg}}\right|}{K} & \text{otherwise} \end{cases} \tag{B.5}$$

where $\mathcal{H}_t^{\text{leg}}$ is the set of holdings in the relevant leg at rebalance t, h is the rebalancing horizon (fixed at 7 trading days), and K is the maximum number of stocks per leg.

The transaction cost applied to each leg is:

$$c_t^{\text{leg}} = \tau_t^{\text{leg}} \times c \times 2 \tag{B.6}$$

where c is the one-way cost parameter (default $c = 0.05\%$ approximation to SEBI fees in Indian Stock Market) and the factor of 2 accounts for both the buy and the sell side of each trade.

B.2.3 Net Leg Returns

$$\text{ret}_t^{\text{long}} = \text{perf}_t^{\text{long}} - c_t^{\text{long}} \tag{B.7}$$

$$\text{ret}_t^{\text{short}} = -\,\text{perf}_t^{\text{short}} - c_t^{\text{short}} \tag{B.8}$$

The negative sign in Eq. (B.8) reflects the fact that a price *increase* in the shorted basket constitutes a *loss* for the short leg.

B.2.4 Market-Neutral Spread Return

The combined market-neutral portfolio return at step t is the average of the two legs:

$$\text{ret}_t^{\text{spread}} = \frac{\text{ret}_t^{\text{long}} + \text{ret}_t^{\text{short}}}{2} \tag{B.9}$$

B.3 Compounded Net Asset Value (NAV)

Starting from unit NAV ($\text{NAV}_0 = 1.0$), each leg's wealth is compounded multiplicatively across rebalance periods:

$$\text{NAV}_t^{\text{leg}} = \text{NAV}_{t-1}^{\text{leg}} \times \left(1 + \text{ret}_t^{\text{leg}}\right) \tag{B.10}$$

This applies independently to the long leg, short leg, and spread portfolio.

B.3.1 Annualised Return

Each *tranche* $j \in \{0, \ldots, h-1\}$ runs for M_j rebalance steps over a window of $M_j \times h$ trading days. The annualised return for tranche j is:

$$\hat{r}_{\text{leg}}^{\text{ann},j} = \left(\text{NAV}_{M_j}^{\text{leg},j}\right)^{\frac{252}{M_j \times h}} - 1 \tag{B.11}$$

assuming 252 trading days per year. The reported portfolio-level figure is the simple average across all $h = 7$ tranches:

$$\bar{r}_{\text{leg}}^{\text{ann}} = \frac{1}{h} \sum_{j=0}^{h-1} \hat{r}_{\text{leg}}^{\text{ann},j} \tag{B.12}$$

B.4 Spearman Information Coefficient

The **Information Coefficient** (IC) at time step t is the Spearman rank correlation between the model's Q50 (median) quantile forecast and the realised returns:

$$\text{IC}_t = \rho_s\big(\hat{q}_t^{50},\ \boldsymbol{\ell}_t\big) \tag{B.13}$$

where $\hat{q}_t^{50} \in \mathbb{R}^N$ is the vector of median predictions across the universe at step t, and $\boldsymbol{\ell}_t \in \mathbb{R}^N$ is the vector of realised log returns. NaN values (arising from degenerate cross-sections) are replaced with 0.

Over the full test window $[t_0, T)$ the mean and standard deviation of IC are:

$$\overline{\text{IC}} = \frac{1}{T - t_0} \sum_{t=t_0}^{T-1} \text{IC}_t, \qquad \sigma_{\text{IC}} = \text{std}\Big(\{\text{IC}_t\}_{t=t_0}^{T-1}\Big) \tag{B.14}$$

B.5 Adaptive EWMA Regime Detection and Flip Signal

B.5.1 EWMA Update Rule

To detect shifts in signal quality, an Exponentially Weighted Moving Average (EWMA) of past ICs is maintained across rebalance steps. Let $\mathrm{IC}_t^{\text{prev}} = \rho_s\big(\hat{q}_{t-h}^{50},\, \boldsymbol{\ell}_{t-h}\big)$ be the IC observed at the *previous* rebalance step. The running EWMA is updated as:

$$\widetilde{\mathrm{IC}}_t = \begin{cases} \mathrm{IC}_t^{\text{prev}} & \text{if } t \text{ is the first rebalance in this tranche} \\ \alpha\, \mathrm{IC}_t^{\text{prev}} + (1-\alpha)\, \widetilde{\mathrm{IC}}_{t-1} & \text{otherwise} \end{cases} \tag{B.15}$$

where $\alpha \in [0, 1]$ is the smoothing coefficient:

- $\alpha \to 0$: EWMA responds very slowly; effectively the signal is never flipped (used for the "no-flip" baseline, $\alpha = 10^{-9}$).
- $\alpha = 0.1$: moderate adaptation; the EWMA gives most weight to the last ≈ 10 rebalance observations.
- $\alpha \to 1$: EWMA collapses to the most recent IC value only (maximum noise sensitivity).

B.5.2 Signal Inversion (Flip) Logic

After updating $\widetilde{\mathrm{IC}}_t$, a binary flip flag is determined:

$$\mathrm{flip}_t = \begin{cases} \texttt{False} & \text{if } \alpha < \epsilon \quad (\text{no-flip baseline, } \epsilon = 10^{-6}) \\ \mathbf{1}\left[\widetilde{\mathrm{IC}}_t < 0\right] & \text{otherwise} \end{cases} \tag{B.16}$$

When $\mathrm{flip}_t = \texttt{True}$, the ranking signal is negated before stock selection:

$$\tilde{\sigma}_{t,n} = \begin{cases} -\hat{q}_{t,n}^{50} & \text{if } \mathrm{flip}_t = \texttt{True} \\ \hat{q}_{t,n}^{50} & \text{otherwise} \end{cases} \tag{B.17}$$

Stocks are then sorted in ascending order of $\tilde{\sigma}_{t,n}$ to form the ranked candidate list.

The **structural inversion rate** for a tranche j summarises how frequently the signal was flipped:

$$p^{\text{flip},j} = \frac{1}{M_j} \sum_t \mathbf{1}[\text{flip}_t = \texttt{True}] \tag{B.18}$$

B.6 Sector-Constrained Stock Selection

Stocks are selected greedily from the ranked list subject to a maximum of $S_{\max}$ stocks per sector:

$$\mathcal{L}_t = \textsc{Select}\Big(\pi_t^{\downarrow},\ K,\ S_{\max}\Big) \tag{B.19}$$

$$\mathcal{S}_t = \textsc{Select}\Big(\pi_t^{\uparrow},\ K,\ S_{\max}\Big) \tag{B.20}$$

where $\pi_t^{\downarrow}$ is the ranking in *descending* signal order (for the long leg) and $\pi_t^{\uparrow}$ is the reverse (for the short leg). The algorithm walks the ranked list and appends each stock to the selected set only if its sector count has not yet reached $S_{\max}$, halting once K stocks have been chosen.

B.7 Equal-Weight Benchmark

The benchmark NAV is constructed by investing equally in all N universe stocks at each rebalance step, with no transaction costs and no signal:

$$r_t^{\text{bench}} = \frac{1}{N} \sum_{n=1}^{N} \big[\exp(\ell_{t,n}) - 1\big] \tag{B.21}$$

$$\text{NAV}_t^{\text{bench}} = \prod_{s=t_0}^{t} \big(1 + r_s^{\text{bench}}\big) \tag{B.22}$$

Alpha is defined as the cumulative return of the long strategy in excess of this benchmark:

$$\alpha_t = \big(\text{NAV}_t^{\text{long}} - 1\big) \times 100\% - \big(\text{NAV}_t^{\text{bench}} - 1\big) \times 100\% \tag{B.23}$$

B.8 Advanced Risk Metrics

Let $\{d_t\}$ denote the time series of daily arithmetic returns for a given strategy leg, derived from the averaged tranche NAV via $d_t = (\text{NAV}_t/\text{NAV}_{t-1}) - 1$.

B.8.1 Maximum Drawdown (MDD)

$$W_t = \prod_{s=1}^{t}(1+d_s), \qquad \text{MDD} = \min_t \frac{W_t - \max_{s\le t} W_s}{\max_{s\le t} W_s} \tag{B.24}$$

B.8.2 Sortino Ratio

Only returns falling below the daily risk-free rate $r^f = r^f_{\text{ann}}/252$ contribute to downside risk:

$$\sigma^- = \sqrt{252} \cdot \sqrt{\frac{1}{T}\sum_{t=1}^{T} \min\big(0,\ d_t - r^f\big)^2} \tag{B.25}$$

$$\text{Sortino} = \frac{\bar{d} \times 252 - r^f_{\text{ann}}}{\sigma^-} \tag{B.26}$$

An annual risk-free rate of $r^f_{\text{ann}} = 7\%$ is used throughout, consistent with Indian government bond yields over the evaluation period.

B.8.3 Calmar Ratio

The Calmar Ratio is a risk-adjusted performance metric. While other Sharpe Ratio measures return against volatility, the Calmar Ratio measures return strictly against the worst-case historical loss. The formula is as follows:

$$\text{Calmar} = \frac{\bar{d} \times 252}{|\text{MDD}|} \tag{B.27}$$

B.9 Notation Summary

Table B.1: Summary of notation used throughout the appendix.

Symbol	Definition
T	Number of time steps in the test window
N	Number of stocks in the investment universe
h	Rebalancing horizon (fixed at 7 trading days)
K	Maximum stocks per portfolio leg (top-K long / bottom-K short)
$S_{\max}$	Maximum stocks per sector per leg
$\ell_{t,n}$	Raw log return of stock n at step t (`actuals`); used for all P&L
$r_{t,n}$	Arithmetic return derived from actuals: $\exp(\ell_{t,n}) - 1$
$y_{t,n}$	Volatility-adjusted target (`vol_adj`); used only for calibration
$\hat{q}^q_{t,n}$	Predicted quantile $q \in \{10, 50, 90\}$ for stock n at step t; in vol-adj space
$\mathbf{IC}_t$	Spearman rank correlation between $\hat{q}^{50}_t$ and $\boldsymbol{\ell}_t$
$\widetilde{\mathbf{IC}}_t$	EWMA-smoothed IC used for regime detection
α	EWMA smoothing coefficient
$\mathbf{flip}_t$	Binary indicator: 1 if signal is inverted at step t, else 0
c	One-way transaction cost rate (default 0.05%)
$\tau_t^{\mathbf{leg}}$	Turnover fraction for a leg at step t
$\mathcal{L}_t,\ \mathcal{S}_t$	Index sets of selected long / short stocks at step t
$\mathbf{NAV}_t^{\mathbf{leg}}$	Compounded net asset value for a strategy leg at step t
MDD	Maximum drawdown
σ^-	Annualised downside standard deviation
$r^f_{\mathbf{ann}}$	Annual risk-free rate (7%)

Appendix C

Dynamic Graph Pruning and Structural Model Comparison

C.1 Dynamic Edge Pruning Mechanism and Loss Formulation

To mitigate the impact of noisy or spurious correlations in highly interconnected financial networks, a dynamic graph pruning step was introduced into the training engine. This was operationalized via a `LearnableEdgeMask` module, which scores each edge based on temporal node features.

For each edge $e_{i,j}$ connecting nodes i and j, the network learns a continuous mask score $m_{i,j} \in [0, 1]$ via a multi-layer perceptron scorer. To force the network to drop low-importance connections and retain only the most robust predictive pathways, a sparsification penalty is applied directly to the loss function. The standard hybrid loss ($\mathcal{L}_{hybrid}$) is modified by adding an L1 regularization term over the learned edge masks:

$$\mathcal{L}_{total} = \mathcal{L}_{hybrid} + \lambda \sum_{(i,j)\in\mathcal{E}} |m_{i,j}|$$

where λ represents the Sparsity Index (penalty weight) optimized during hyperparameter tuning. Edges with a mask score $m_{i,j}$ falling below a predefined 50% percentile threshold are actively pruned from the graph before the message-passing layers.

The objective of this modified loss function is to allow the network topology to adapt fluidly to changing market regimes while preserving fundamental structural priors. Notably, the pruning mechanism acts as an asymmetric filter: highly volatile price-

correlation edges are actively pruned or added depending on rolling market conditions, while robust, deterministic connections (such as intra-sector groupings) remain structurally intact.

C.2 Comparative Structural Results

Two distinct spatio-temporal architectures were evaluated under this dynamic pruning regime: a HYBRID model (combining GATv2 with LSTM layers) and a ROLAND model (combining GATv2 with GRU cells). During the tuning phase, the optimal Sparsity Index (λ) diverged between the architectures: the HYBRID model settled on a lower penalty ($\lambda \approx 0.00039$), while the ROLAND model required a stricter sparsification regime ($\lambda \approx 0.00253$).

Table C.1 provides a deep-dive evaluation of the models on the unseen test set, demonstrating the profound impact of portfolio allocation strictness by comparing an unconstrained regime ($\alpha = 1e^{-9}$) against a rank-aware regime ($\alpha = 0.1$).

C.3 Inference and Discussion

The integration of the dynamic pruning engine yielded several critical insights regarding network topology and predictive behavior:

- **Topological Fluidity vs. Strictness:** ROLAND's stricter sparsity penalty ($\lambda \approx 0.00253$) resulted in slightly higher structural volatility (Jaccard of 0.401 compared to HYBRID's 0.425), highlighting that aggressive pruning forces the model to constantly rewire its non-deterministic edges to adapt to rolling windows.
- **Sensitivity to Allocation Constraints (α):** The table exposes a profound sensitivity to the α constraint. Under the near-unconstrained regime ($\alpha = 1e^{-9}$), both models suffered severely negative combined returns (-8.9% and -6.2%) with heavy drawdowns. However, tightening the constraint to $\alpha = 0.1$ drastically improved performance across the board, pushing HYBRID's long leg from -0.4% to +13.4%, and ROLAND's from +9.1% to +17.2%. This empirically demonstrates that raw model signal must be strictly coupled with rank-aware portfolio weighting to be viable for any live trading tasks.
- **Signal Asymmetry and Inversion:** The IC Quintile Diagnostics expose a stark contrast in signal origin. ROLAND derived the vast majority of its signal

from a strong positive asymmetry (+0.1187), performing well at identifying long targets. HYBRID exhibited an inverted negative asymmetry (-0.0460). However, at $\alpha = 0.1$, both models takes hit from high "Structural Inversion" (38.6% for HYBRID; 52.5% for ROLAND) where targeted bottom-quintile shorts generated positive average returns (+0.0072 for HYBRID; +0.0055 for ROLAND) due to a prevailing market-wide bull regime (+14.5% benchmark).

- **Risk-Adjusted Performance:** While ROLAND's Long Leg achieved superior raw returns (+17.2% cumulative at $\alpha = 0.1$) and a higher HAC Sharpe ratio (1.370 vs 1.126), its Short Leg suffered a severe max drawdown of -27.11%. Ultimately, HYBRID provided a marginally better combined L/S cumulative return (+3.6% vs +0.9%) by maintaining tighter drawdowns on the short side. Notably, none of the L/S Spread Sharpe ratios achieved statistical significance under Newey-West bounds ($p > 0.05$).

Table C.1: Comprehensive Topological, Predictive, and Financial Evaluation (Test Set) Max_Sector=2, Top_K=5

Metric	**HYBRID (GAT+LSTM)**		**ROLAND (GAT+GRU)**	
	$\alpha = 1e^{-9}$	$\alpha = 0.1$	$\alpha = 1e^{-9}$	$\alpha = 0.1$
Panel A: Model & Structural Characteristics (Pre-Allocation)				
Sparsity Index (λ)	0.00039		0.00253	
Mean Jaccard Similarity	0.425		0.401	
Panel B: IC Diagnostics & Distribution Coverage (Pre-Allocation)				
Spearman Mean IC (Tradeable Window)	-0.0354 (Std: 0.1611)		-0.0180 (Std: 0.1308)	
IC Q50 (Native Vol-Adj)	-0.0349 ± 0.1659		-0.0119 ± 0.1311	
IC Q50 (Raw Net Returns)	-0.0270 ± 0.1667		-0.0162 ± 0.1301	
Coverage Q10 / Q50 / Q90	0.9733 / 0.7684 / 0.1651		0.9116 / 0.5811 / 0.1120	
Coverage (Ideal Target)	0.9000 / 0.5000 / 0.1000		0.9000 / 0.5000 / 0.1000	
Top Quintile IC (Hit Rate)	-0.0069 (45.7%)		+0.0171 (46.7%)	
Bottom Quintile IC (Hit Rate)	+0.0391 (53.8%)		-0.1016 (39.1%)	
Signal Asymmetry	-0.0460		+0.1187	
Panel C: Cumulative & Annualized Returns				

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Table C.1 – continued from previous page

Metric	HYBRID (GAT+LSTM)		ROLAND (GAT+GRU)	
	$\alpha = 1e^{-9}$	$\alpha = 0.1$	$\alpha = 1e^{-9}$	$\alpha = 0.1$
Avg. Annual Return (Long)	-0.25%	+17.75%	+11.79%	+22.71%
Avg. Annual Return (Short)	-22.63%	-8.77%	-26.36%	-19.07%
Avg. Annual Return (Neutral)	-11.20%	+4.76%	-7.81%	+1.25%
Cum. Return (Long Leg)	-0.4%	+13.4%	+9.1%	+17.2%
Cum. Return (Short Leg)	-18.3%	-7.1%	-21.3%	-15.3%
Cum. Return (L/S Combined)	-8.9%	+3.6%	-6.2%	+0.9%
Market Benchmark Return	+14.5%	+14.5%	+14.5%	+14.5%
Structural Inversion (%)	0.00%	38.63%	0.00%	52.55%
Panel D: Statistical Significance & Hit Rates				
Hit Rate (Long Leg)	50.3%	54.8%	54.8%	58.9%
Hit Rate (Short Leg)	41.6%	46.7%	38.1%	41.6%
Hit Rate (L/S Spread)	38.6%	51.8%	44.7%	51.3%
HAC Sharpe (Long Leg)	0.030	1.126	0.859	1.370
NW p-value	*0.937*	*0.291*	*0.410*	*0.204*
HAC Sharpe (Short Leg)	-1.998	-0.647	-1.839	-1.507

Continued on next page

Table C.1 – continued from previous page

Metric	**HYBRID (GAT+LSTM)**		**ROLAND (GAT+GRU)**	
	$\alpha = 1e^{-9}$	$\alpha = 0.1$	$\alpha = 1e^{-9}$	$\alpha = 0.1$
NW p-value	***0.089*** †	*0.611*	*0.116*	*0.204*
HAC Sharpe (L/S Spread)	-1.557	0.587	-1.445	-0.003
NW p-value	*0.205*	*0.512*	*0.257*	*0.837*
Panel E: Advanced Risk Metrics				
Max Drawdown (Long)	-21.50%	-15.98%	-11.98%	-15.19%
Max Drawdown (Short)	-19.08%	-16.36%	-33.50%	-27.11%
Max Drawdown (Combined)	-14.95%	-6.31%	-9.70%	-3.95%
Sortino Ratio (Combined)	-5.221	-0.841	-4.890	-2.275
Calmar Ratio (Combined)	-0.769	0.705	-0.812	0.294

C.4 Training Dynamics and Cumulative Returns

The empirical dynamics of the pruning regime can be visually validated through the convergence behavior of the models and their resulting financial trajectories. The following figures illustrate the training loss and validation Information Coefficient (IC) curves across epochs, alongside the cumulative return distributions for the optimized Long/Short portfolios.

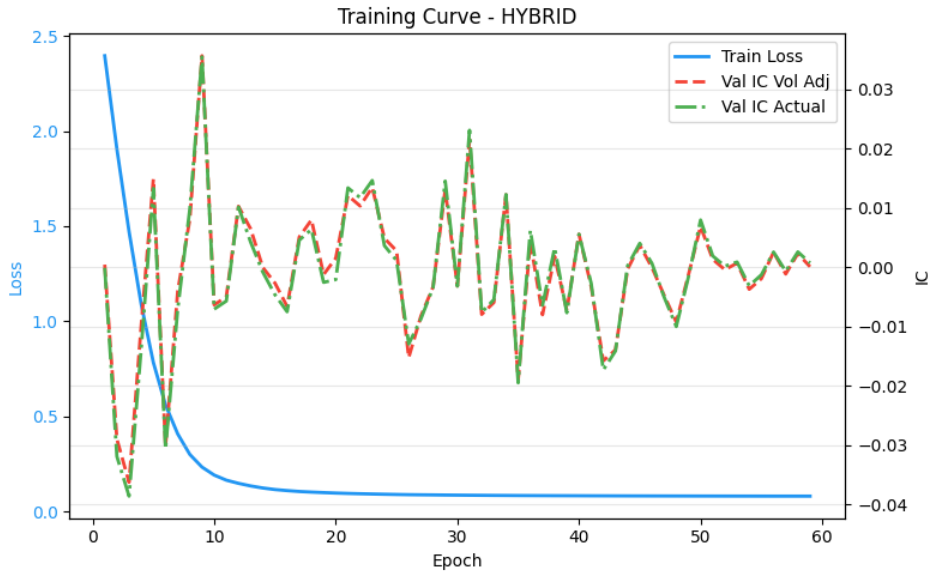


Figure C.1: Training Loss and Validation IC convergence for the HYBRID Architecture.

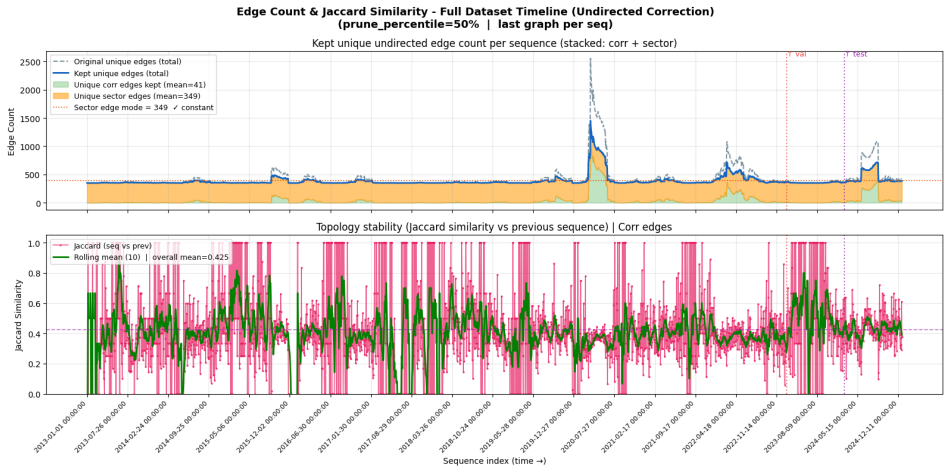
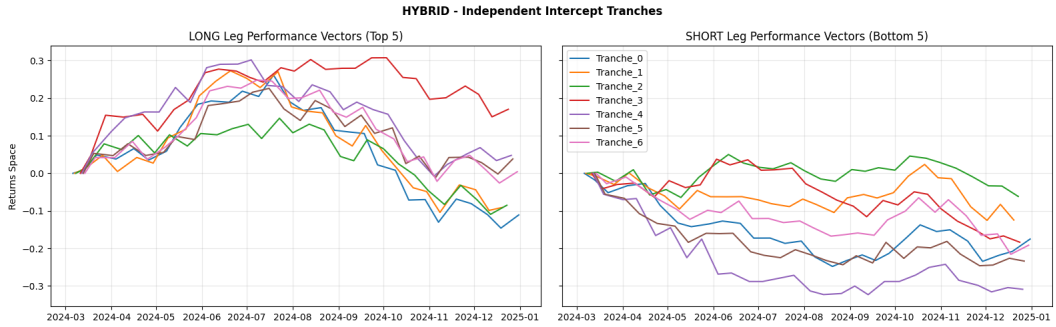
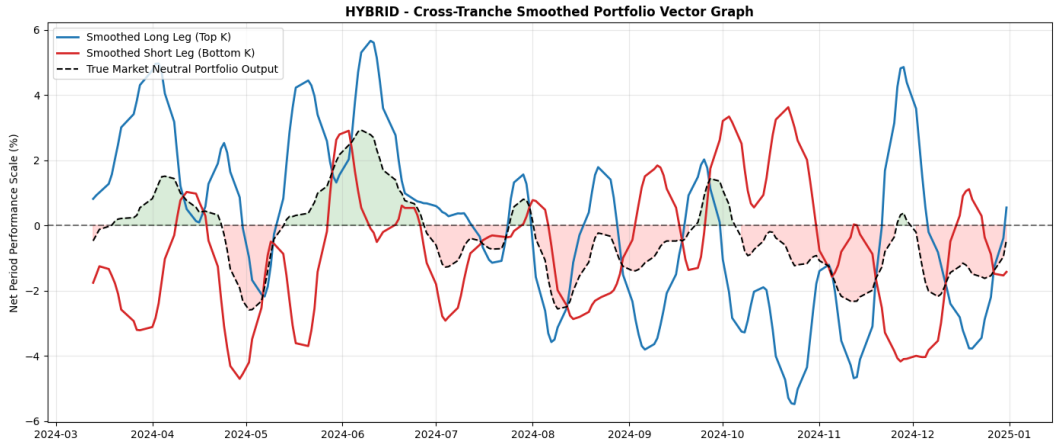


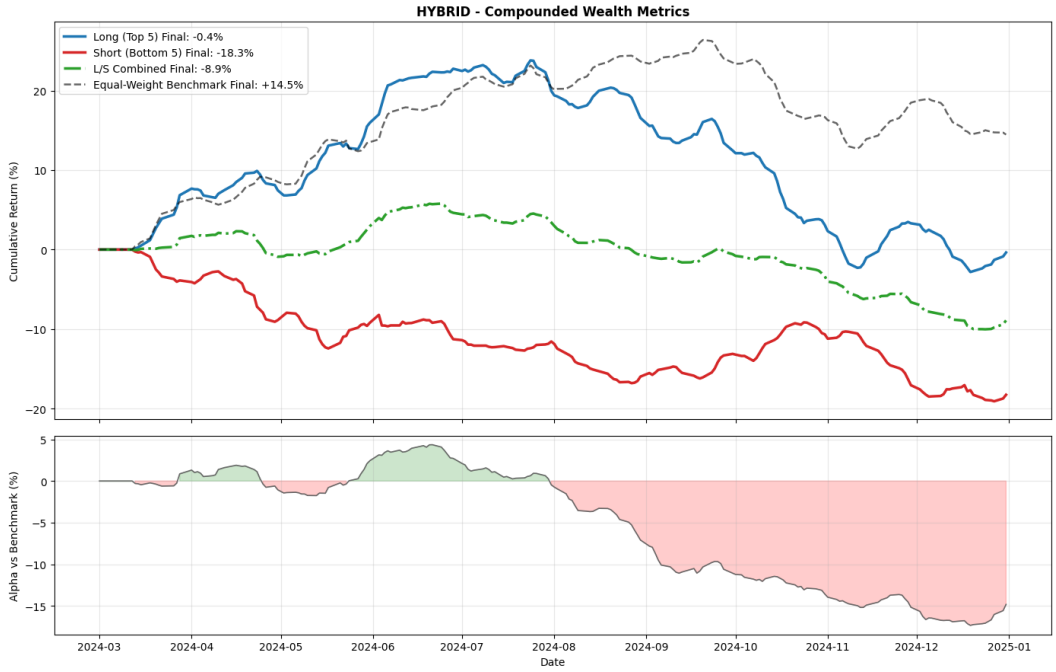
Figure C.2: Edge Distribution Before and After Pruning for GAT Hybrid Model



(a) Independent Intercept Tranches (Long and Short Legs)

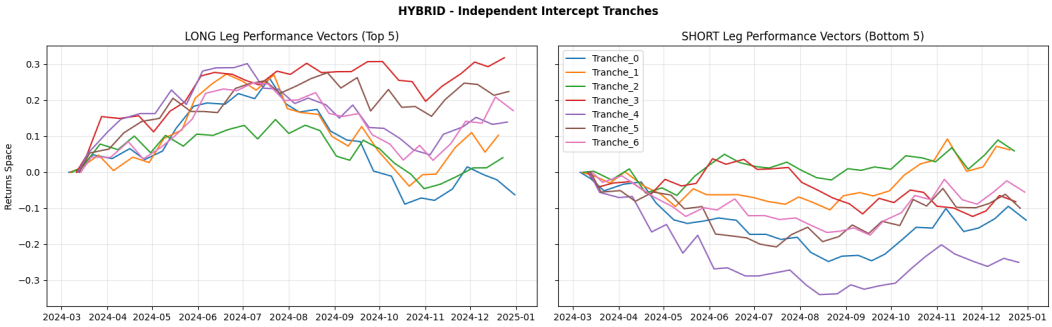


(b) Cross-Tranche Smoothed Portfolio Vector Graph

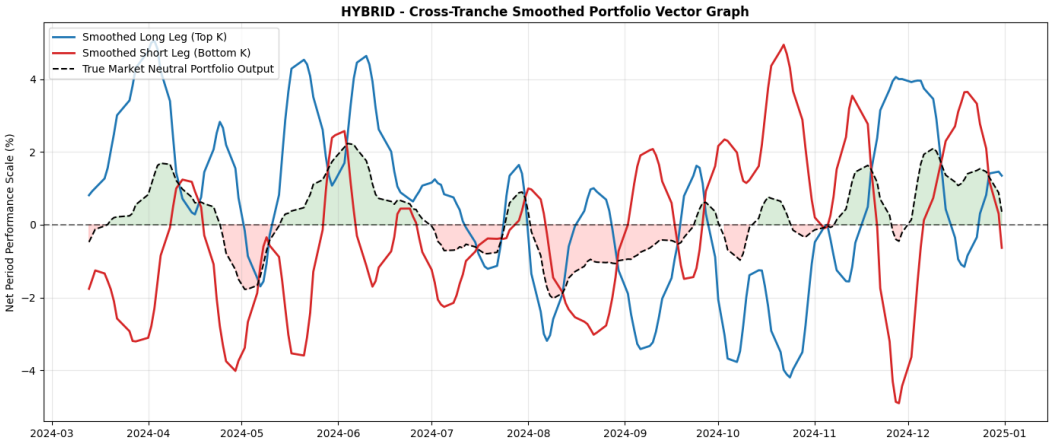


(c) Compounded Wealth Metrics and Alpha vs. Benchmark

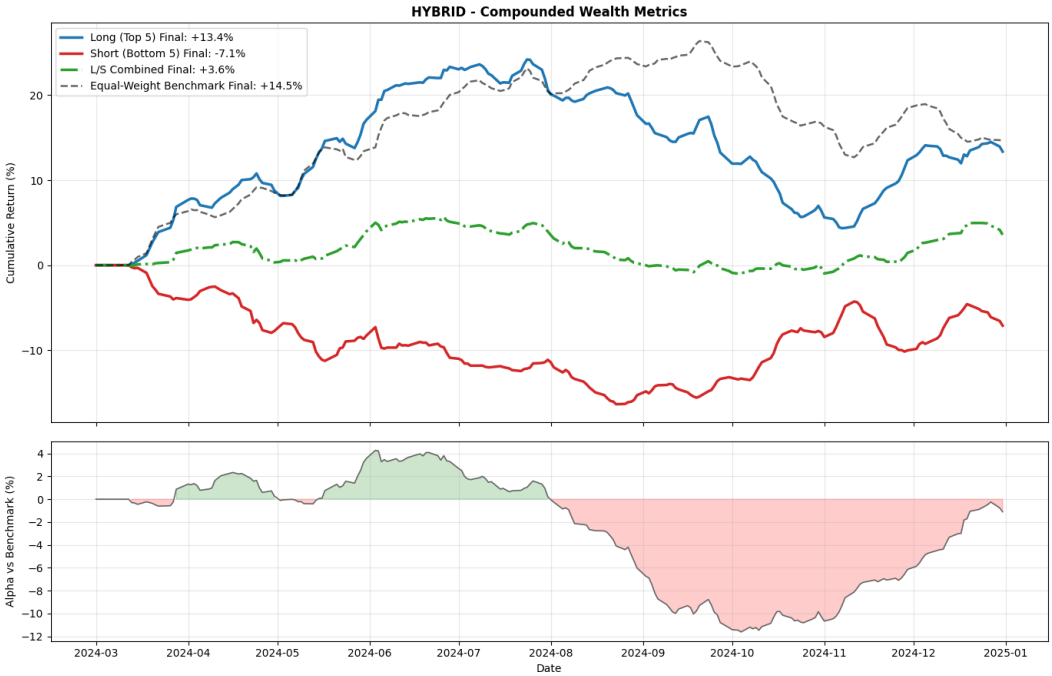
Figure C.3: Comprehensive Portfolio Performance Analysis for the HYBRID Architecture (No Rank Flip for Trading $\alpha = 1e - 9$)



(a) Independent Intercept Tranches (Long and Short Legs)



(b) Cross-Tranche Smoothed Portfolio Vector Graph



(c) Compounded Wealth Metrics and Alpha vs. Benchmark

Figure C.4: Comprehensive Portfolio Performance Analysis for the HYBRID Architecture (Rank Flip for Trading $\alpha = 0.1$)

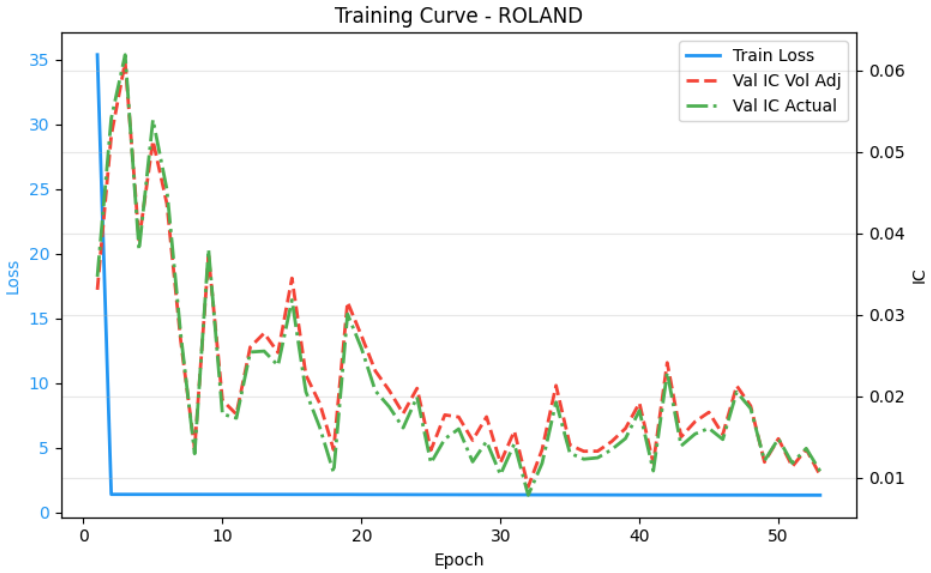


Figure C.5: Training Loss and Validation IC convergence for the ROLAND Architecture.

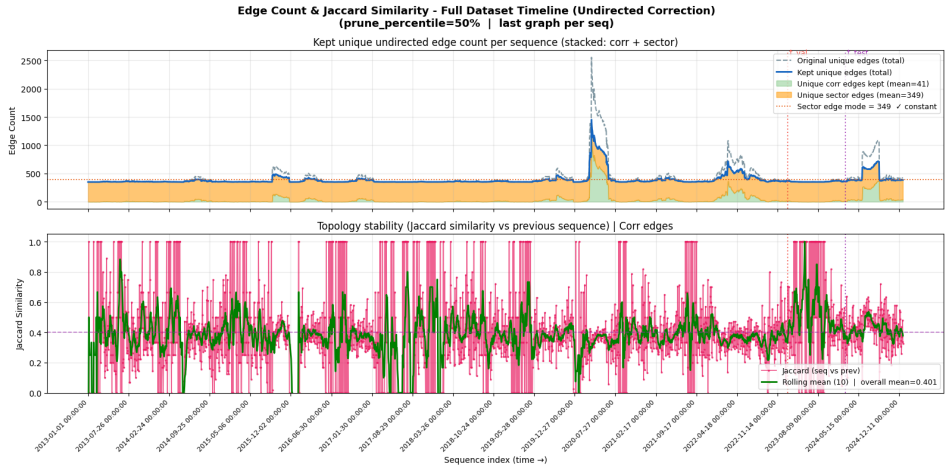
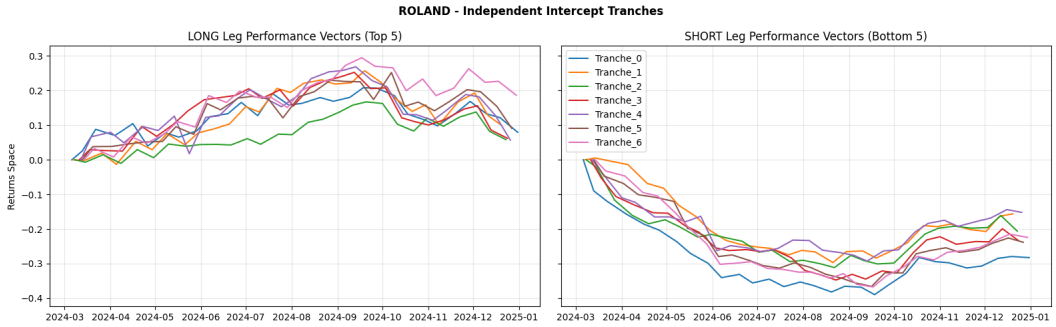
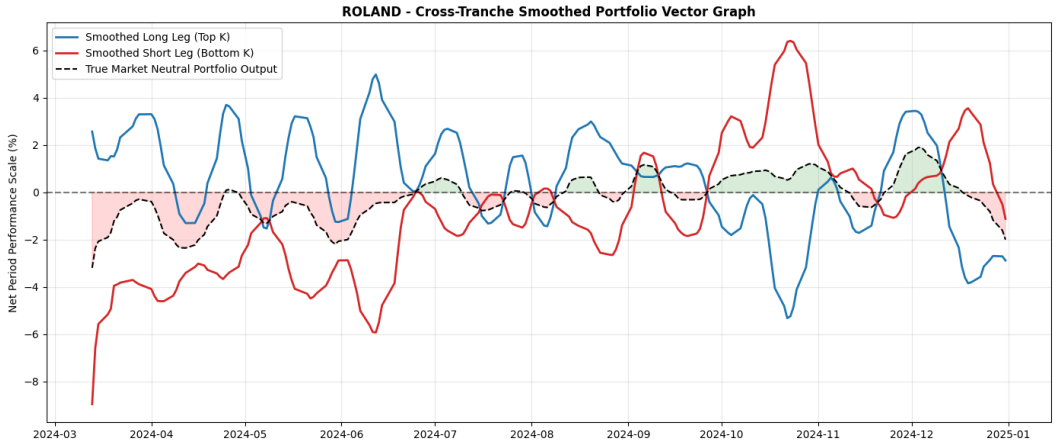


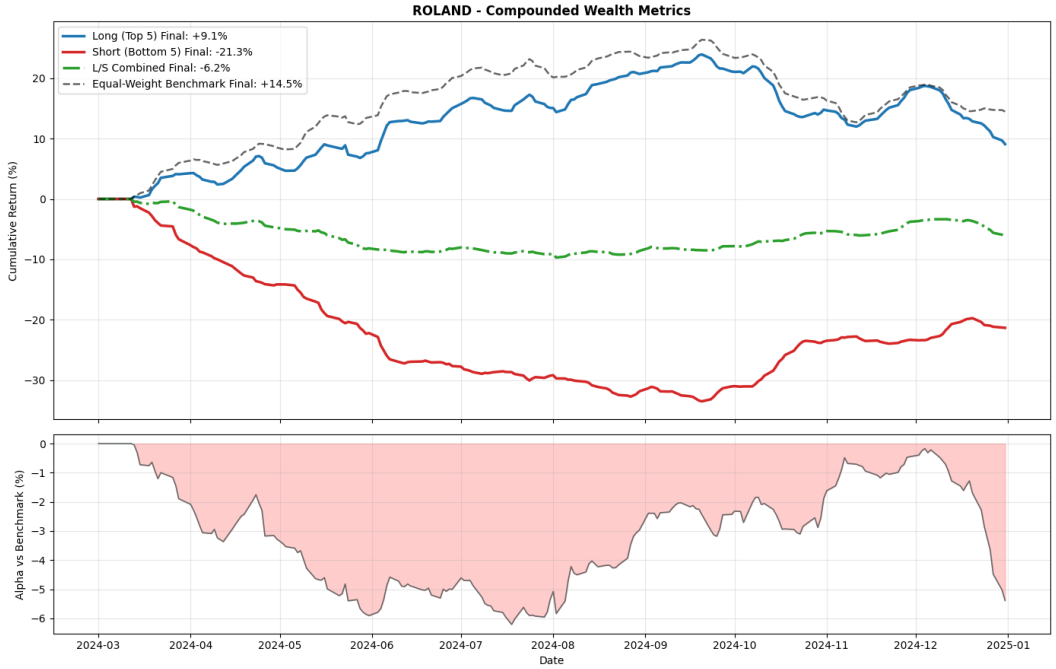
Figure C.6: Edge Distribution Before and After Pruning for ROLAND Model



(a) Independent Intercept Tranches (Long and Short Legs)

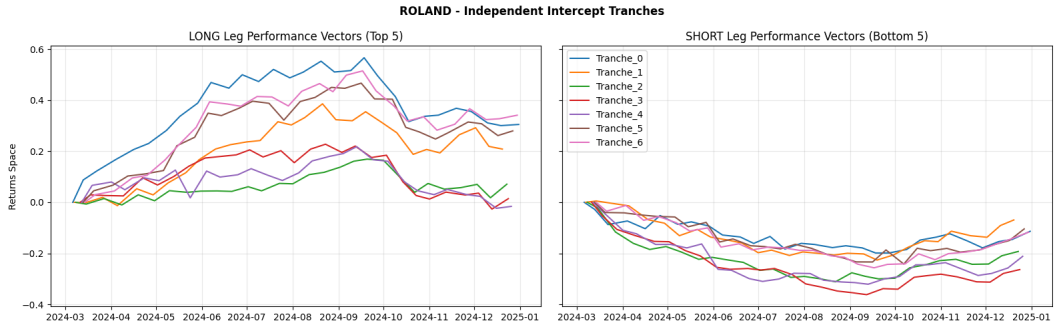


(b) Cross-Tranche Smoothed Portfolio Vector Graph

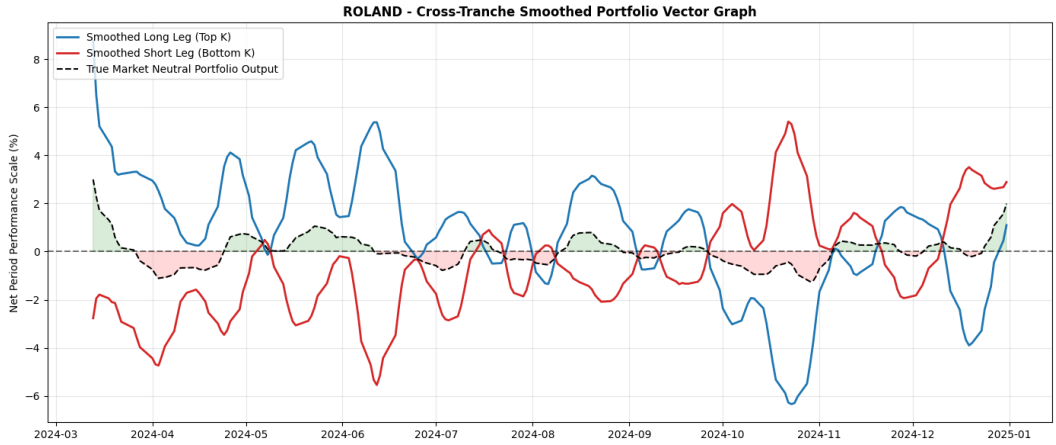


(c) Compounded Wealth Metrics and Alpha vs. Benchmark

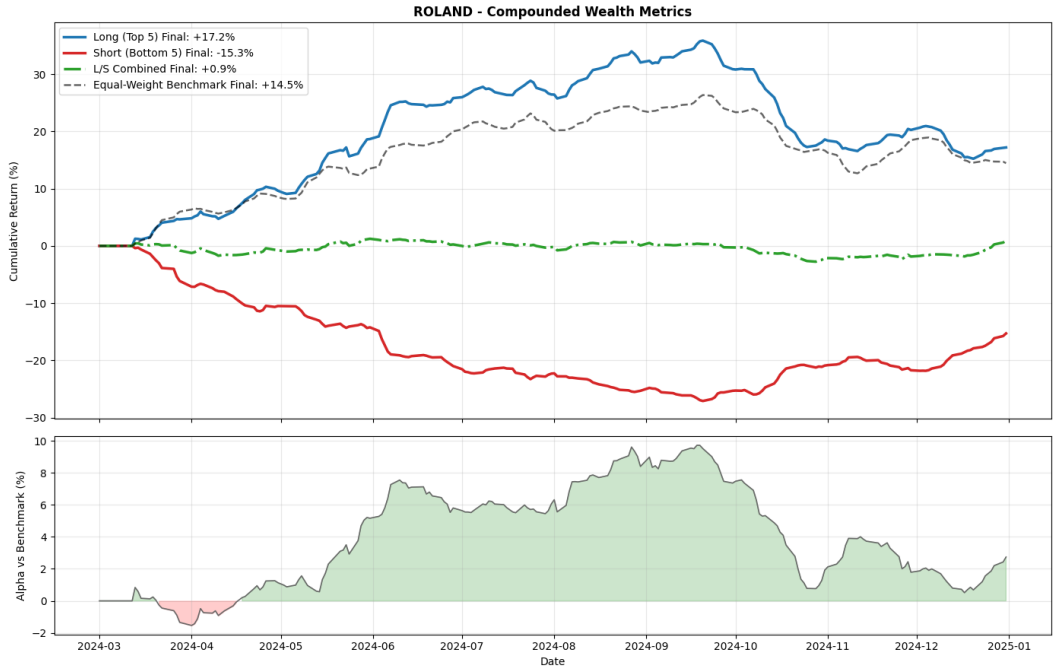
Figure C.7: Comprehensive Portfolio Performance Analysis for the ROLAND Architecture (No Rank Flip for Trading $\alpha = 1e - 9$).



(a) Independent Intercept Tranches (Long and Short Legs)



(b) Cross-Tranche Smoothed Portfolio Vector Graph



(c) Compounded Wealth Metrics and Alpha vs. Benchmark

Figure C.8: Comprehensive Portfolio Performance Analysis for the ROLAND Architecture (Rank Flip for Trading $\alpha = 0.1$)

Bibliography

[1] Gandhmal, D. P., & Kannan, K. (2019). Systematic analysis and review of stock market prediction techniques. *Computer Science Review*, 34. https://doi.org/10.1016/j.cosrev.2019.08.001

[2] Senapati, M. R., Das, S., & Mishra, S. (2018). A novel model for stock price prediction using hybrid neural network. *Journal of the Institution of Engineers (India): Series B*, 99(6), 555–563. https://doi.org/10.1007/s40031-018-0343-7

[3] Thakkar, A., & Chaudhari, K. (2021). Fusion in stock market prediction: A decade survey on the necessity, recent developments, and potential future directions. *Information Fusion*, 65, 95–107. https://doi.org/10.1016/j.inffus.2020.08.019

[4] Ran, X., Shan, Z., Fan, Y., & Gao, L. (2024). A model based LSTM and graph convolutional network for stock trend prediction. *PeerJ Computer Science*, 10, e2326. https://doi.org/10.7717/peerj-cs.2326

[5] Tadoori, G., & Yakaiah, G. (2020). An introduction to quantamental investing. In *Two Day National Seminar on "Fin-tech Adoption in the Indian BFSI Sector-Opportunities and Challenges", 19-20 March 2019* (pp. 208–212). Professional Book Publishers. https://doi.org/10.2139/ssrn.3704670

[6] Xie, C., Zhang, Y., Wang, M., & Liu, Z. (2023). Quantamental trading: Fundamental and quantitative analysis with multi-factor regression model strategy. In C. T. Dang, J. Cifuentes-Faura, & X. Li (Eds.), *Proceedings of the 2nd International Conference on Business and Policy Studies (CONF-BPS 2023)* (pp. 1455–1470). Springer. https://doi.org/10.1007/978-981-99-6441-3_131

[7] Rahi, P., Patel, D., Prabhakar, S., Srivastava, R., Bhargav, & Singh, P. (2024). Predictive analytics for stock markets using graph neural networks (GNNs). In *2024 International Conference on Progressive Innovations in Intelligent Systems and Data Science (ICPIDS)* (pp. 531–538). IEEE. https://doi.org/10.1109/ICPIDS65698.2024.00088

[8] Sonani, M. S., Badii, A., & Moin, A. (2025). Stock price prediction using a hybrid LSTM-GNN model: Integrating time-series and graph-based analysis. arXiv. https://arxiv.org/abs/2502.15813

[9] Korablyov, M., Fomichov, O., Kobzev, I., Antonov, D., & Tkachuk, O. (2025). Stock market price forecasting using evolving graph neural network. In *Proceedings of the International Workshop on Computational Intelligence, co-located with the IV International Scientific Symposium "Intelligent Solutions" (IntSol-2025)*. CEUR-WS. https://ceur-ws.org/Vol-4035/Paper6.pdf

[10] You, J., Du, T., & Leskovec, J. (2022). ROLAND: Graph learning framework for dynamic graphs. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)* (pp. 2358–2366). Association for Computing Machinery. https://doi.org/10.1145/3534678.3539300

[11] Bukhari, M., Maqsood, M., & Sattar, A. (2025). A novel inter-intra graph neural networks for stock price forecasting modeling cross-border relationships. *Expert Systems with Applications*, 286, 127907. https://doi.org/10.1016/j.eswa.2025.127907

[12] Jia, S., Gao, H., Huang, J., Liu, Y., & Li, S. (2025). CIRGNN: Leveraging Cross-Chart Relationships with a Graph Neural Network for Stock Price Prediction. *Mathematics*, 13(15), 2402. https://doi.org/10.3390/math13152402

[13] Chavan, R. (2025). Multimodal agentic AI architecture for high frequency trading using reinforcement learning and temporal graph encoders. TechRxiv. https://doi.org/10.36227/techrxiv.175037093.37731185/v1

[14] Karpoff, J. M. (1987). The relation between price changes and trading volume: A survey. *The Journal of Financial and Quantitative Analysis*, 22(1), 109–126. http://www.jstor.org/stable/2330874

[15] Bouchaud, J. P., Gefen, Y., Potters, M., & Wyart, M. (2004). Fluctuations and response in financial markets: the subtle nature of 'random' price changes. *Quantitative Finance*, 4(2), 176–190. https://doi.org/10.1080/14697680400000022

[16] Andersen, T. G. (1996). Return Volatility and Trading Volume: An Information Flow Interpretation of Stochastic Volatility. *The Journal of Finance*, 51(1), 169–204. https://doi.org/10.2307/2329306

[17] Lillo, F., Farmer, J. D., & Mantegna, R. N. (2003). Econophysics: Master curve for price-impact function. *Nature*, 421(6919), 129–130. https://doi.org/10.1038/421129a

[18] Zhang, F., Guo, R., & Cao, H. (2020). Information coefficient as a performance measure of stock selection models. arXiv. https://arxiv.org/abs/2010.08601

[19] Financial Edge Team. (2025). Information coefficient (IC). FE Training. Retrieved February 25, 2026, from https://www.fe.training/free-resources/portfolio-management/information-coefficient-ic/

[20] Kirsten, E. (2026, March 5). Mastering the information coefficient: Your key to smarter factor investing. *Coding Nexus*. Retrieved March 10, 2026, from https://medium.com/coding-nexus/mastering-the-information-coefficient-your-key-to-smarter-factor-investing-244531e45538

[21] AnalystPrep. (2024, June 4). Quantitative investing. Retrieved February 10, 2026, from https://analystprep.com/study-notes/cfa-level-iii/quantitative-investing/

[22] Bajaj Finserv Asset Management Company. (2025, November 12). Information coefficient. *Bajaj Finserv AMC Knowledge Centre*. Retrieved February 10, 2026, from https://www.bajajamc.com/knowledge-centre/information-coefficient

[23] Brody, S., Alon, U., & Yahav, E. (2022). How attentive are graph attention networks? arXiv preprint arXiv:2105.14491. https://doi.org/10.48550/arXiv.2105.14491

[24] Velazquez Bustamante, A. (2025, March 25). KFold cross validation with purging and embargo: The ultimate cross validation technique for time series. *Medium*. https://antonio-velazquez-bustamante.medium.com/kfold-cross-validation-with-purging-and-embargo-the-ultimate-cross-validation-technique-for-time-2d656ea6f476

[25] Lo, A. W. (2002). The Statistics of Sharpe Ratios. Financial Analysts Journal, 58(4), 36–52. https://doi.org/10.2469/faj.v58.n4.2453